

# Careers in Firms: The Role of Learning about Ability and Human Capital Acquisition

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Elena Pastorino

*Stanford University, Hoover Institution, and National Bureau of Economic Research*

Job and wage mobility can arise from firms and workers learning about workers' ability and from workers acquiring human capital with experience. To date, the relative importance of these two mechanisms is debated. Using administrative data from one firm, I estimate a structural model that integrates them. I find the direct effect of beliefs about ability on wages, which existing work has focused on, to be small. However, by improving the sorting of workers to the firm's jobs, learning about ability is indirectly a crucial determinant of wage growth and dispersion.

## I. Introduction

A major question in labor economics is how wages are determined and evolve over time. Wage growth is known to be closely related to job mobility (Topel and Ward 1992; Keane and Wolpin 1997) not just between firms, as workers pursue better employment opportunities through turnover (Buchinsky et al. 2010; Bagger et al. 2014), but also within firms, as workers advance through a firm's internal hierarchy of jobs (Lazear 1992, 2009;

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Gibbons and Waldman 1999a, 1999b, 2006; Lazear and Shaw 2007). For instance, Baker, Gibbs, and Holmström (1994a) document that promotions on average entail an immediate 6% increase in wages relative to not being promoted. Baker, Gibbs, and Holmström (1994b) further estimate that after 10 years, a promotion amounts to a wage premium of 20% compared with the wages of those who remain at the lowest hierarchical level (see also Waldman 2013).<sup>1</sup> Workers who are promoted or receive wage increases once are more likely to experience future promotions and wage increases. Thus, understanding the determinants of job mobility in firms is central to explaining wage growth.

Yet the dynamics of wages within firms cannot be described solely by the progression of workers through a firm's job hierarchy. For example, a large fraction of the changes in real wages that workers experience yearly in a firm are negative, which leads both to a sizable variability in individual wages over time and to a significant dispersion in wages at any level of the job hierarchy (Waldman 2013). Uncovering the sources of this variability is key to interpreting the forces that shape wage profiles, the nature of labor income risk, and ultimately the origins of inequality.

A common interpretation of these features of careers is that they arise from a gradual process of learning about workers' ability by firms and workers, as ability may be uncertain when workers enter the labor market, and of human capital acquisition by workers with experience in the labor market. Intuitively, as workers discover their talents through success and failure on the job and accumulate new skills, they eventually settle in the jobs and firms that best match and most reward their productivity. Indeed, learning and human capital acquisition have long been recognized as crucial sources of job and wage mobility. To date, though, their relative importance is debated (Neal and Rosen 2000; Rubinstein and Weiss 2006; Neal 2017).

**The goal of this paper is to assess the importance for careers of learning about ability and human capital acquisition within a model that accounts for detailed patterns of job and wage mobility in firms.** Specifically, I estimate how workers sort into firms and jobs and the extent to which the resulting assignment and wage process is governed by the acquisition of new information about workers' ability and of new skills by workers. I exploit rich panel data on the wages, jobs, and job performance of all managers of a US firm first analyzed by Baker, Gibbs, and Holmström (1994a, 1994b). From these data, I infer how the organization of production within a firm, as reflected in its job hierarchy, and the allocation of workers to jobs affect the accumulation of information about workers' ability and of

<sup>1</sup> Baker, Gibbs, and Holmström (1994b) further estimate that after 10 years, two promotions are worth a wage premium of about 40%, three promotions of 78%, and four promotions of 127%.

workers' human capital and, conversely, how this process of information and human capital acquisition influences managers' job and wage paths.<sup>2</sup>

One reason why little is known about the relative role of these mechanisms of wage growth and dispersion is the challenge of the required measurement exercise. Conceptually, when jobs differ in the output they generate and in the opportunities they offer for information and human capital acquisition, firms choose jobs for workers, and workers sort into firms and jobs, in the face of repeated trade-offs between current and future output and wages. As a result, firms and workers might be willing to *experiment—that is, sacrifice current output and wages in order to acquire more precise information about ability*. This problem of information acquisition is non-standard whenever ability is at least partly transferable across firms or jobs. Empirically, in these instances, job and wage mobility are the result of a complex dynamic selection process, as workers match with the firms and jobs that best fit their accumulating information and skills and in turn determine their future information and skills. In this paper, I show that such a process, largely ignored by the empirical literature, is the critical channel through which learning affects wages.

Three main findings emerge from the analysis. First, in contrast to the literature (Gibbons et al. 2005; Lluís 2005; Hunnes 2012), I find the contribution of learning to wages to be sizable: the additional wage growth and dispersion due to it amount to one-quarter of the cumulative wage growth and dispersion over the first 7 years at the firm—the rest is due to human capital acquisition. This novel result follows from accounting for both the direct and the indirect effects of learning on wages. I estimate that the direct effect—capturing the impact of beliefs about ability in a period on the period wages—is small, as is consistent with the literature, which has focused solely on it. But by estimating the processes of beliefs, human capital, jobs, and wages, I can also measure the indirect effect of learning on wages resulting from its impact on the dynamics of promotions. I find that this effect is large and responsible for almost the entire impact of learning on wages.

This indirect effect, which is reminiscent of the sorting mechanisms in Heckman and Singer (1984) and Cameron and Heckman (1998, 2001), operates through the endogenous process by which managers are progressively selected to higher levels of the job hierarchy, at which higher ability and acquired skills tend to be more valuable.<sup>3</sup> Intuitively, as both

<sup>2</sup> The BGH data are well known in the literature because they contain information on performance and many of their features, which have been replicated by other studies, are now considered stylized facts about careers. See Waldman (2013) and Frederiksen, Kriechel, and Lange (2017) on this point.

<sup>3</sup> These authors emphasize the role of dynamic selection on observed and unobserved attributes of individuals and available alternatives for educational attainment and careers. The idea that selection is a major force shaping careers builds on the original insight of

information about managers' ability and their human capital accumulate and are revealed by performance, managers who perform well advance to the jobs that are most suited to their productive characteristics. As a result, wages at higher-level jobs are higher and, by reflecting managers' differing information and human capital paths, more dispersed than those at lower-level jobs. This sorting process, of which learning is a key source, is critical to the growth and dispersion of wages with tenure in the firm. The patterns of careers in the Baker, Gibbs, and Holmström (BGH) data suggest that this indirect effect is present: higher performance leads to promotions to jobs at which managers are paid higher and more variable wages.

The second main finding is that the differential informativeness of jobs about ability, typically absent from models of careers, is a crucial determinant of wage profiles. To see why, note that according to my estimates, the firm's lowest-level job, at which managers are hired, is the most informative, but wages at this level are the lowest. Given the informational advantage of this level, the firm employs managers at this level even when beliefs about ability imply that they would be more productive at higher levels—this force *depresses* wages in *early* tenures relative to the wages in the standard case of equally informative jobs. Yet as learning about ability is faster at this entry level, when managers are promoted to higher levels, beliefs about their ability and their wages are on average higher than in the case of equally informative jobs—this force *increases* wages in *later* tenures. Hence, the different informativeness of jobs helps explain why wage returns may be small in early tenures; accounts for the markedly convex relationship between wages and job levels, or tenure, in firms (Waldman 2013); and rationalizes the wage increases paid at promotion even to experienced workers—a puzzle for learning models (Gibbons and Waldman 2006)—as compensating differentials for switching from more to less informative jobs.<sup>4</sup>

The third main finding concerns the measurement of the role of learning, especially when jobs vary in their informativeness about ability and firms differ in their productivity. To elaborate, the firm I study acquires information about managers' ability by first employing them at jobs that, although they contribute little to output, are particularly informative about

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Heckman (1979) and on Heckman (1981, 1990), Heckman and Sedlacek (1985, 1990), Heckman and Honoré (1990), and Sauer (1998). Search frictions and performance incentives are also sources of wage growth and dispersion. Like most work on the careers of nonexecutives, my analysis focuses on learning about ability and human capital acquisition. See Gayle, Golan, and Miller (2015) for a model of promotions, turnover, and performance incentives that accounts for the relationship between executive pay and firm size.

<sup>4</sup> In this case, wage premia at promotion compensate workers for the discrete loss in the option value of future learning about ability implied by a promotion and so can be large even for very small changes in uncertainty about ability, which are the most likely for experienced workers. See also Pastorino (2015).

ability. Since such jobs pay low wages, learning delays promotions to higher-paying positions. The speed of promotions then does not necessarily reflect the speed or amount of learning that takes place before a promotion occurs, contrary to what is often argued. In particular, it may take time for learning to have a positive effect on wages through promotions. As a result, to measure the impact of learning on wages, it is critical to capture its cumulative effect on job assignment.<sup>5</sup>

Popular instrumental variable (IV) methods developed by Gibbons et al. (2005) for models of comparative advantage and learning cannot be used for this purpose though, as they abstract from the estimation of the dynamics of beliefs about ability and of job assignment. When firms idiosyncratically or systematically differ in their productivity, I show that in general, these methods cannot be applied to estimate the direct effect of learning on wages either. The reason is as follows. These methods exploit variables such as a worker's past jobs as instruments for a worker's current job in the estimated wage equation, once beliefs about ability—the persistent unobserved individual effect—are quasi-differenced out from it by analogy with usual panel data techniques. Indeed, past jobs provide natural instruments. First, they are correlated with a worker's current job, as job choices depend on ability and thus on beliefs about it. Second, past jobs are uncorrelated with the error term of the wage equation, although new performance information—which is part of this error term, together with measurement error—determines future beliefs about ability, on which future job choices are based. This absence of correlation is due to any new performance information used to update beliefs being random conditional on a worker's current job, since beliefs are martingales, and hence uncorrelated with any previous information or outcome. Critically, however, when firms are asymmetric, the labor market is imperfectly competitive, so the assignment of workers to jobs involves strategic considerations that entail intertemporal trade-offs between output and wages today and information and human capital tomorrow. In such instances, the wage equation becomes nonlinear in beliefs, thus precluding the differencing strategies on which these IV methods rely.

Formally, I consider a labor market in which a worker's ability is symmetrically unobserved by all and correlated across firms and jobs, as in Gibbons and Waldman (1999b, 2006). Production in firms is organized into jobs that differ in the output they generate, the human capital workers

<sup>5</sup> To estimate the cumulative impact of learning on wages accounting for the endogeneity of employment, job assignment, and beliefs about ability, the wage equation needs to be augmented with equations for the processes of employment, job assignment, and beliefs. As this cumulative impact arises from the sorting of managers to jobs, which is also affected by the human capital process, the role of learning for wages could be small even once its indirect effect on promotions is taken into account. I find instead that learning has a large impact on wages through this effect.

acquire when assigned to them, and the information they provide about workers' ability. For instance, performance at entry-level jobs, which usually entail simple tasks of limited value to a firm, may provide a less noisy signal about ability than performance at higher-level jobs, which normally involve more complex tasks of greater value to a firm.<sup>6</sup> In contrast to influential learning models of the labor market (Jovanovic 1979; Miller 1984; Flinn 1986), since ability is transferable across jobs and firms, job transitions can occur after both high and low performance, which respectively provide "good" and "bad" news about ability. Firms compete à la Bertrand in wages and jobs for workers who differ in their ability and human capital, which stochastically fluctuates over time and is job and firm specific to varying degrees. As technologies can differ across firms, equilibrium implies a flexible sharing rule of the surplus from a match between a firm and a worker, which leads to wages that in general differ from expected output and reflect differences in the opportunities for information and human capital acquisition across jobs.

Central to the identification of the model is that the process of learning about ability and human capital acquisition can be recovered from information on managers' jobs and wages at the firm I study, as I formally establish. **Importantly for the validation of this main model mechanism, the BGH data inform not only about individual managers' jobs and wages but also about the yearly evaluation of their job performance, which provides direct evidence on the signals at each job that the firm and its managers can use to learn about ability.**<sup>7</sup> I estimate the model by maximum likelihood using 8 years of observations on managers entering the firm between 1970 and 1979, imposing all the model restrictions. The model captures well the rich patterns of managers' job and wage paths, including the tenure profiles of job-to-job transitions at the firm and of separations from it and the distributions of performance and wages at the main job levels in each tenure.

The estimates shed light on several features of the process of information acquisition at the firm. First, uncertainty about managers' ability at entry into the firm is substantial, and the ordering of job levels by their

<sup>6</sup> Acquiring information is then an investment activity that makes job assignment a truly dynamic decision, in contrast to common learning models of careers, such as Gibbons and Waldman (1999b, 2006), in which job choices effectively solve a sequence of static problems. See Prescott and Visscher (1980), Holmström and Tirole (1989), and Jovanovic and Nyarko (1997) on this trade-off and Harris and Holmström (1982) and MacDonald (1982) for related models of learning about ability.

<sup>7</sup> Even in the absence of information on managers' performance, which is the outcome dependent on ability on which agents' decisions are based, identification can still be established. The reason is that the learning process can be inferred by linking information on the distributions of managers' jobs and wages across periods (sec. IV.B). By relying solely on information on managers' jobs, the learning and human capital process can be identified under suitable invertibility conditions that exploit the panel structure of the data, as shown in the online appendix.

informativeness about ability is the opposite of their ordering by a manager's probability of high performance. Hence, the firm necessarily experiments through job assignment. Second, comparing the estimated wage process with that in a counterfactual scenario in which learning about ability is precluded, so beliefs about it are never updated, reveals that learning contributes to about 25% of wage growth and dispersion over the first 7 years at the firm. Without learning, wages would increase much more slowly, because of the lower speed of promotions, and be much less dispersed. Third, learning is gradual: uncertainty about ability declines slowly over time, significantly reducing the pace of transitions to higher-level jobs relative to that in a counterfactual case in which ability is mostly learned in one period. **Both this persistent uncertainty and the differential informativeness of jobs lead to a substantial compression of wage growth in early tenures, and together they account for the curvature of wages in job levels and tenure.**

As for the human capital acquired at the firm, I estimate that it accounts for most of the growth and dispersion in managers' wages, but much like learning, it affects wages primarily through its impact on the dynamics of managers' job assignments, which is twofold. First, as managers acquire skills that make them on average more productive at higher levels, they are promoted to jobs at which they are paid higher and—because of the accumulating differences in the beliefs about their ability and in their human capital—more dispersed wages. Second, since human capital improves managers' productivity, it makes demotions less likely for any given decrease in beliefs about ability after low performance.

Overall, many features of careers in the BGH data are similar to those documented in the literature—for instance, the patterns of job assignment; the size of wage increases at promotion; the relationship between wages, job levels, and tenure in a firm; and the magnitude of wage growth on the job (Topel 1991; Main, O'Reilly, and Wade 1993; Gibbons and Waldman 1999a; Belzil and Bognanno 2008; Buchinsky et al. 2010; Waldman 2013; Frederiksen, Kriechel, and Lange 2017; Song et al. 2019). In these dimensions, the firm I study is comparable to those studied in other work. By focusing on detailed data from one firm, I can examine aspects of careers that are central to gauging the determinants of job and wage mobility but are difficult to measure in more representative worker or even matched employer-employee data—for example, the nexus between jobs, wages, and performance and the role of a firm's internal organization as embedded in its job hierarchy. Since my analysis is limited to one firm, however, results must be subject to further verification to establish their generality. The availability of data on multiple firms would also make it possible to relax the assumptions on the labor market structure, the wage setting protocol, and the symmetry in the law of motion of beliefs about ability and of human capital across firms that I have maintained in my empirical



analysis. The linkage of the firm I study to the rest of the market crucially relies on them. With multifirm data, additional sources of uncertainty—for instance, about firms' demand conditions—could be incorporated. My estimates provide just a peek into one firm's wage policy that hopefully offers a first step toward a reassessment of the role of learning for wages.

In the literature, job mobility in the labor market and learning about productivity have been examined by Berkovec and Stern (1991), Farber and Gibbons (1996), Altonji and Pierret (2001), Antonovics and Golan (2012), and Sanders (2016), in addition to the papers cited earlier.<sup>8</sup> Pries and Rogerson (2005) propose a model with learning and matching frictions for the differences in worker turnover between the United States and Europe. Moscarini (2005) nests the job-matching model of Jovanovic (1979)—extended in Jovanovic (1984) to allow for the random arrival of job offers—into an equilibrium model with search frictions and shows that idiosyncratic productivity risk dampens learning and worker sorting, thus compressing wage inequality.<sup>9</sup> Nagypál (2007) estimates a model of worker turnover that integrates two match-specific learning processes—namely, learning about match productivity and learning about the best use of an output technology in a match—and finds support for a gradual version of the first process from the response of worker turnover to price shocks.<sup>10</sup> Papageorgiou (2014) develops a search model of learning and occupational mobility in which ability is correlated across occupations and documents the importance of comparative advantage for occupational sorting. All this work, though, abstracts from within-firm job and wage mobility. Papageorgiou (2018) enriches Papageorgiou (2014) with occupational mobility in firms to account for the relationship between firm size and wages.

<sup>8</sup> Building on Farber and Gibbons (1996), Altonji and Pierret (2001) show how the importance of learning about productivity can be inferred from the impact on wages of individual characteristics that are not observed by employers—such as Armed Forces Qualification Test scores—but, because of learning, should be progressively reflected in wages. In contrast, the impact on wages of early signals about productivity, such as schooling, should correspondingly decline over time.

<sup>9</sup> Relatedly, Lochner, Park, and Shin (2018) measure the role of changes in the returns to unobserved skills and in the variance of unobserved skills and of transitory nonskill shocks for the increase in US residual wage inequality from the 1980s onward. See also Lochner and Shin (2014) on the importance of unobserved skills for the evolution of log earning residuals. Ghosh (2007) proposes a model of learning and human capital acquisition in which worker turnover across firms arises from disutility shocks to continuing employment at a same firm.

<sup>10</sup> The progressively greater selectivity implied by the first learning process makes both low- and high-tenure matches vulnerable to termination. As the author argues, this result is due to the two countervailing effects of a higher match quality and a lower option value of learning on match-specific informational capital as tenure increases. By contrast, under the second learning process, matches with the shortest tenure are the most vulnerable to termination.



Building on Gibbons and Waldman (1999b, 2006), Gibbons et al. (2005) explore the determinants of wage differentials across occupations and industries. By following a similar methodology, Lluís (2005) and Hunnes (2012) assess the importance of comparative advantage and learning for job and wage mobility within firms. This work on careers is based on models of job assignment with general rather than job-, firm-, occupation-, or industry-specific ability, unlike nearly all of the literature mentioned in the previous paragraph. As discussed, only when ability is correlated across jobs, firms, occupations, or industries do workers turn over across them not only after low performance—as in Jovanovic (1979), Miller (1984), and Flinn (1986)—but also after high performance, as in the data.

Chiappori, Salanié, and Valentin (1999) provide evidence of learning and downward wage rigidity for executives of a French state-owned firm.<sup>11</sup> Using information on wages and performance to estimate a perfectly competitive model of the labor market without job assignment, Kahn and Lange (2014) document that learning and stochastic productivity changes are important determinants of the variance of wages at the BGH firm. My results, based on a framework that extends Gibbons and Waldman (1999b, 2006) to asymmetric firms and differentially informative jobs, imply that the dynamic selection of managers to jobs through promotions is central to both the growth and dispersion of wages at the BGH firm.

As for the rest of the paper, section II examines the data, section III describes the model, section IV establishes its identification, section V presents its empirical specification and discusses the model estimates, section VI illustrates the counterfactual exercises, and section VII concludes. See appendixes A–C and the online appendix for omitted details.

## II. Data

The data, first analyzed by Baker, Gibbs, and Holmström (1994a), consist of the personnel records of all management employees (supervisory workers) of a medium-sized US firm in a service industry, observed between 1969 and 1988. The firm's job hierarchy inferred by Baker, Gibbs, and Holmström (1994a) consists of eight levels but is clearly divided into two parts: levels 1–4, which contain nearly all observations (97.6%), and levels 5–8 (chairman–chief executive officer [CEO]), which consist of top management positions (see app. B). Given the small number of observations at high levels, I aggregate levels 3–8 into a single one, henceforth referred to as level 3.<sup>12</sup> I use information on managers' year of entry into

<sup>11</sup> See also Garicano and Van Zandt (2013) on the role of organizations for acquiring information.

<sup>12</sup> Note that if levels were determined by pay, then the firm would promote managers once their pay reached a certain threshold. Thus, managers would always be promoted from the top percentiles of the wage distributions at their previous levels, which is not

the firm, age, education, job level, salary, and job performance. Ratings of performance are available for about two-thirds of the records and range from 1 (highest) to 5 (lowest). As 36,750 of the 45,673 rating observations are ratings of 1 or 2, ratings of 3–5 represent only a small fraction. I combine ratings of 2–5 into a single measure of 0, thus obtaining a binary classification of high and low performance.<sup>13</sup>

I focus on 1,426 managers who enter the firm between 1970 and 1979 at level 1 with at least 16 years of education, experience no change in their education during their first 10 years at the firm, and have no missing level information. No such manager is missing age or education information or is older than 45 years at entry into the firm. Because of the high separation rate from the firm each year, I restrict attention to their first 8 years at the firm. For completeness, I also examine a larger sample that includes entrants at the original levels 1–4, which displays very similar features to those of the sample of entrants at level 1. As shown in the online appendix, the estimates of the key parameters of the model presented in section III based on this larger sample are very close to those discussed in section V.B, which are based on the baseline sample just described.<sup>14</sup>

*Separations and job assignments.*—Table 1 illustrates two main patterns for managers' separations from the firm and job assignments at the firm. First, separation rates are high in all tenures. By the seventh year, over half

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the case in the BGH data (Baker, Gibbs, and Holmström 1994a, table VII). Although the relative size of these three levels is quite stable across the sample years, during which the firm experienced employment growth (Baker, Gibbs, and Holmström 1994a, their fig. 1c), the number of managers at each such level substantially varies over time, suggesting that capacity constraints on managerial employment at these levels, if present, may not be too stringent (see also app. C).

<sup>13</sup> Like Baker, Gibbs, and Holmström (1994a), I do not use information on bonus pay, as it is unavailable before 1981; see Ekinci, Kauhanen, and Waldman (2019) for an analysis of it. I treat ratings as year-end variables, as Baker, Gibbs, and Holmström (1994a) do. Since managers with a performance of 2 display assignment and wage profiles similar to those of managers with a performance of 3, 4, or 5, performance worse than 2 may not convey information different from that of a performance of 2, as is consistent with the evidence that supervisors are averse to negative evaluations. Thus, intermediate evaluations may be analogous to poor ones.

<sup>14</sup> In the BGH data, the average age of managers is 39 years, with a standard deviation of 10 years, and their average number of years of education is 15, with a standard deviation of 2 years, ranging from 12 years (high school) to 23 years (PhD). Age and education display little variation across entrant cohorts. Baker, Gibbs, and Holmström (1994a) report that the share of minorities and women at the firm, who cannot be identified in my copy of the data, increased over time. I consider entrants at level 1 between 1970 and 1979 to be able to compare my results with those of Baker, Gibbs, and Holmström (1994a), as most of their analysis concerns this group, and to avoid an excessive right censoring of the careers of later entrants. I exclude entrants in 1969, since it is unclear in which year managers observed in 1969 entered the firm. Although it is impossible to determine whether new entrants in any year are new hires, since they could have been promoted from clerical to managerial positions, Baker, Gibbs, and Holmström (1994a) argue that workers promoted into managerial ranks are very likely to be treated similarly to outside hires, since such a promotion entails a major shift in job tasks and a change from hourly to salaried employment.

TABLE 1  
PERCENTAGE DISTRIBUTIONS OF MANAGERS ACROSS LEVELS BY TENURE

| TENURE | SEPARATION |       | LEVEL 1 |       | LEVEL 2 |       | LEVEL 3 |       |
|--------|------------|-------|---------|-------|---------|-------|---------|-------|
|        | Data       | Model | Data    | Model | Data    | Model | Data    | Model |
| 1      | .0         | .0    | 100.0   | 100.0 | .0      | .0    | .0      | .0    |
| 2      | 14.5       | 14.5  | 45.6    | 45.7  | 39.9    | 39.8  | .0      | .0    |
| 3      | 27.7       | 26.5  | 16.8    | 17.2  | 46.8    | 47.3  | 8.7     | 8.9   |
| 4      | 37.9       | 37.1  | 7.6     | 8.1   | 29.2    | 29.2  | 25.3    | 25.6  |
| 5      | 46.1       | 45.3  | 4.6     | 5.3   | 18.1    | 18.3  | 31.1    | 31.2  |
| 6      | 52.0       | 51.5  | 2.9     | 3.4   | 12.3    | 12.6  | 32.8    | 32.5  |
| 7      | 57.6       | 56.9  | 2.1     | 2.7   | 7.7     | 8.3   | 32.6    | 32.1  |

the managers hired at level 1 have separated from the firm. Second, the percentage of managers assigned to level 1 rapidly decreases, whereas that of managers assigned to levels 2 and 3 first increases and then decreases (at level 2) or plateaus (at level 3) with tenure. Table B1 converts these distributions into level- and tenure-specific hazard rates of separation from the firm, retention at the same level, and promotion to the next level. Whereas separation hazards at each level are approximately constant with tenure, promotion hazards initially increase and then decrease with tenure.<sup>15</sup> Interpreting the data through the lens of a learning and stochastic human capital acquisition model helps rationalize all these features. For instance, when higher ability and acquired human capital are more valuable at higher-level jobs, managers advance through a firm's job hierarchy and turn over across firms as information about their ability and their human capital accumulate. Such a process can account for the nonmonotone tenure profiles of promotion rates, especially their eventual decrease with tenure, through two mechanisms that the model below will incorporate. First, as the best managers, according to their perceived ability and human capital, are promoted out of a level, the remaining ones necessarily face worse promotion prospects. Second, if acquired human capital makes a manager on average more productive at a given level than at any other levels, then the probability of promotion out of that level eventually decreases with tenure.

*Performance ratings.*—Table 2 shows two features of the profiles of ratings at the firm. First, the percentages of high ratings at levels 1 and 2 decrease with tenure in the firm. Second, high ratings are more likely at level 2 than at level 1 in any tenure.<sup>16</sup> As shown in the online appendix, the percentage of high ratings is higher among promoted managers

<sup>15</sup> As is common, promotions are mostly by one level, and demotions almost never occur in the BGH data.

<sup>16</sup> As ratings at level 3 are missing for more than 50% of managers, I do not use this information in estimation and so omit statistics on them. See corollary 1 for the identification of the model in sec. III in this case.

TABLE 2  
PERCENTAGES OF HIGH RATINGS AT LEVELS 1 AND 2

| TENURE | LEVEL 1 |       | LEVEL 2 |       |
|--------|---------|-------|---------|-------|
|        | Data    | Model | Data    | Model |
| 1      | 52.7    | 51.7  | ...     | ...   |
| 2      | 34.8    | 34.9  | 58.4    | 56.1  |
| 3      | 19.6    | 21.2  | 43.9    | 42.7  |
| 4      | 11.8    | 11.9  | 26.2    | 30.4  |
| 5      | 2.4     | 6.2   | 18.7    | 20.4  |
| 6      | 3.7     | 3.2   | 12.5    | 13.0  |
| 7      | .0      | 1.6   | 13.0    | 8.0   |

than among unpromoted ones and higher among those promoted early in their tenure in a level relative to those promoted later. These patterns are consistent with ability and acquired human capital leading to high performance and being more valuable at higher levels. For instance, as managers with higher ability and acquired human capital progress through the job hierarchy, the performance of managers at higher levels is naturally higher on average than that of managers at lower levels and higher for promoted than for unpromoted managers. By such a selection process, the performance of unpromoted managers worsens with tenure in a level, so the performance of those promoted early out of a level tends to be higher than the performance of those promoted later. As a result, the performance of managers at levels 1 and 2 who are not promoted to level 3 declines with tenure in the firm.

*Wages.*—Table 3 displays the distribution of wages (in 1988 US dollars) at the firm by level and tenure, and table B2 reports statistics on wage changes by tenure. Three features emerge. First, wages are on average higher at higher levels, as the percentage of managers earning wages above \$40,000 increases with the job level, and the spread of the distribution of wages tends to increase with the job level (see note to table 3). Since, as discussed, managers at higher levels on average also receive higher performance ratings, this evidence is consistent with a sorting process whereby managers with higher perceived ability and acquired human capital, who tend to perform better, are promoted to higher levels over time and compensated more. As beliefs about managers' ability and managers' realized human capital increasingly differ over time, so do their wages, which are correspondingly more variable at higher levels. Second, negative wage changes are quite frequent—over 20% in each tenure by table B2—as implied by a learning and stochastic human capital process. Intuitively, as low performance lowers beliefs about ability and implies lower levels of realized human capital, it reduces the value of managers' contribution to output and thus their wages. Third, although wages tend to increase with tenure, their growth is nonmonotone, as shown in table B2, which confirms

TABLE 3  
PERCENTAGE WAGE DISTRIBUTIONS BY LEVEL AND TENURE

| LEVEL AND<br>TENURE | \$20,000–\$40,000 |       | \$40,000–\$60,000 |       | \$60,000–\$80,000 |       |
|---------------------|-------------------|-------|-------------------|-------|-------------------|-------|
|                     | Data              | Model | Data              | Model | Data              | Model |
| Level 1:            |                   |       |                   |       |                   |       |
| 1                   | 59.1              | 57.5  | 40.5              | 42.0  | .4                | .5    |
| 2                   | 54.4              | 57.4  | 44.8              | 41.7  | .8                | .8    |
| 3                   | 55.6              | 56.6  | 44.4              | 42.2  | .0                | 1.2   |
| 4                   | 53.8              | 56.7  | 46.2              | 41.9  | .0                | 1.4   |
| 5                   | 64.1              | 68.1  | 35.9              | 31.1  | .0                | .7    |
| 6                   | 69.2              | 69.5  | 30.8              | 29.8  | .0                | .7    |
| 7                   | 75.0              | 70.6  | 25.0              | 28.5  | .0                | .8    |
| Level 2:            |                   |       |                   |       |                   |       |
| 2                   | 35.1              | 34.3  | 63.3              | 63.8  | 1.6               | 1.8   |
| 3                   | 31.3              | 36.1  | 65.7              | 61.7  | 2.9               | 2.2   |
| 4                   | 36.3              | 36.9  | 60.3              | 60.5  | 3.4               | 2.5   |
| 5                   | 37.1              | 37.5  | 59.6              | 59.8  | 3.3               | 2.7   |
| 6                   | 42.4              | 38.1  | 53.3              | 59.0  | 4.2               | 2.9   |
| 7                   | 41.3              | 38.8  | 55.8              | 58.2  | 2.9               | 3.0   |
| Level 3:            |                   |       |                   |       |                   |       |
| 3                   | 2.8               | 8.2   | 84.9              | 82.4  | 12.3              | 9.3   |
| 4                   | 4.5               | 10.3  | 85.3              | 80.5  | 10.1              | 9.2   |
| 5                   | 5.3               | 11.3  | 84.2              | 79.1  | 10.5              | 9.5   |
| 6                   | 6.1               | 12.4  | 84.4              | 77.6  | 9.5               | 9.9   |
| 7                   | 4.5               | 13.4  | 81.1              | 76.4  | 14.4              | 10.0  |

NOTE.—Means of wages across tenures are \$39,584 at level 1, \$43,179 at level 2, and \$48,963 at level 3. Standard deviations of wages across tenures are \$6,924 at level 1, \$7,377 at level 2, and \$7,270 at level 3.

that wages are not governed simply by managers' progression through the firm's job hierarchy over time.

*Evidence of learning and human capital acquisition.*—As is commonly observed in firm-level data, in the BGH data, workers who perform well advance from lower- to higher-level jobs of the firm's job hierarchy; wages increase with tenure in the firm, the level of the job hierarchy, and performance; promotions to higher levels and wage increases occur after high performance and are correlated over time; demotions to lower levels (almost) never occur, despite wage decreases being frequent after low performance; and wage dispersion at each level is substantial.<sup>17</sup> I interpret these patterns as resulting from a stochastic process of information and human capital acquisition, whereby managers perceived to be likely of high ability, as they accumulate new skills, progressively advance to the jobs (and firms) at which their ability and human capital are most valuable; see appendix C for potential alternative interpretations. Without human capital acquisition, ability alone would need to be sufficiently general across jobs for managers to be promoted after high performance. Were ability purely job

<sup>17</sup> See Baker, Gibbs, and Holmström (1994a) on the relationship between promotions, demotions, performance, and wages.

specific instead, job transitions would occur only after low performance, as in bandit models of learning with independent arms (Miller 1984), by their stay-on-a-winner property. Hence, a learning model with general ability is a natural starting point.

To see the importance of combining learning and stochastic human capital acquisition, observe that learning and job assignment can account for managers' mobility across firms and jobs and, unlike random productivity shocks, also for the serial correlation in wage increases, promotions, and performance (Gibbons and Waldman 1999b, 2006; Waldman 2013). Learning further leads to wage increases and promotions after high performance and wage decreases and demotions after low performance. By contrast, acquired human capital tends to give rise to promotions and wage increases by improving managers' productivity and so accounts for the low frequency of demotions relative to promotions.

Intuitively, then, the two mechanisms can be distinguished as follows. For any given frequency of wage increases and promotions, the higher their serial correlation, the larger the role of unobserved ability by analogy with the runs test for the nonrandomness of data. In the presence of unobserved ability, the higher the frequency of wage decreases or demotions, the greater the scope for learning. Yet for any given frequency of wage decreases, the lower the frequency of demotions relative to that of promotions, the larger the role of human capital acquisition: by increasing productivity on average, higher human capital offsets the adverse impact of low performance on beliefs about ability, so that managers are not demoted even after repeated low performance.

The patterns of job assignments, performance ratings, and wages in the BGH data are consistent with learning affecting wages not only directly—as, say, managers with higher performance ratings and thus more likely to be of higher ability are paid more on average—but also indirectly. That is, higher performance ratings and so higher beliefs about ability are associated with a greater chance of promotions, which lead to higher and more dispersed wages. Hence, job assignment importantly mediates the impact of learning on wages, as the analysis below will illustrate.

### III. A Model of Careers

I consider a labor market for an occupation (managers) in discrete time  $t \geq 1$ , in which two firms,  $f = A, C$ , compete for workers; see the online appendix for the case of multiple firms. I will occasionally refer to firm A as *my firm*, the firm in my data, and firm C as firm A's or the *competitor*.<sup>18</sup>

<sup>18</sup> Such a setting nests perfect competition when firms have identical technologies, a differentiated duopoly when firms have different technologies, or a virtual monopoly when, say, A's technology is more productive than C's.

Firms and workers discount the future by the factor  $\delta \in (0, 1)$ . Workers differ in their ability and human capital. Human capital evolves stochastically according to a process that depends on a worker's ability, employing firm, and assigned job. **Since ability, unlike human capital, is unobserved by all, firms and workers learn about ability over time from a worker's output.**

#### A. Environment

In this section, I describe the labor market, define equilibrium, and characterize employment, job assignment, and wages.

*Firms and workers.*—Firms produce a homogeneous good sold in a perfectly competitive market at a price normalized to 1. Production in firm  $f$  is organized in  $K$  jobs, which amount to groupings of productive tasks that are possibly different across firms. For instance, tasks are grouped into jobs by function, such as accounting or marketing, in some firms and by type of consumer, such as individuals or businesses, in others. Production at each job is governed by a constant returns to scale technology in workers' effective labor, or *human capital*, which is the only input. As in Gibbons and Waldman (1999b, 2006), a worker is characterized by a level of ability  $\theta$ , which is either high ( $\alpha$ ) or low ( $\beta$ ) and unobserved by all, including the worker, and a stock of human capital, which is described next.<sup>19</sup>

*Human capital and output.*—In period  $t$ , a worker is endowed with human capital acquired before entry into the market,  $H_t$ ; human capital acquired with general experience at the firms in the market,  $H_t^G = \{H_{fkt}^G\}$ ; and human capital acquired with specific experience at their jobs,  $H_t^S = \{H_{fkt}^S\}$ . The (log) production functions of human capital  $H_t^G$  ( $j = G$ ) and  $H_t^S$  ( $j = S$ ) are

$$h_{fkt}^j = \ln H_{fkt}^j = a_{fkt}^j + g_k^j(i_{kt}^j, \dots, i_{k,t-1}^j) + \varepsilon_{fkt}^j \quad (1)$$

at job  $k$  of firm  $f$ , with  $h_{fkt}^j = a_{fkt}^j + \varepsilon_{fkt}^j$ . The term  $a_{fkt}^j$  is stochastic and its distribution depends on a worker's ability,  $i_{kt}^j$  denotes the investment in period  $\tau \leq t - 1$ , and  $\varepsilon_{fkt}^j$  is a zero mean independently and identically distributed (i.i.d.) human capital or productivity shock, realized at the beginning of  $t$ , which captures any temporary unexpected change in human capital unrelated to ability. Whereas human capital  $H_t^G$  accumulates with time spent in the labor market, human capital  $H_t^S$  accumulates with time spent in a job. Specifically,  $i_{kt}^G$  equals the constant  $\bar{i}$  if a worker is employed at any job of either firm in period  $\tau$  and zero otherwise. By contrast,  $i_{kt}^S$  equals the constant  $\bar{i}_{k\tau}$  if a worker is employed at job  $k_\tau$  of either firm in period  $\tau$  and zero otherwise. Investments as well as  $a_{fkt}^j$  and  $\varepsilon_{fkt}^j$  are observed by all.

<sup>19</sup> I assume that the value of not working in the market is low enough that a worker is employed each period.



When a worker is employed at job  $k$  of firm  $f$  in period  $t$ ,  $(a_{fkt}^G, a_{fkt}^S)$  equals  $(\bar{a}_{fkt}^G, \bar{a}_{fkt}^S)$  with probability  $\pi_{fk}(\theta)$ , which is  $\alpha_{fk}$  for a high-ability worker and  $\beta_{fk}$  for a low-ability worker, and  $(\underline{a}_{fkt}^G, \underline{a}_{fkt}^S)$  otherwise. That is, each ability level corresponds to a vector of probabilities of high productivity of human capital at the jobs of each firm,  $\alpha = \{\alpha_{fk}\}$  and  $\beta = \{\beta_{fk}\}$ , with  $\alpha_{fk} \geq \beta_{fk}$ . Thus,  $\theta$  influences the evolution of human capital by affecting the probabilities of the realized  $a_{fkt}^G$  and  $a_{fkt}^S$ —note that if ability instead affected the support of  $a_{fkt}^G$  and  $a_{fkt}^S$ , it could be inferred in one period. Here  $a_{fkt}^G$  and  $a_{fkt}^S$  are assumed to have a common Bernoulli distribution for consistency with the one-dimensional and essentially binary measure of job performance in the data.<sup>20</sup> As  $\alpha_{fk} \geq \beta_{fk}$ , the human capital of a high-ability worker is on average higher than that of a low-ability worker with the same initial human capital and experience. Denote by  $(\bar{h}_{fkt}^G, \bar{h}_{fkt}^S)$  and  $(\underline{h}_{fkt}^G, \underline{h}_{fkt}^S)$  realized human capital when  $(\bar{a}_{fkt}^G, \bar{a}_{fkt}^S)$  (“success”) and  $(\underline{a}_{fkt}^G, \underline{a}_{fkt}^S)$  (“failure”) occur, respectively, which I refer to as *high performance* and *low performance*.<sup>21</sup>

With labor supply normalized to 1, a worker’s (log) output at job  $k$  of firm  $f$  in period  $t$  is

$$y_{fkt} = h_{fkt}^p + h_{fkt}^G + h_{fkt}^S, \quad (2)$$

where  $h_{fkt}^p$  is a (log) function of  $H_1$ . Denote output at the end of  $t$  after  $(\bar{h}_{fkt}^G, \bar{h}_{fkt}^S)$  and  $(\underline{h}_{fkt}^G, \underline{h}_{fkt}^S)$  are realized, respectively, by  $\bar{y}_{fkt}$  and  $\underline{y}_{fkt}$  and, net of productivity shocks, by  $\bar{y}_{fkt}$  and  $\underline{y}_{fkt}$ . At the start of  $t$ , let  $E(y_{fkt}|\theta, \kappa_t)$  be expected output at job  $k$  of firm  $f$  conditional on  $(\theta, \kappa_t)$ , where  $\kappa_t = (h_1, \{\kappa_{\tau}^j\}_{\tau=1}^{t-1})$  summarizes the deterministic component of human capital, and  $p_t$  be the prior that a worker is of high ability. By (1) and (2), expected output conditional on  $(p_t, \kappa_t)$ , which is a weighted average of  $E(y_{fkt}|\theta, \kappa_t)$  across the two possible values of  $\theta$ , is given by

$$y_f(p_t, \kappa_t, k) = \underbrace{\underline{y}_{fkt} + \beta_{fk}(\bar{y}_{fkt} - \underline{y}_{fkt})}_{E(y_{fkt}|\beta, \kappa_t)} + \underbrace{(\alpha_{fk} - \beta_{fk})(\bar{y}_{fkt} - \underline{y}_{fkt})}_{E(y_{fkt}|\alpha, \kappa_t) - E(y_{fkt}|\beta, \kappa_t)} p_t = d_{fk}(\kappa_t) + e_{fk}(\kappa_t) p_t. \quad (3)$$

<sup>20</sup> This setup differs from one with normal priors and signals mainly in that the variance of posterior beliefs does not deterministically decline with the number of signals. Otherwise, characterization results analogous to those derived below apply when ability and performance signals are continuously distributed. When performance is continuously distributed, its support can be made to depend on ability without trivializing the inference process about it. If performance were continuously distributed but its reporting binary, then its distribution would reduce to the one described. Formally, a supervisor can be thought of as performing a likelihood ratio test of the null hypothesis that a worker is of high ability against the alternative that the worker is of low ability, with beliefs updated according to test outcome.

<sup>21</sup> This process generalizes that in Bagger et al. (2014),  $h_t = a + g(t) + \varepsilon_t$ —where  $a$  is known worker productivity,  $g(t)$  is a deterministic trend for acquired human capital, and  $\varepsilon_t$  is a zero mean shock—to a setting with multijob firms and learning about ability. In light of the data, like these authors, I abstract from differences in  $g_k^i(\cdot)$  across firms. See app. sec. B2 for common laws of motion of human capital whose cumulative forms are instances of (1).

A high-ability worker has an absolute advantage at job  $k$ , since  $E(y_{fkt}|\alpha, \kappa_t) \geq E(y_{fkt}|\beta, \kappa_t)$ , and a comparative advantage at job  $k'$  over  $k$  if  $E(y_{fkt}|\alpha, \kappa_t)/E(y_{fkt}|\beta, \kappa_t) \geq E(y_{fk't}|\beta, \kappa_t)/E(y_{fk't}|\alpha, \kappa_t)$ .

*Information.*—Firms and workers share a common initial prior  $p_t$  that a worker is of high ability and update their beliefs about a worker's ability at the end of each period according to Bayes's rule; in the empirical analysis, I allow for multiple initial priors. Both firms and workers learn about a worker's ability from the performance signals  $\{a_{fkt}^i\}$  extracted from the worker's realized output, which are observed by all and whose distribution depends, as described, on the worker's unobserved ability. Hence, learning is symmetric in that firms and workers share the same information and thus beliefs in each period.<sup>22</sup> Given  $p_t$ , the two possible values of  $p_{t+1}$  are

$$P_{fHk}(p_t) = \frac{\alpha_{fk}p_t}{\alpha_{fk}p_t + \beta_{fk}(1 - p_t)} \text{ or } P_{fLk}(p_t) = \frac{(1 - \alpha_{fk})p_t}{(1 - \alpha_{fk})p_t + (1 - \beta_{fk})(1 - p_t)} \quad (4)$$

after high ( $H$ ) and low ( $L$ ) performance, respectively. As  $\alpha_{fk}$  and  $\beta_{fk}$  vary across jobs, the *informativeness* of jobs and so the *speed of learning* about ability differ across them, ranging from the case of no learning ( $\alpha_{fk} = \beta_{fk}$ ), in which posterior beliefs always equal prior beliefs, to that of complete learning ( $\alpha_{fk} = 1$  and  $\beta_{fk} = 0$ ), in which the posterior belief that a worker is of high ability is 1 or 0, depending on the worker's actual ability, after just one period of employment.<sup>23</sup> Note that jobs with the same expected output can be differentially informative about ability: as  $\alpha_{fk}$  and  $\beta_{fk}$  change, expected output in (3) can be kept constant by adjusting  $\bar{y}_{fkt}$  and  $\underline{y}_{fkt}$ . Also, jobs characterized by a low variability of output (with  $\bar{y}_{fkt}$  close to  $\underline{y}_{fkt}$ ) can be more or less informative than jobs characterized by a high variability of output (with  $\bar{y}_{fkt}$  much larger than  $\underline{y}_{fkt}$ ).

<sup>22</sup> As discussed in app. C, Baker, Gibbs, and Holmström (1994a) find evidence in support of symmetric learning in their data. The notion of symmetric learning captures the idea that résumés, references, interviews, and other information gathered by human resource departments and recruiters accurately convey performance at previous and current jobs in professional labor markets (Oyer and Schaefer 2011). Maintaining that performance is commonly observed is less restrictive than it may seem: firms other than a worker's current employer can infer the worker's performance in the previous period from the worker's wage in the current period. Then, observing wages is analogous to observing performance.

<sup>23</sup> By Blackwell's informativeness criterion, job  $k'$  is more informative than job  $k$  if posterior beliefs reached after performance is observed at job  $k'$  second-order stochastically dominate those reached at job  $k$ . Relative to the perfectly competitive setup of Gibbons and Waldman (1999b, 2006), where human capital  $h_{fkt} = \theta f(x_t) + \varepsilon_{fkt}$  depends on ability  $\theta$ , experience  $x_t$ , and learning noise  $\varepsilon_{fkt}$ , the setup here differs in that the speed of learning varies across jobs, both expected and realized human capital and output differ across firms and jobs, and productivity shocks are present in addition to learning noise.

*Separations.*—As information about ability and human capital accumulate, workers' perceived productivity and actual productivity at the jobs of the two firms evolve, leading to endogenous workers' turnover across jobs and firms. I also allow for exogenous separations between workers and firms by assuming that a worker leaves the labor market at the end of each period with probability  $1 - \eta_{fk}(\kappa_t)$ , which depends on the worker's employing firm  $f$ , job  $k$ , and human capital as captured by  $\kappa_t$ .<sup>24</sup>

*Timing.*—At the start of period  $t$ , the productivity shocks  $\{\varepsilon_{fht}\}$  are realized. Next, firms simultaneously submit wage and job offers to workers for the period.<sup>25</sup> Then, each worker decides which offer to accept. Last, the offered wage is paid; performance, human capital, and output are realized; beliefs about ability are updated; and separation shocks occur. Observe that as firms commit to the period offers they make, the timing of wage payments in a period is immaterial. I refer to  $y_f(p_t, \kappa_t, k) + \varepsilon_{fht}$  with  $\varepsilon_{fht} = \varepsilon_{fht}^G + \varepsilon_{fht}^S$  as a worker's *conditional expected output* before performance is realized. Without loss, I focus on the competition between the two firms for one worker. Denote by  $(w_t, k_t) = \{w_{ft}, k_{ft}\}$  the vector of each firm  $f$ 's wage and job offer and by  $l_t = \{l_{ft}\}$  the vector of the worker's decisions to accept ( $l_{ft} = 1$ ) or reject ( $l_{ft} = 0$ ) each offer in period  $t$ .

## B. Equilibrium

I restrict attention to robust Markov perfect equilibria (MPE). The state that firms face when making their wage and job offers is  $(s_t, \varepsilon_t)$ , with  $s_t = (p_t, \kappa_t)$ , where  $p_t$  is the prior that the worker is of high ability,  $\kappa_t = (h_t, \{i_k^t\}_{k=1}^{t-1})$  captures the deterministic component of the worker's human capital, and  $\varepsilon_t = \{\varepsilon_{fht}\}$  collects all realized productivity shocks. The state that the worker faces when choosing among offers is given by  $(s_t, \varepsilon_t)$  and firms' wage and job offers,  $(w_t, k_t)$ . An equilibrium consists of wage and job offer strategies  $w_{ft} = w_f(s_t, \varepsilon_t)$  and  $k_{ft} = k_f(s_t, \varepsilon_t)$  for each firm  $f$ , an acceptance strategy  $l_t = l(s_t, \varepsilon_t, w_t, k_t)$  for the worker with typical element  $l_{ft} = l_f(s_t, \varepsilon_t, w_t, k_t)$  for firm  $f$ , and belief-updating rules  $P_{fHk}(p_t)$  and  $P_{fLk}(p_t)$  for each firm  $f$  and job  $k$  such that at each state (i) the worker maximizes the (expected present discounted) value of wages, (ii) each firm maximizes the (expected present discounted) value of profits, (iii) the nonemploying firm is indifferent between employing and not employing the worker at the job that maximizes its (expected present discounted) value of profits, and (iv) beliefs are updated as in (4).

<sup>24</sup> Note that provided that new workers enter the market in each period, the market is always populated by workers.

<sup>25</sup> The presence of productivity shocks implies that the variability of wage and job offers over time and across workers is not due to just the process of learning about ability and human capital acquisition.

Given the firms' strategies, the worker's strategy satisfies

$$\tilde{W}(s_t, \varepsilon_t, w_t, k_t) = \max_{\{l_t\}} \sum_f l_f \left\{ w_{ft} + \delta \eta_{fk_t}(\kappa_t) \int_{\varepsilon_{t+1}} E \tilde{W}(s_{t+1}, \varepsilon_{t+1}, w_{t+1}, k_{t+1} | s_t, k_{ft}) dG \right\}, \quad (5)$$

where the expectation  $E \tilde{W}(\cdot)$  is over performance at job  $k_{ft}$  ( $s_{t+1}, \varepsilon_{t+1}$ ) equals  $(P_{fHk_{ft}}(p_t), \kappa_{t+1}, \varepsilon_{t+1})$  or  $(P_{fLk_{ft}}(p_t), \kappa_{t+1}, \varepsilon_{t+1})$  depending on realized performance,  $(w_{t+1}, k_{t+1})$  is the future set of offers, and  $G$  is the cumulative distribution function of the future productivity shocks  $\varepsilon_{t+1}$ . By (5), in evaluating an offer, the worker weighs the wage  $w_{ft}$  against the prospect of acquiring information and human capital at job  $k_{ft}$ . Given the worker's and the competitor's strategies, firm  $f$ 's strategy satisfies

$$\begin{aligned} \Pi^f(s_t, \varepsilon_t) = \max_{w,k} \left( l_{ft} \left\{ y_f(s_t, k) + \varepsilon_{fkt} - w + \delta \eta_{fk}(\kappa_t) \int_{\varepsilon_{t+1}} E \Pi^f(s_{t+1}, \varepsilon_{t+1} | s_t, k) dG \right\} \right. \\ \left. + l_{f't} \delta \eta_{f'k_{f't}}(\kappa_t) \int_{\varepsilon_{t+1}} E \Pi^f(s_{t+1}, \varepsilon_{t+1} | s_t, k_{f't}) dG \right), \quad (6) \end{aligned}$$

$f' \neq f$ , where  $l_{f't} = l_{f'}(s_t, \varepsilon_t, w_t, k_t)$  is the worker's acceptance decision about the offer by firm  $f'$  and  $k_{f't}$  is the job offered by firm  $f'$  in period  $t$ . By the second line in (6), each firm takes into account the option value of not employing the worker in a period and attracting the worker in some future period, which arises from the information revealed by performance and the human capital acquired by the worker at the competing firm. In particular, by (5) and (6), workers and firms experiment in that they contemplate sacrificing current wages and profits, respectively, to acquire more information about ability. Recall that in a standard (multiarmed bandit) problem of experimentation, a decision maker repeatedly chooses among alternatives (arms) with uncertain independent rewards, thereby learning about their distribution. Here, instead, rewards in terms of wages and profits associated with employment at the firms' jobs (arms) are dependent because they are governed by the worker's ability, which is common across them.

By point iii, I require an MPE to be robust in that if firm  $f'$  employs the worker, then the offer by firm  $f$ ,  $f \neq f'$ , must make  $f$  indifferent between not employing the worker (left-hand side of (7)) and employing the worker (right-hand side of (7)); that is,  $\Pi^f(s_t, \varepsilon_t | f') = \Pi^f(s_t, \varepsilon_t | f)$ , or

$$\begin{aligned} \delta \eta_{f'k_{f't}}(\kappa_t) \int_{\varepsilon_{t+1}} E \Pi^f(\cdot | \cdot, k_{f't}) dG \\ = \max_{w,k} \left\{ y_f(s_t, k) + \varepsilon_{fkt} - w + \delta \eta_{fk}(\kappa_t) \int_{\varepsilon_{t+1}} E \Pi^f(\cdot | \cdot, k) dG \right\}. \quad (7) \end{aligned}$$

This refinement implies that wages and so equilibrium are unique by ruling out unappealing offers by firm  $f$  if it employed the worker—namely,

offers with a present value of wages higher than  $f$ 's present value of output net of that of its outside option (left-hand side of (7)).<sup>26</sup>

### 1. Employment and Job Assignment

As argued, since a worker's ability is correlated across firms and jobs, each firm's job offer policy and the worker's acceptance policy solve a dependent bandit problem. These policies do not admit a simple index form, as policies for independent bandit problems do. Employment and job assignment can be nonetheless characterized as the solution to a pseudo-planning problem that is the natural analog of the stopping problem leading to an optimal (Gittins) index policy for independent bandits. Formally, denote by  $V^f(s_t, \varepsilon_t)$  the maximal value of firm  $f$ 's match surplus at state  $(s_t, \varepsilon_t)$ , or *match surplus value*, which is defined as the sum of the worker's value function  $W(s_t, \varepsilon_t) = \tilde{W}(s_t, \varepsilon_t, w_t, k_t)$ , or *value of wages*, expressed using the equilibrium dependence of wage and job offers,  $(w_t, k_t)$ , on  $(s_t, \varepsilon_t)$ , and firm  $f$ 's value function  $\Pi^f(s_t, \varepsilon_t)$ , or *value of profits*. Let  $V^f(s_t, \varepsilon_t|f')$ ,  $W(s_t, \varepsilon_t|f')$ , and  $\Pi^f(s_t, \varepsilon_t|f')$  be, respectively, firm  $f$ 's match surplus value, the worker's value of wages, and firm  $f$ 's value of profits conditional on firm  $f'$  employing the worker.

I establish two properties of equilibrium. First, given each firm's choice of job, the employing firm at each state is the one that generates the largest sum of values to all,  $S(s_t, \varepsilon_t) = \Pi^A(s_t, \varepsilon_t) + W(s_t, \varepsilon_t) + \Pi^C(s_t, \varepsilon_t)$ . That is, the employing firm is determined by a planning problem in which the planner's choice of job is restricted to the two jobs that maximize each firm's value of profits. Thus, conditional on a firm's choice of job, a worker's choice of firm is efficient. Second, each firm's choice of job at each state maximizes the value of its output  $\tilde{V}^f(s_t, \varepsilon_t)$ , defined in proposition 1, regardless of whether the firm employs the worker, as if it were the only firm in the market. This result is due to the worker's indifference between the two firms' offers and the nonemploying firm's indifference between employing and not employing the worker in equilibrium.<sup>27</sup>

**PROPOSITION 1.** The employing firm and assigned job are determined by the policy that solves

<sup>26</sup> Offers with a present value of wages lower than this net present value of output are already ruled out in an MPE, as  $f$  must obtain a higher value of profits by not employing the worker than it would by employing the worker—the left-hand side of (7) must weakly exceed the right-hand side of (7). This notion of MPE generalizes that of cautious MPE (Bergemann and Välimäki 1996) to the case of firm competition in products—here, jobs—as well as prices—here, wages.

<sup>27</sup> These efficiency properties do not imply that equilibrium is efficient, as neither firm necessarily internalizes the impact of its assignment decisions on its competitor's profits. In app. sec. B4, I consider an extension of the model that allows for the free entry of firms into the labor market and derive a simpler version of proposition 1 for this case.

$$S(s_t, \varepsilon_t) = \max_{f \in \{A, C\}} \left\{ y_f(s_t, k_{ft}) + \varepsilon_{fkt} + \delta \eta_{fkt}(\kappa_t) \int_{\varepsilon_{t+1}} ES(s_{t+1}, \varepsilon_{t+1} | s_t, k_{ft}) dG \right\}, \quad (8)$$

where  $k_{ft} = k_f(s_t, \varepsilon_t)$  solves  $\bar{V}^f(s_t, \varepsilon_t) = \max_{k \in K^f} \{y_f(s_t, k) + \varepsilon_{fkt} + \delta \eta_{fkt}(\kappa_t) \int_{\varepsilon_{t+1}} E \bar{V}^f(\cdot | s_t, k) dG\}$ .

## 2. Wages

The logic behind equilibrium wages generalizes the one familiar from a static model of Bertrand competition. In such a model, the wage that the employing firm pays is sufficiently high that a competitor cannot match it and obtain positive profits. For example, suppose that the worker's expected output at firms A and C is  $y_{At}$  and  $y_{Ct}$ , respectively, in period  $t$ . If  $y_{At} \geq y_{Ct}$ , then the worker is employed by A at wage  $w_{At} = w_{Ct} = y_{Ct}$ , whereas if  $y_{At} < y_{Ct}$ , then the worker is employed by C at wage  $w_{Ct} = w_{At} = y_{At}$ . Thus, a worker is typically paid less than output, unless the two firms share the same technology, in which case  $w_{At} = w_{Ct} = y_{At} = y_{Ct}$ , as under perfect competition.

Similarly, in the dynamic case considered here, the employing firm must pay a wage sufficiently high that the competitor cannot match it and obtain a positive value of profits—net of the value of its outside option, namely, its continuation value from not employing the worker. The implied wage rule, though, is more general than the one in the static case: a worker can be paid more or less than (conditional) expected output. For instance, firm A's value of profits can be high enough to justify paying a wage higher than expected output in a period, if A can sufficiently improve its future employment and assignment decisions on the basis of the information or the human capital acquired when it employs the worker. Whenever firms have the same number of jobs, wages take a simple and convenient form if the productivity shocks  $\{\varepsilon_{fkt}\}$  are Gumbel distributed (maximum) and any job  $k$  entails the same opportunity for information acquisition and risk of exogenous separation across firms, as I assume in section IV (see sec. 2 in the online appendix for the general case).<sup>28</sup> In what follows, I denote by  $\{f_t = f\}$  the event  $\{l_{ft} = 1\}$  that the worker is employed by firm  $f$  in  $t$ .

**PROPOSITION 2.** Let  $\{\varepsilon_{fkt}\}$  be independent Gumbel distributed, each with mean zero and variance  $\pi^2/6$ . If firms have the same number of jobs and  $\alpha_{Ak} = \alpha_{Ck}$ ,  $\beta_{Ak} = \beta_{Ck}$ , and  $\eta_{Ak}(\kappa_t) = \eta_{Ck}(\kappa_t)$  for each job  $k$ , then the worker's wage at state  $(s_t, \varepsilon_t)$  when employed by firm  $f$  is

$$w_f(s_t, \varepsilon_t) = y_{f'}(s_t, k_{ft}) - \ln \Pr(k_{f't} = k_{ft} | f_t = f', s_t) + \varepsilon_{f't}, \quad (9)$$

<sup>28</sup> See Galichon (2022) for examples of Gumbel distributed random variables of the same form as  $\varepsilon_{fkt} = \varepsilon_{fkt}^G + \varepsilon_{fkt}^S$ .

where  $\Pr(k_{f't} = k_{\bar{f}} | f_t = f', s_t)$  is the probability that firm  $f' \neq f$  offers the same job as firm  $\bar{f}$ , conditional on employing the worker at state  $s_t$ , and  $\varepsilon_{f't}$  is an independent Gumbel distributed shock with mean zero and variance  $\pi^2/6$ .

By proposition 2, the wage paid by, say, firm A can be expressed as the sum of the worker's expected output at firm C at the same job offered by A,  $y_C(s_t, k_{At})$ , the negative (log) probability that C offers the same job as A (the event  $\{k_{Ct} = k_{At}\}$ ) conditional on employing the worker (the event  $\{f_t = C\}$ ),  $-\ln \Pr(k_{Ct} = k_{At} | f_t = C, s_t)$ , and the random component of C's match surplus value,  $\varepsilon_{Ct}$  (see the proof of the result for details). The second term of this sum is a premium that accounts for C's job offer typically differing from A's and is higher the less likely firm C is to choose the same job as A—hence, the more different the employment experience at C that the worker forgoes by working at A is. Intuitively, by (7), C bids its entire match surplus value from employing the worker at its offered job, net of its continuation value from not employing the worker. This match surplus value can be expressed in terms of C's value had C offered the same job that A offers, which I observe in my data, up to an adjustment term for the possibility that A's preferred job may not coincide with C's. By the properties of Gumbel distributions, this term amounts to the negative (log) probability that C offers the same job as A. By the symmetry of the law of motion of the state across firms, the remaining terms reduce to  $y_C(s_t, k_{At}) + \varepsilon_{Ct}$ .

Note that if firms shared the same technologies and a worker experienced the same shocks across them each period, then the market would be perfectly competitive and firms would offer the same jobs. In this case, the worker's wage would equal (conditional) expected output,  $w_A(s_t, \varepsilon_t) = y_A(s_t, k_{At}) + \varepsilon_{A k_{At}}$ , as in the static case of the earlier example with  $y_{At} = y_A(s_t, k_{At}) + \varepsilon_{A k_{At}}$ .

#### IV. Empirical Analysis

In this section, I discuss the assumptions maintained in the empirical analysis and examine the identification of the model. As I observe workers—henceforth, managers—only since they are first employed by firm A to up to 8 years later, I denote by  $t = 1$  the first year at A from now on.

##### A. Preliminaries

I maintain that the discount factor  $\delta$  is known, productivity shocks  $\{\varepsilon_{jkt}\}$  are Gumbel distributed (maximum) each with mean zero and variance  $\pi^2/6$ , and firms have three jobs or *levels*, with common probabilities of high performance  $\{\alpha_{jk}, \beta_{jk}\}$  and exogenous separation  $\{1 - \eta_{jk}(\kappa_t)\}$  across them, which I will refer to as  $\{\alpha_{k}, \beta_{k}\}$  and  $\{1 - \eta_{kt}\}$ . Recall that



$\kappa_t = (h_1, \{i_k^j\}_{j=1}^{t-1})$  summarizes the deterministic component of human capital, which is known from the data.<sup>29</sup>

*Unobserved heterogeneity.*—As managers may differ along dimensions known to them and firms—say, through job interviews—but unobserved by the econometrician, in the spirit of Heckman (1981), I assume that each manager is of latent skill type  $i = 1, \dots, I$ . Skill types affect the distributions of the initial priors about managers' ability, their output, and their wages.<sup>30</sup> Let  $p_{i1}$  be the initial prior that a manager of skill type  $i$  is of high ability;  $q_i = \Pr(i|L_{A1} = 1, h_1) = \Pr(i|L_{A1} = 1)$  be the probability that a manager is of skill type  $i$  conditional on the assignment to level 1 at entry into firm A, where level  $L_{jt}$  at firm  $j$  in  $t$  in the data corresponds to job  $k_{jt} = k_j(s_{it}, \varepsilon_t)$  in the model; and  $s_{it} = (p_{it}, \kappa_t, i)$  be a manager's state at the beginning of  $t$ . Since firm A may face different competitors for managers of different skill types, I let performance  $a_{Ckt}^j$  in (1) depend on  $i$  and denote its two possible values for each  $j, k, t$ , and  $i$  by  $\bar{a}_{Ckt}^j(i)$  and  $\underline{a}_{Ckt}^j(i)$ . Then, (3) becomes

$$y_t(s_{it}, k) = \underbrace{y_{fkt}(i) + \beta_k[\bar{y}_{fkt}(i) - y_{fkt}(i)]}_{d_{fk}(\kappa_t, i)} + \underbrace{(\alpha_k - \beta_k)[\bar{y}_{fkt}(i) - y_{fkt}(i)]}_{e_{fk}(\kappa_t, i)} p_{it}, \quad (10)$$

with  $d_{fk}(\kappa_t, i) = E(y_{fkt}(i)|\beta, \kappa_t, i)$  and  $e_{fk}(\kappa_t, i) = E(y_{fkt}(i)|\alpha, \kappa_t, i) - E(y_{fkt}(i)|\beta, \kappa_t, i)$ . Through this formulation, the model can account for managers' selection into firms and jobs on the basis of their unobserved ability ( $\theta$ ), measured human capital ( $\kappa_t$ ), and other unmeasured characteristics ( $i$ ).

*Performance ratings.*—It is well known that performance ratings measure performance imperfectly and that the associated error may be systematic, since supervisors tend to assign uniform ratings to employees regardless of their actual performance, potentially leading to its repeated misreporting (Baker, Jensen, and Murphy 1988; Murphy 1992; Prendergast 1999; Lazear, Shaw, and Stanton 2015). I interpret recorded ratings as noisy measures of managers' performance, as Baker, Gibbs, and Holmström (1994a) do, and allow their error to be both biased and persistent over time (Keane and Sauer 2009). Formally, denote by  $E_0(k, t) = \Pr(R_{At}^o = 1|L_{At} = k, R_{At} = 0, t)$  the probability of a recorded high rating,  $R_{At}^o = 1$ , when low performance,  $R_{At} = 0$  corresponding to  $(\underline{a}_{Akt}^G, \underline{a}_{Akt}^S)$  in the model,

<sup>29</sup> See sec. 2 in the online appendix for the identification of the model when the information and human capital process and the risk of exogenous separation differ across firms, provided that data from multiple firms are available.

<sup>30</sup> On the basis of changes in likelihood values and of information criteria, I set  $I = 4$  in estimation; see Bruni and Koch (1985) for the identification of the number of latent classes in the case of finite mixtures of mixtures of normal distributions considered below. In the online appendix, I consider a version of the model in which the initial priors  $\{p_{i1}\}$  about managers' ability vary with a manager's entry level. Their estimated distribution is similar to the one in panel A of table 4.

is realized at level  $k$  in  $t$ . Here,  $R_{\Lambda t}^o$  is the performance rating observed by the econometrician, whereas  $R_{\Lambda t}$  is the true performance observed by firms and managers. Similarly, denote by  $E_1(k, t) = \Pr(R_{\Lambda t}^o = 0 | L_{\Lambda t} = k, R_{\Lambda t} = 1, t)$  the probability of a recorded low rating,  $R_{\Lambda t}^o = 0$ , when high performance,  $R_{\Lambda t} = 1$  corresponding to  $(\bar{a}_{\Lambda kt}^G, \bar{a}_{\Lambda kt}^S)$  in the model, is realized at level  $k$  in  $t$ . I allow error rates to vary across levels and tenures, as performance appraisal is often job specific and may be more thorough at certain stages of a manager's career at firm A. I specify these rates as functions of potential tenure in each level,

$$E_0(k, t) = \frac{e^{\rho_0 + \rho_z(k)[1\{k=1\}t + 1\{k=2\}(t-1) + 1\{k=3\}(t-2)]}}{1 + e^{\rho_0 + \rho_z(k)[1\{k=1\}t + 1\{k=2\}(t-1) + 1\{k=3\}(t-2)]}} \text{ and} \quad (11)$$

$$E_1(k, t) = \frac{1}{1 + e^{\rho_0 + \rho_1 + \rho_z(k)[1\{k=1\}t + 1\{k=2\}(t-1) + 1\{k=3\}(t-2)]}},$$

where  $\rho_1 > 0$  so that the probability of a high performance rating increases with performance,  $\Pr(R_{\Lambda t}^o = 1 | L_{\Lambda t} = k, R_{\Lambda t} = 1, t) > \Pr(R_{\Lambda t}^o = 1 | L_{\Lambda t} = k, R_{\Lambda t} = 0, t)$ .<sup>31</sup>

*Wages.*—I allow for measurement error in wages as follows. Define  $\varepsilon_{\Lambda it} = \lambda_{ik}\varepsilon_{Ct} + \varepsilon_{it}^m$ , where  $\varepsilon_{Ct}$  is firm C's shock from proposition 2,  $\lambda_{ik}$  is given by  $\sqrt{6}\sigma_{\Lambda ik}/\pi$ , and  $\varepsilon_{it}^m$  is a zero mean i.i.d. Gumbel distributed (minimum) shock independent of any other shocks, with variance  $\sigma_{\Lambda ik}^2$ . Then,  $\varepsilon_{\Lambda it}$  is a zero mean i.i.d. logistically distributed shock with variance  $\sigma_{\Lambda ik}^2$ , as it is the difference between two independent Gumbel distributed (maximum) shocks with the same variance, and

$$w_{\Lambda it} = Ew_{\Lambda it}(k) + \varepsilon_{\Lambda it} = y_C(s_{it}, k) - \ln \Pr(L_{Ct} = k | f_t = C, s_{it}) + \varepsilon_{\Lambda it} \quad (12)$$

is the recorded wage of a manager of skill type  $i$  at level  $k$  of firm A at state  $(s_{it}, \varepsilon_t)$  by (9). As a result, the observed distribution of wages at level  $k$  of A in any tenure is a mixture of the logistic distributions of wages of managers with all possible states.

### B. Identification

Identifying a model of the labor market from data on one firm is a known challenge. The constructive approach that I follow relies on combining information on managers' wages, level assignments, and performance ratings and exploits two features of the model: (i) the symmetry in the opportunities for information acquisition across firms and (ii) the relationship between wages and firm productivity implied by Bertrand competition. By this symmetry, the initial distribution of beliefs about ability and

<sup>31</sup> Since no manager in the sample is observed at level 2 before  $t = 2$  or at level 3 before  $t = 3$ , I let error rates at these levels depend on  $t - 1$  and  $t - 2$ , respectively.

their law of motion can be recovered from just the distribution of managers' wages and level assignments at firm A—the law of motion of beliefs is also identified from the distribution of performance ratings at A. Under Bertrand competition, the wage process at firm A is directly informative about the productivity process at its competitor—namely, firm C—or, in general, the firm offering a manager the second-highest present value of wages in each period.

As is common to dynamic discrete choice models, a key step consists of the recovery of the distribution of unobserved heterogeneity—here, of managers' unobserved skill types and beliefs about their ability—and of conditional choice probabilities—here, conditional job assignment probabilities. This step is based on the information provided by managers' wages at firm A. Intuitively, by (12), the distribution of managers' wages at any of A's job levels in any tenure, conditional on their histories of assignments and performance ratings at A, is a finite mixture of the logistic distributions of wages of managers of each skill type with all possible histories of performance, conditional on their histories of assignments and performance ratings at A. Since these mixtures are identifiable, the weights of such a mixture in managers' first year at A, which capture the proportions of managers of each skill type, pin down the probability masses  $\{q_j\}$  of skill types. In subsequent years, as performance accumulates, the weights of these finite wage mixtures identify the proportions of managers of each skill type with all possible histories of performance—which pin down managers' probabilities of high performance by skill type and ability as well as the initial priors  $\{p_n\}$  about ability—conditional on managers' histories of assignments and performance ratings at A.<sup>32</sup> As these latter histories are observed, the joint probabilities of managers' histories of assignments, performance ratings, and performance at A by skill type are identified. From these joint probabilities linked across years, managers' assignment probabilities at A by skill type, conditional on their past histories at A, can be easily recovered. From these conditional assignment probabilities, in turn, managers' expected output and human capital acquired at A, up to a level normalization, can be pinned down by standard arguments for discrete choice models.

Another key step is that by (12), a manager's expected wage at A at any level in each tenure equals the difference between C's expected output at the same level and the (log) probability that C would assign the manager to it, if it employed the manager. Thus, when conditional assignment probabilities are equal across firms, once these probabilities are recovered for A, C's expected output is identified from managers' wages at A. In general,

<sup>32</sup> The probabilities of high performance by manager ability, which describe the law of motion of beliefs, can also be recovered from the probabilities of high performance ratings at A's levels.

since conditional assignment probabilities at C are determined by C's expected match surplus value, which is a function of C's expected output, it is simple to show that this value is recursively identified from managers' wages at A. Before turning to the formal argument, I provide an overview of it next.

### 1. Overview of the Argument

I first establish that the finite mixture distributions of wages of managers of each skill type with all possible histories of performance at A, conditional on their observed histories at A, and so the probability masses  $\{q_{ij}\}$  of managers' skill types are identified from managers' wages at A. The rest of the argument proceeds in four sequential steps. In step 1, I prove that classification error rates in performance ratings at each level  $k$ ,  $\{E_0(k, t), E_1(k, t)\}$ , and their parameters are identified from the distributions of managers' wages and performance ratings at A. Then, the level probabilities of high performance by manager ability,  $\{\alpha_{ik}, \beta_{ik}\}$ , are identified from the level probabilities of high performance ratings at A. In step 2, I establish that from the weights of the identified finite mixture distribution of managers' wages in their second year at A at any level, the initial priors  $\{p_{in}\}$  can be recovered. The weights of these identified finite mixture distributions in subsequent years also pin down the probabilities  $\{\alpha_{ik}, \beta_{ik}\}$ . By these two steps, the learning process is known.

In step 3, I show that managers' conditional assignment probabilities at A by skill type in each tenure are pinned down by the weights of the identified conditional finite mixture distributions of managers' wages at A's levels in the corresponding tenure. Expected output at A up to that at one level and, under conditions, the exogenous separation rates can then be recovered from managers' conditional assignment probabilities and wages at A. In step 4, I show that firm C's expected match surplus value is identified from managers' wages at A, and so is C's expected output at each level up to managers' conditional assignment probabilities at C—these latter probabilities can be recovered from those at A when identical across firms. The human capital process is thus pinned down by the two firms' expected output processes.<sup>33</sup>

The formal argument presented next applies to both the case of a long panel covering the entire duration of a manager's employment at firm A and the case of a short panel of length  $T$  ( $T = 8$  in my sample). When the argument for the latter case differs from that for the former case, it holds up to knowledge of firms' expected match surplus values in  $T + 1$

<sup>33</sup> As will become apparent in sec. V.A, the parameters of human capital and output of the estimated specification of the model are straightforwardly identified by the two firms' expected output processes.

or under suitable exclusion or parametric restrictions after period  $T$ , as proved below.

## 2. Formal Argument

The observed density of wages of managers with initial human capital  $h_1$  at level 1 in the first year at A is a finite mixture of the logistic densities of the wages of such managers of each skill type,

$$f(w_{A1}|L_{A1} = 1, h_1) = \sum_i \Pr(i|L_{A1} = 1, h_1)f(w_{A1}|L_{A1} = 1, h_1, i), \quad (13)$$

by (12), where  $q_i = \Pr(i|L_{A1} = 1, h_1) = \Pr(i|L_{A1} = 1)$  is the typical weight,  $f(w_{A1}|L_{A1} = 1, h_1, i)$  is the typical component density with standard deviation  $\sigma_{A1i}$ , and  $w_{A1i}$  in (12) with  $t = 1$  denotes a draw from this component density according to the model. Such a mixture is identified up to the labeling of its components by  $i$  by the results in Holzmann, Munk, and Stratmann (2004). In any subsequent year, managers' wages depend on the updated beliefs about their ability based on their performance. Then, the density of wages of managers with initial human capital  $h_1$  and first-year performance rating  $R_{A1}^o$  at level  $L_{A2}$  in the second year at A is a finite mixture of the logistic densities of the wages of such managers of each skill type with high and low first-year performance  $R_{A1}$ ,  $f(w_{A2}|L_{A1} = 1, R_{A1}^o, L_{A2}, h_1) = \sum_{R_{A1}} \sum_i \Pr(R_{A1}, i|L_{A1} = 1, R_{A1}^o, L_{A2}, h_1)f(w_{A2}|L_{A1} = 1, R_{A1}^o, R_{A1}, L_{A2}, h_1, i)$ , and so it is identified up to the labeling of its components by  $R_{A1}$  and  $i$ . In turn, the products of the identified weights  $\{\Pr(R_{A1}, i|L_{A1} = 1, R_{A1}^o, L_{A2}, h_1)\}$  and the known probabilities  $\{\Pr(L_{A1} = 1, R_{A1}^o, L_{A2}, h_1)\}$  pin down the probabilities  $\{\Pr(L_{A1} = 1, R_{A1}^o, R_{A1}, L_{A2}, h_1, i)\}$  of managers' histories at A by skill type. A similar argument applies to the remaining tenure years and to the densities of managers' wages conditional on only their initial human capital and histories of assignments at A. From now on, I maintain that the component densities of all these wage mixtures are distinct for distinct values of the state  $s_{ib}$  as is necessary for their identifiability.<sup>34</sup>

LEMMA 1. Let  $\mathcal{H}_L(t) = (L_{A1} = 1, \dots, L_{At-1})$  and  $\mathcal{H}_{LR'}(t, \tau) = (L_{A1} = 1, \dots, R_{A\tau}^o, \dots, L_{At-1}), \tau = 1, \dots, t - 1$ . The densities of managers' wages at level 1 in  $t = 1$  in (13) and level  $L_{At}$  in  $t > 1$ ,

<sup>34</sup> Period  $t$  histories of assignments and performance with different numbers of high performance realizations by level are assumed to lead to different component densities and so to be distinguishable. For period  $t$  histories of assignments and performance with the same number of high performance realizations by level, which lead to the same period  $t$  state and so component densities, the corresponding mixtures can be appropriately reduced. Weights and component densities can be nonetheless identified for each possible history of performance by manager skill type, since the repeated densities have identical weights. See the proof of lemma 1.

$$f(w_{A_t}|\mathcal{H}_{t\tau}, L_{A_t}, h_1) \\ = \sum_{R_{A_1}} \cdots \sum_{R_{A_{t-1}}} \sum_i \Pr(R_{A_1}, \dots, R_{A_{t-1}}, i|\mathcal{H}_{t\tau}, L_{A_t}, h_1) f(w_{A_t}|R_{A_1}, \dots, R_{A_{t-1}}, i, \cdot),$$

with  $\mathcal{H}_{t\tau} \in \{\mathcal{H}_L(t), \mathcal{H}_{LR}(t, \tau)\}$ , given managers' initial human capital  $h_1$ , are identifiable mixtures up to the labeling of their components by managers' skill type  $i$  and performance  $\{R_{A\tau}\}_{\tau=1}^{t-1}$ .

As argued in the proof of the lemma, ordering the standard deviation parameters  $\{\sigma_{Aik}\}$  by managers' skill type  $i$  for each level  $k$  is sufficient to label mixture components by  $i$ . Alternatively, components can be labeled by  $i$  by ordering their weights or parameters by  $i$ . Given  $i$ , components can also be labeled by performance  $\{R_{A\tau}\}_{\tau=1}^{t-1}$ . Intuitively, consider the simple case in which only high or only low performance is realized up to  $t-1$ . Then, the ratio of the probabilities  $\Pr(L_{A1} = 1, R_{A1}, \dots, R_{A\tau}^o, \dots, L_{A_{t-1}}, R_{A_{t-1}}, L_{A_t}, h_1, i)$  for  $\tau = 1, \dots, t-1$  and  $\Pr(L_{A1} = 1, R_{A1}, \dots, L_{A_{t-1}}, R_{A_{t-1}}, L_{A_t}, h_1, i)$ —which are identified, respectively, from the weights of the finite wage mixtures  $f(w_{A_t}|\mathcal{H}_{LR}(t, \tau), L_{A_t}, h_1)$  and  $f(w_{A_t}|\mathcal{H}_L(t), L_{A_t}, h_1)$ —equals  $\Pr(R_{A\tau}^o|L_{A\tau}, R_{A\tau}, \tau)$ .<sup>35</sup> The value of  $R_{A\tau}$  of this latter probability can be easily determined, since  $\Pr(R_{A\tau}^o = 1|L_{A\tau} = k, R_{A\tau} = 1, \tau) > \Pr(R_{A\tau}^o = 1|L_{A\tau} = k, R_{A\tau} = 0, \tau)$  for any level  $k$ , and the opposite inequality holds for  $R_{A\tau}^o = 0$ , with  $\rho_1 > 0$  in (11)—this argument also provides the basis for the identification of the distribution of classification error in performance ratings. More generally, mixture components can be labeled by performance by ordering the means of their component densities by the period  $t$  prior  $p_{it}$ .

*Step 1: rating error and performance.*—Consider the distributions of classification error in performance ratings and of managers' performance. As just argued, the probabilities  $\{\Pr(R_{A1}^o|L_{A1} = 1, R_{A1}, 1)\}$  and  $\{\Pr(R_{A2}^o|L_{A2} = 1, R_{A2}, 2)\}$  of managers' performance ratings at level 1 of firm A, given their true performance, are identified by lemma 1. Then, the error rates  $E_0(1, t)$  and  $E_1(1, t)$  at level 1 in  $t = 1, 2$  and so the parameters  $(\rho_0, \rho_1, \rho_2(1))$  of the distribution of the error in performance ratings at level 1 can be recovered by (11). An analogous argument holds for  $\{\rho_2(k)\}_{k=2,3}$ .

Now the distribution of performance ratings of managers at level  $k$  of A is a mixture of the distributions of performance of managers of high and low ability with parameters  $(\alpha_k, \beta_k)$ , subject to classification error. Once this error distribution is recovered, such a mixture distribution is identified

<sup>35</sup> To see how these probabilities can be paired, consider, say,  $t = 2$ . Note that as  $f(w_{A2}|L_{A1} = 1, R_{A1}^o, R_{A1}, L_{A2}, h_1, i)$  is independent of  $R_{A1}^o$ , the same densities  $\{f(w_{A2}|L_{A1} = 1, R_{A1}, L_{A2}, h_1, i)\}$  can be recovered from the mixtures  $f(w_{A2}|L_{A1} = 1, R_{A1}^o, L_{A2}, h_1)$  and  $f(w_{A2}|L_{A1} = 1, L_{A2}, h_1)$ . Through these common component densities, the weights  $\{\Pr(R_{A1}, i|L_{A1} = 1, R_{A1}^o, L_{A2}, h_1)\}$  and  $\{\Pr(R_{A1}, i|L_{A1} = 1, L_{A2}, h_1)\}$  can be matched to obtain  $\Pr(R_{A1}^o|L_{A1} = 1, R_{A1}, 1)$ .

as follows. Note that for managers with initial human capital  $h_1$  at level 1 of A (at least) up to  $t = 3$ , the probability of high performance in  $t = 3$ ,  $\Pr(R_{A3} = 1 | L_{A1} = 1, L_{A2} = 1, L_{A3} = 1, h_1)$ , can be recovered from that of a high rating in  $t = 3$ ,  $\Pr(R_{A3}^o = 1 | L_{A1} = 1, L_{A2} = 1, L_{A3} = 1, h_1)$ , which equals  $E_0(1, t) + [1 - E_0(1, t) - E_1(1, t)] \Pr(R_{A3} = 1 | L_{A1} = 1, L_{A2} = 1, L_{A3} = 1, h_1)$ , if error rates are known. More generally, these managers' probability of high performance in  $t \leq 3$  satisfies

$$\Pr(R_{A1} = 1, \dots, R_{At} = 1 | \cdot, h_1) = \sum_i q_{i3}(h_1) p_{i3}(h_1) \alpha'_i + [1 - \sum_i q_{i3}(h_1) p_{i3}(h_1)] \beta'_1, \quad (14)$$

where  $q_{i3}(h_1)$  denotes  $\Pr(i | L_{A1} = 1, L_{A2} = 1, L_{A3} = 1, h_1)$  and  $p_{i3}(h_1)$  is the probability that any such manager of skill type  $i$  is of high ability,  $\Pr(\bar{\theta} | L_{A1} = 1, L_{A2} = 1, L_{A3} = 1, h_1, i)$ . The probability in (14) can be recovered from the probability of a high rating in  $t \leq 3$ ,  $\Pr(R_{A1}^o = 1, \dots, R_{At}^o = 1 | L_{A1} = 1, L_{A2} = 1, L_{A3} = 1, h_1)$ . This logic extends to the probability of any sequence of high and low performance in  $t \leq 3$ . Thus, the distribution of performance of managers at level 1 of A (at least) up to  $t = 3$  is a finite binomial mixture with component probabilities  $(\alpha_1, \beta_1)$  and weights  $(\sum_i q_{i3}(h_1) p_{i3}(h_1), 1 - \sum_i q_{i3}(h_1) p_{i3}(h_1))$ , which is identified (Blishke 1964, 1978). An analogous argument applies to the distributions of managers' performance at levels 2 and 3.

**PROPOSITION 3.** The classification error parameters  $(\rho_0, \rho_1, \rho_2(1))$  at level 1 are identified from the distributions of wages of managers at any level of A in  $t = 2, 3$  who were at level 1 of A and received a high or low performance rating in  $t - 1$ . The classification error parameter  $\rho_2(k)$  at level  $k = 2, 3$  is identified from the distribution of wages of managers at any level of A in  $t = k + 1$  who were at level  $k$  of A and received a high or low performance rating in  $t - 1$ . The probabilities  $(\alpha_k, \beta_k)$  of high performance by manager ability at level  $k \geq 1$  are identified from three periods of observations on the distribution of managers' performance ratings at level  $k$  of A.

The argument so far has relied on wage and rating distributions being identifiable logistic mixture distributions of the wages of managers of each skill type with each possible performance and binomial mixture distributions of the performance of managers of each ability, respectively. Many other specifications, including less parametric as well as continuous ones, are also identifiable. See, for instance, Hunter, Wang, and Hettmansperger (2007) for semiparametric symmetric finite mixtures, Lindsay (1995) for general mixing distributions, and Bruni and Koch (1985) for continuous normal mixtures.

*Step 2: initial priors.*—Recall that the probabilities  $(\alpha_1, \beta_1)$  of high performance by manager ability at level 1 are identified by step 1. Also, the probabilities  $\{\Pr(L_{A1} = 1, R_{A1}, L_{A2}, h_1, i)\}$  of the histories at firm A of



managers employed by the firm (at least) up to  $t = 2$  are identified by lemma 1. From each of these probabilities, expressed as  $\Pr(L_{A2}|L_{A1} = 1, R_{A1}, h_1, i) \Pr(R_{A1}|L_{A1} = 1, h_1, i) \Pr(i|L_{A1} = 1, h_1) \Pr(L_{A1} = 1, h_1)$ , the product of the first two probabilities can be recovered, since  $q_i = \Pr(i|L_{A1} = 1, h_1)$  is identified by lemma 1 and  $\Pr(L_{A1} = 1, h_1)$  is known from the data. Summing the identified product  $\Pr(L_{A2}|L_{A1} = 1, R_{A1}, h_1, i) \Pr(R_{A1}|L_{A1} = 1, h_1, i)$  with  $R_{A1} = 1$  over  $L_{A2}$  yields  $\Pr(R_{A1} = 1|L_{A1} = 1, h_1, i) = \alpha_1 p_{i1} + \beta_1(1 - p_{i1})$ , which pins down the initial prior  $p_{i1}$  for each  $i$ . Once  $\{\alpha_k, \beta_k, p_{i1}\}$  are identified, the process of learning about ability is known.

**PROPOSITION 4.** If the probability masses  $\{q_i\}$  of the initial priors  $\{p_{i1}\}$  about managers' ability and the probabilities  $(\alpha_1, \beta_1)$  of high performance by manager ability at level 1 are identified, then  $\{p_{i1}\}$  are identified from the distribution of managers' wages at any level of A in  $t = 2$ .

The logic of proposition 4 can be extended to establish that the initial priors  $\{p_{i1}\}$  and the probabilities  $(\alpha_k, \beta_k)$  of high performance at level  $k \geq 1$  are identified even without information on performance ratings—that is, when agents learn and make decisions on the basis of information not observed in the data—as the next corollary shows. Section 2 in the online appendix presents an alternative result, based on Hu and Shum (2012), that does not require information on either wages or performance ratings. As noted in section II, I do not use information on ratings at level 3 in estimation.

**COROLLARY 1.** The initial priors  $\{p_{i1}\}$  about managers' ability and the probabilities  $(\alpha_k, \beta_k)$  of high performance by manager ability at level  $k \geq 1$  are identified from the distributions of wages of managers at any level of A in  $t = k + 3$  assigned to level  $k \geq 1$  of A in  $t = k, k + 1, k + 2$ .

*Step 3: expected output at the firm and exogenous separation rates.*—Note first that conditional assignment probabilities at A are identified. Intuitively, the probabilities  $\{\Pr(i|L_{A1} = 1, h_1)\}$  and  $\{\Pr(L_{A1} = 1, R_{A1}, L_{A2}, h_1, i)\}$  are identified by lemma 1, and the probabilities  $\{\Pr(R_{A1}|L_{A1} = 1, h_1, i)\}$  are identified by step 2. Thus, the conditional assignment probability  $\Pr(L_{A2}|L_{A1} = 1, R_{A1}, h_1, i)$  is identified for all possible  $L_{A2}, R_{A1}, h_1$ , and  $i$  from the ratio of  $\Pr(L_{A1} = 1, R_{A1}, L_{A2}, h_1, i)$  to the product  $\Pr(R_{A1}|L_{A1} = 1, h_1, i) \Pr(i|L_{A1} = 1, h_1) \Pr(L_{A1} = 1, h_1)$ . This logic can be adapted to any tenure.

As for expected output at A, abstract first from exogenous separations. The expected value of  $\bar{V}^A(s_{it}, \varepsilon_t)$  from proposition 1 conditional on  $s_{it}$  can be expressed as  $\ln \sum_k \exp\{\bar{v}^A(s_{it}, k')\}$ , where

$$\bar{v}^A(s_{it}, k) = y_A(s_{it}, k) + \delta E[\bar{v}^A(s_{it+1}, k) - \ln \Pr(L_{A,t+1} = k | f_{t+1} = A, s_{it+1}) | s_{it}, k], \quad (15)$$

the second term on the right-hand side of (15) equals  $\delta E(\ln \sum_{k'} \exp\{\bar{v}^A(s_{it+1}, k')\} | s_{it}, k)$ , and  $\Pr(L_{At} = k | f_t = A, s_{it}) = \exp\{\bar{v}^A(s_{it}, k)\} / \sum_{k'} \exp\{\bar{v}^A(s_{it}, k')\}$  is the probability of assignment to level  $k$  conditional on employment at A at state  $s_{it}$  by the properties of Gumbel distributions. For fixed probabilities, (15) is a functional equation defining a contraction mapping and so admits a unique solution. Since the process of  $s_{it}$  and conditional assignment probabilities at A are identified, it thus follows that if  $y_A(s_{it}, k)$  is assumed to be known at each  $s_{it}$  for one level, as is standard, say,  $k = 2$ , then  $\bar{v}^A(s_{it}, 2)$  is also identified by (15). At levels 1 and 3,  $\bar{v}^A(s_{it}, k)$  is identified at each  $s_{it}$  by the sum of  $\bar{v}^A(s_{it}, 2)$  and  $\ln \Pr(L_{At} = k | f_t = A, s_{it}) - \ln \Pr(L_{At} = 2 | f_t = A, s_{it})$ , and so is  $y_A(s_{it}, k)$  as the difference between  $\bar{v}^A(s_{it}, k)$  and the term  $\delta E[\cdot]$  on the right-hand side of (15).

With exogenous separations, a similar argument applies. The rates  $\{1 - \eta_{kt}\}$  are identified from the distributions of managers' wages at A's levels whenever they are equal between any two levels and constant from period  $T$  on, and the process for  $s_{it}$  is *level symmetric* after period  $T$ . Namely, I require the distribution of, say,  $s_{iT+2}$  to be the same regardless of whether a manager is assigned to levels  $k$  and  $k'$  or to levels  $k'$  and  $k$  in  $T$  and  $T + 1$ . This property ensures that the wage moment conditions from which I show that  $\{\eta_{kt}\}$  can be recovered are polynomials of low order in  $\{\eta_{kt}\}$ .<sup>36</sup>

**PROPOSITION 5.** Let  $\{y_A(s_{it}, 2)\}$  be known. If (i)  $\{y_C(s_{it}, 2)\}$  are known in  $t \geq T + 1$ , (ii)  $\eta_{kt} = \eta_{k't}$  for some  $k$  and  $k'$  in  $t \geq T$  and  $\eta_{kt} = \eta_{kT}$  in  $t \geq T + 1$ , and (iii) the process for  $s_{it}$  is level symmetric in  $t \geq T + 1$ , then  $\{\eta_{kt}\}$  and, for  $k = 1, 3$ ,  $\{y_A(s_{it}, k)\}$  are identified from the distributions of managers' level assignments and wages at A, respectively, once  $\{\alpha_k, \beta_k, p_{i1}, q_i\}$  are identified.

Alternatively,  $\{\eta_{kt}\}$  and  $\{y_A(s_{it}, k)\}_{k=1,3}$  are identified from a panel of length  $T$  until  $T$  up to  $\{\ln \sum_{k'} \exp\{\bar{v}^C(s_{iT+1}, k')\}\}$  and  $\{\ln \sum_{k'} \exp\{\bar{v}^A(s_{iT+1}, k')\}\}$ , respectively; see the proof of proposition 5 for details.<sup>37</sup>

*Step 4: wages at the firm.*—It is immediate that the expected wage at level  $k$  of A in (12) satisfies

$$Ew_{Ait}(k) = \ln \sum_{k'} \exp\{\bar{v}^C(s_{it}, k')\} - \delta \eta_{kt} E(\ln \sum_{k'} \exp\{\bar{v}^C(s_{it+1}, k')\} | s_{it}, k) \quad (16)$$

<sup>36</sup> The process for  $s_{it}$  is level symmetric after  $T$  if, e.g., human capital acquisition no longer occurs from  $T$  on, in which case the state in  $t \geq T + 1$  reduces to  $(p_{it}, \kappa_T, i)$ , or more generally, human capital acquisition no longer depends on the assigned level from  $T$  on. To see why, consider the former case and note that since the process of performance on which the updating of beliefs is based is exchangeable, the same distribution of posterior beliefs is reached in  $T + 2$ , given any prior in  $T$ , if a manager's assigned levels in  $T$  and  $T + 1$  are permuted.

<sup>37</sup> That  $\{y_C(s_{it}, 2)\}$  are known past  $T$  helps pin down how  $\{\eta_{kt}\}$  affect wages at A, as is apparent from (16). See app. sec. B4 for the estimation of firms' expected match surplus values.

by the properties of Gumbel distributions. Once rearranged, (16) defines a functional equation that pins down firm C's expected match surplus value,  $\ln \Sigma_{k'} \exp\{\bar{v}^C(s_{it}, k')\}$ , since mean wages,  $\{Ew_{\lambda it}(k)\}$ , are identified by lemma 1, the process of  $s_{it}$  is identified by lemma 1 and steps 1 and 2, and exogenous separation rates,  $\{1 - \eta_{it}\}$ , are identified by step 3. Suppose that conditional assignment probabilities are equal across firms, as is the case when differences in expected output across firms can be represented as described in the next proposition.<sup>38</sup> Then, C's expected output at each level is identified from the distribution of managers' wages at A at the corresponding level.

**PROPOSITION 6.** Firm C's expected match surplus value is identified from the level distributions of managers' wages at A, once  $\{\alpha_k, \beta_k, p_{it}, \eta_{it}\}$  are identified. Define  $\gamma(s_{it}, k) = \phi(s_{it}) - \delta \eta_{it} E[\phi(s_{it+1}) | s_{it}, k]$  for some measurable function  $\phi(\cdot)$ . If  $y_C(s_{it}, k) = y_A(s_{it}, k) + \gamma(s_{it}, k)$  for each  $s_{it}$  and  $k$ , then  $\{y_C(s_{it}, k)\}$  are also identified from the level distributions of managers' wages at A.

Since the panel length of my data is  $T = 8$ , firm C's expected match surplus value is identified until period  $T$  up to its value in  $T + 1$ .

## V. Estimation

In this section, I present the empirical specification of the model and the estimation results.

### A. Empirical Specification

I first discuss the specification of the process for human capital and output and then that of wages.

*Human capital and output.*—In (1), I let  $g_k^G(i_k^G, \dots, i_{k-1}^G) = g_k^G(t - 1)$  for human capital  $j = G$ , so only the sum of past investments matters for acquiring human capital with experience in the labor market, as is commonly assumed. Similarly, I let  $g_k^S(i_k^S, \dots, i_{k-1}^S) = g_k^S(i_{k-1}^S)$  for human capital  $j = S$ , so only the last-period investment is relevant for acquiring human capital with experience at specific jobs. This latter process is flexible enough to capture, for instance, the benefit of training before a promotion occurs when  $g_k^S(i_{k-1}^S) > 0$  for  $k > k_{t-1}$  in  $t$ , as is consistent with the evidence that training often occurs right before a promotion or is a

<sup>38</sup> To see how flexible this condition is, note that if the function  $\phi(\cdot)$  defined in proposition 6 is such that  $\phi(s_{it}) = \phi_0(h_1, t - 1, k_{t-1}, i) + \phi_1(h_1, t - 1, k_{t-1}, i)p_{it}$  so that  $\gamma(s_{it}, k) = \phi_0(h_1, t - 1, k_{t-1}, i) - \delta \eta_{it} \phi_0(h_1, t, k, i) + [\phi_1(h_1, t - 1, k_{t-1}, i) - \delta \eta_{it} \phi_1(h_1, t, k, i)]p_{it}$ , then the difference in expected output across firms at any level  $k$  depends not only on  $k$  but also on a manager's previous level  $k_{t-1}$ . I estimate the parameters of the wage process at A on the basis of (12) by exploiting the relationship between  $\{y_C(s_{it}, k)\}$  and  $\{y_A(s_{it}, k)\}$  posited in proposition 6.

prerequisite for it (Cobb-Clark and Dunlop 1999); the potential loss of job-specific human capital upon a promotion when  $g_k^S(i_{k-1}^S) = 0$  for  $k > k_{t-1}$  in  $t$ ; or any constraints that may lead to a cost of reallocating managers across levels when  $g_k^S(i_{k-1}^S) < 0$  for  $k \neq k_{t-1}$  in  $t$ .<sup>39</sup> Then, the state  $s_{it} = (p_{it}, \kappa_t, i)$  reduces to  $s_{it} = (p_{it}, h_1, t - 1, k_{t-1}, i)$  and the human capital functions in (1) to

$$h_{jkt}^G = a_{jkt}^G(i) + g_k^G(t - 1) + \varepsilon_{jkt}^G \text{ and } h_{jkt}^S = a_{jkt}^S(i) + g_k^S(i_{k-1}^S) + \varepsilon_{jkt}^S, \quad (17)$$

where I specify  $a_{jkt}^j(i)$  as  $[a_{jkt}^j + a_f^j(i)]/(\alpha_k - \beta_k)$ , with  $a_{jkt}^j \in \{\bar{a}_{jkt}^j, \underline{a}_{jkt}^j\}$ ,  $a_f^j(i) \in \{\bar{a}_f^j(i), \underline{a}_f^j(i)\}$ , and  $a_A^j(i) = 0$  for each  $i$ .<sup>40</sup> Thus, as I show in appendix section B3, expected output in (10) simplifies to

$$y_f(s_{it}, k) = b_k(h_1) + d_{jkt}(k_{t-1}, i) + e_{jkt}(i)p_{it}, \quad (18)$$

where  $b_k(h_1) + d_{jkt}(k_{t-1}, i)$  is the form of  $d_{jk}(\kappa_t, i)$  and  $b_k(h_1)$ , which captures  $h_{jkt}^P$ , is general across firms and constant over time, with  $h_1 = \ln H_1 = (e_1, x_1, y_1)$  denoting education at entry  $e_1$ , experience (age) at entry  $x_1$ , and the year of entry  $y_1$  into firm A. I specify  $b_k(h_1)$  as  $b_k^x x_1 + b_k^{xx} x_1^2 + b_k^e e_1 + \sum_m b_{ym} 1\{y_1 = m\}$ , with  $m$  ranging from 0 for the year 1970 to 9 for the year 1979. As discussed, I estimate  $d_{Akt}(k_{t-1}, i)$  and  $e_{Akt}(i)$  for firm A as  $d_{Akt}(k_{t-1})$  and  $e_{Akt}$ .

This formulation allows for different degrees of transferability of the human capital of both type G and type S across job levels that can be measured. Define the human capital acquired at firm  $f$  (type G) to be task general if the tenure profiles of the parameters  $\{d_{jkt}(k_{t-1}, i)\}$  and  $\{e_{jkt}(i)\}$  are positively correlated across levels ( $k$ ) and task specific otherwise (Gibbons and Waldman 2006). This notion captures the idea that managers may become more productive at all levels with experience at a firm. Define instead the human capital acquired at level  $k$  of firm  $f$  (type S) to be task general if it is at least as productive at level  $k'$  as it is at level  $k$ ,  $d_{jkt}(k, i) \geq d_{jkt}(k', i)$ , and task specific otherwise. This notion captures the idea that experience at a firm's level ( $k$ ) may increase productivity at a firm's other levels ( $k'$ ) even more. As in Cameron and Heckman (2001), to conserve on parameters, for transitions with relatively few observations, I do not estimate any of the associated parameters, and I set them equal to the reference value of zero. For transitions with a similar

<sup>39</sup> Baker et al. (1994b) argue that training occurs at the firm, as high performance is rewarded primarily with salary raises rather than bonuses, suggesting that these raises reward permanent rather than transitory increases in productivity.

<sup>40</sup> Scaling  $a_{jkt}^j(i)$  by the informativeness of level  $k$ , as captured by  $\alpha_k - \beta_k$ , allows for a convenient form for (10) for firm C and so for the wage equation in (19), in which the coefficient on the prior  $p_{it}$  is separable in the term capturing the impact of a manager's skill type  $i$  and the term capturing the impact of a manager's level and tenure at A.

number of observations, I restrict their parameters to be equal (see app. sec. B5).<sup>41</sup>

*Wages.*—By (12), managers' wages at firm A depend on the distribution of their assignments when employed by firm C, which are not recorded in my data. Whenever the difference  $y_C(s_{it}, k) - y_A(s_{it}, k)$  for each state  $s_{it}$  and level  $k$  satisfies the condition in proposition 6, however, conditional assignment probabilities are equal across firms in that  $\Pr(L_{Ct} = k | f_i = C, s_{it}) = \Pr(L_{At} = k | f_i = A, s_{it})$  for each  $s_{it}$  and  $k$ . Thus, given the parameterization of the human capital process, (12) reduces to

$$w_{\Lambda it} = \omega(h_i, k, i) + \omega_{4kt} \cdot (t - 1) + (\omega_{5i} + \omega_{5kt})p_{it} - \ln \Pr(L_{\Lambda t} = k | f_i = A, s_{it}) + \varepsilon_{\Lambda it}, \quad (19)$$

where  $\omega(h_i, k, i) = \omega_{0ik} + \omega_{1k}x_1 + \omega_{2k}x_1^2 + \omega_{3k}e_1 + \sum_m \omega_{ym}1\{y_1 = m\}$ , with  $(\omega_{1k}, \omega_{2k}, \omega_{3k}, \omega_{ym}) = (b_k^x, b_k^{xx}, b_k^e, b_{ym})$  and  $(\omega_{1k}, \omega_{2k}, \omega_{3k})$ , denoted by  $(\omega_1, \omega_2, \omega_3)$ , common between levels 1 and 2 for parsimony.<sup>42</sup> See appendix section B7 for the simple mapping between the parameters of C's expected output and those in (19). These latter parameters are easy to interpret:  $\omega_{1k}$ ,  $\omega_{2k}$ ,  $\omega_{3k}$ , and  $\omega_{ym}$  measure the impact of human capital at entry into the firm, allowing through  $\omega_{ym}$  for aggregate conditions at the time of hiring, which Baker, Gibbs, and Holmström (1994a, 1994b) find important, to persistently affect the value of a manager reflected in wages;  $\omega_{4kt}$  measures the impact of the deterministic component of the human capital acquired at the firm; and  $\omega_{5i} + \omega_{5kt}$  measures the impact of its systematic stochastic component governed by a manager's ability and so of beliefs about it.<sup>43</sup>

## B. Estimation Results

I estimate the model by full-information, full-solution maximum likelihood, with the discount factor  $\delta$  set to 0.95. I simplify the solution of equilibrium by extending the model to allow for the free entry of firms

<sup>41</sup> For instance, I set level 2 as the reference level so that  $d_{\Lambda 2t}(L2) = 0$ , where  $Lk = k_{t-1}$ , and restrict  $d_{\Lambda kt}(L1) = 0$  for each  $k$  in  $t > 1$ , as the number of managers at level 1 rapidly declines with tenure. For level 3, given the near absence of demotions from it and the high separation rate from the firm, I impose a form of symmetry in that  $d_{\Lambda 3t}(Lk) = -d_{\Lambda kt}(L3) = 0$ ,  $k < 3$ . The parameters  $\{d_{\Lambda kt}(k_{t-1})\}$  to estimate then reduce to  $\{d_{\Lambda 1t}(L2)\}$  and  $\{d_{\Lambda 3t}(L3)\}$ .

<sup>42</sup> To conserve on parameters, I impose the restriction that  $d_{Ck't}(k, i) = -d_{Ckt}(k', i) = 0$ ,  $k < k'$ , which is analogous to that for level 3 at firm A. Thus, only the parameters  $\{d_{Ckt}(k, i)\}$  among  $\{d_{Ckt}(k_{t-1}, i)\}$  for firm C are nonzero.

<sup>43</sup> I restrict  $\{\sigma_{\Lambda B}\}$  to be constant and denote their common value by  $\sigma_{\Lambda B}$  because of the relatively small number of observations on managers' wages at level 3 compared with that at levels 1 and 2. The components of the finite mixture distributions of managers' wages at firm A at any level and tenure can be labeled by  $i$  by imposing that  $\sigma_{\Lambda ik} \geq \sigma_{\Lambda(i+1)k}$  for level 1 and ordering  $\{\omega_{0ik}\}$  across  $i$  the same way for levels 2 and 3 (see panels A and C of table 5). For the estimates presented below, I approximate the logistic mixture distributions of managers' wages at A with normal ones.

into the labor market and, as is consistent with my sample, by maintaining the no recall of previously employed managers (see app. sec. B4 for details). All parameter estimates are significant at the 1% level. As tables 1–3 and B1 show, the model captures well the patterns of managers’ separations from the firm, their transitions across the firm’s levels, the distributions of performance ratings at levels 1 and 2, and the distributions of wages at each level by tenure. The online appendix presents very similar estimates from a larger sample that includes entrants in the firm at levels higher than level 1.

1. Uncertainty and Learning

The estimated initial prior distribution in panel A of table 4, with  $\{p_{i1}\}$  calculated as  $\exp(\phi_{i1})/[1 + \exp(\phi_{i1})]$ , implies a large degree of uncertainty about managers’ ability and dispersion in the information about it at managers’ entry into the firm. Indeed, the average initial prior probability that a newly hired manager is of high ability is 0.473, but initial priors range from 0.338 for the first skill type to 0.607 for the fourth skill type, with a standard deviation of 0.102.

In terms of the learning parameters, the estimated  $\{\alpha_k, \beta_k\}$  in panel B of table 4 imply that a high-ability manager is more likely to succeed ( $\alpha_k > 0.5$ ), whereas a low-ability manager is more likely to fail ( $\beta_k < 0.5$ ) at any job level ( $k$ ). Also, since the estimated  $\{\alpha_k, \beta_k\}$  are far from 0 and 1, learning about ability is gradual. By Blackwell’s informativeness criterion, the ordering of job levels by their informativeness about ability, from

TABLE 4  
ESTIMATES OF INITIAL PRIOR DISTRIBUTIONS AND PROBABILITIES OF HIGH PERFORMANCE

| Parameters                           | (1)                 | (2)                 | (3)                 | (4)                |
|--------------------------------------|---------------------|---------------------|---------------------|--------------------|
| A. Initial Prior Distributions       |                     |                     |                     |                    |
|                                      | Type 1 ( $i = 1$ )  | Type 2 ( $i = 2$ )  | Type 3 ( $i = 3$ )  | Type 4 ( $i = 4$ ) |
| Prior: $\phi_{i1}$                   | -.672<br>(.075)     | -.484<br>(.016)     | -.141<br>(.006)     | .435<br>(.008)     |
| Mass: $q_i$                          | .155<br>(.012)      | .211<br>(.004)      | .313<br>(.002)      | .321<br>...        |
| B. Probabilities of High Performance |                     |                     |                     |                    |
|                                      | Level 1 ( $k = 1$ ) | Level 2 ( $k = 2$ ) | Level 3 ( $k = 3$ ) |                    |
| High ability: $\alpha_k$             | .514<br>(.062)      | .5437<br>(.006)     | .5435<br>(.002)     |                    |
| Low ability: $\beta_k$               | .456<br>(.014)      | .491<br>(.013)      | .490<br>(.019)      |                    |

NOTE.—I estimate  $\phi_{i1} = \ln[p_{i1}/(1 - p_{i1})]$ , which implies that  $p_{11} = 0.338$ ,  $p_{21} = 0.381$ ,  $p_{31} = 0.465$ , and  $p_{41} = 0.607$ , with an average initial prior of 0.473 and a standard deviation of 0.102. Asymptotic standard errors are in parentheses.

lowest to highest, is 2, 3, and 1. Interestingly, the ordering of job levels by the probability of high performance of a manager of either high or low ability, from lowest to highest, is the opposite—that is, 1, 3, and 2. Hence, when deciding on a manager's job assignment, the firm faces a clear trade-off between a manager's probability of success and the information it can acquire about a manager's ability, especially at level 1. Namely, although employment at level 1 generates the most information about ability, a manager of any ability has the lowest probability of high performance at this level. The difference in the implied learning speed across levels is key to the shape of wage profiles at the firm, as the counterfactual experiments in section VI will confirm. In particular, the greater informativeness of level 1 largely accounts for the convexity of wages in job levels and tenure—wages would increase much faster in early tenures if jobs were equally informative—and contributes to the wage premium paid to managers at promotion to level 2, measured by the mean percentage wage raise relative to unpromoted managers, as is consistent with the estimates in Baker, Gibbs, and Holmström (1994a, table VI). Such a premium compensates managers for switching to a less informative level when promoted out of level 1 and so for a lower speed of future learning about their ability. The estimates of the distribution of classification error in performance ratings are discussed in appendix section B8.<sup>44</sup>

## 2. Human Capital and Output

The estimates of the parameters  $\{d_{Akl}(k_{l-1})\}$  and  $\{e_{Akl}\}$  for firm A, with  $k_{l-1} = Lk$ , are reported in table B4, expressed in thousands; all unreported parameters are set to zero, the value to which  $\{d_{A2l}(L2)\}$  are normalized, as discussed in section V.A.<sup>45</sup> From the table, three features emerge. First, both the parameters  $\{e_{A2d}\}$  and  $\{e_{A3d}\}$  start at low values, increase to peak at intermediate tenures—namely,  $t = 2$  and  $t = 4$  for  $\{e_{A2d}\}$  and  $t = 4$  for  $\{e_{A3d}\}$ —and essentially flatten out thereafter. This positive correlation in the tenure profiles of  $\{e_{A2d}\}$  and  $\{e_{A3d}\}$  suggests that the human capital

<sup>44</sup> To measure the speed of learning, I calculate the number of consecutive years of high performance that is necessary to infer with a 90% chance that a manager is of high ability. This speed differs across levels and managers' skill types. Starting at the average initial prior, it takes 20 years to reach this level of certainty at level 1, whereas this process takes 23 years at levels 2 or 3. For managers of the fourth skill type, this level of certainty is reached after just 15, 18, and 17 years, respectively, at levels 1, 2, and 3. That learning is gradual is consistent with the results in Nagypál (2007), where figure 7 shows that the convergence of beliefs, as reflected in the tenure profiles of match quality and output, occurs past the tenth year of tenure. Such a speed of learning is also in line with the large degree of uncertainty about ability at entry into the firm that I estimate, even for managers with several years of experience.

<sup>45</sup> I set the parameters  $d_{A11}(L0) = d_{A11}$  and  $e_{A11}$ , respectively, to 1,000 and zero, since managers in the estimation sample are first observed at level 1. See later the estimates of  $(\omega_{1k}, \omega_{2k}, \omega_{3k}, \{\omega_{ym}\})$  that parametrize  $b_k(h_i)$ .



acquired with experience at firm A (of type G) is task general between levels 2 and 3 by the definition in section V.A. Moreover, since  $e_{Akt}$  equals  $E(y_{Akt}|\alpha, \kappa_t) - E(y_{Akt}|\beta, \kappa_t)$  by (3) and so measures the output return to high ability, this return tends to increase at levels 2 and 3 with tenure.

Second, the human capital acquired with experience at level 3 of firm A (of type S) is task specific and more productive at level 3 than that acquired at level 2, which is also task specific, or at level 1. Thus, demoting managers to lower levels is costly, which helps explain why demotions are rare. To elaborate, recall from section V.A that the human capital acquired at level  $k$  is task general when  $d_{Akt}(Lk) \geq d_{Akt}(Lk)$  and that  $d_{A3t}(Lk)$  and  $d_{Aht}(L3)$ ,  $k = 1, 2$ , are set to zero (see n. 41). Consider first level 3. Since  $d_{A3t}(L3)$  is large and positive between  $t = 4$  and  $t = 7$ , managers at level 3 in  $t - 1$  and  $t$  produce more in  $t$  between  $t = 4$  and  $t = 8$  than they would if they were demoted in  $t$  from level 3 to level 2 at many priors, as  $d_{A3t}(L3) \geq d_{A2t}(L3)$ , or from level 3 to level 1 at all priors, as  $d_{A3t}(L3) \geq d_{A1t}(L3)$  and  $e_{A3t} > e_{A1t}$ . In particular, the human capital acquired at level 3 is task specific in mid to high tenures. Furthermore, managers at level 3 in  $t - 1$  and  $t$  between  $t = 4$  and  $t = 8$  produce more in  $t$  than they would if they were promoted in  $t$  from levels 1 or 2 to level 3, as  $d_{A3t}(L3) \geq d_{A3t}(L1)$  and  $d_{A3t}(L3) \geq d_{A3t}(L2)$ . The human capital acquired at level 3 is then more productive at level 3 than that acquired at levels 1 or 2. Consider now level 2. As  $d_{A1t}(L2)$  is estimated to be negative between  $t = 4$  and  $t = 7$  so that  $d_{A2t}(L2) \geq d_{A1t}(L2)$  in addition to  $e_{A2t} > e_{A1t}$  between  $t = 4$  and  $t = 8$ , managers at level 2 in  $t - 1$  and  $t$  produce more in  $t$  in these tenures than they would if they were demoted in  $t$  from level 2 to level 1. Since  $d_{A2t}(L2) \geq d_{A1t}(L2)$  and  $d_{A2t}(L2) = d_{A3t}(L2)$  between  $t = 4$  and  $t = 8$ , the human capital acquired at level 2 tends to be task specific in mid to high tenures.

Third, high-ability managers have an absolute advantage at each level in most tenures, as  $e_{Akt} \geq 0$  except at level 3 in  $t \leq 3$ ; a comparative advantage at level 2 over level 1, as  $e_{A2t} > e_{A1t}$  in almost all tenures; and a comparative advantage at level 3 over level 1 in mid to high tenures, as  $e_{A3t} > e_{A1t}$  between  $t = 4$  and  $t = 8$ , and over level 2 in  $t = 8$ .<sup>46</sup> These features help rationalize why managers reach higher levels over time as their perceived ability and human capital increase.

Overall, these estimates suggest that expected output tends to be more sensitive to ability at higher levels in mid to high tenures when  $e_{Akt} > e_{A1t}$ ; to increase at levels 2 and 3 over time, as  $\{d_{Akt}(k_{t-1})\}$  are fairly stable in mid to high tenures and  $\{e_{Akt}\}$  tend to increase at levels 2 and 3 with tenure; and to be higher at levels 2 and 3 than at level 1 in mid to high tenures when  $\{d_{Akt}(k_{t-1})\}$  and  $\{e_{Akt}\}$  at levels 2 and 3 are higher than at level 1. Thus,

<sup>46</sup> Since comparative advantage at level  $k'$  over level  $k$  amounts to  $[b_k(h_1) + d_{kt}(k_{t-1}, i) + e_{kt}(i)]/[b_k(h_1) + d_{kt}(k_{t-1}, i) + e_{kt}(i)] \geq [b_{k'}(h_1) + d_{k't}(k_{t-1}, i)]/[b_{k'}(h_1) + d_{k't}(k_{t-1}, i)]$  by (18), an approximately equivalent condition is  $e_{Akt} > e_{A1t}$ , in light of the estimates.

higher ability and acquired human capital make managers more productive at higher levels, which contribute more to output than lower levels when managers in mid to high tenures are assigned to them.

3. Wages

The estimates in table 5 are consistent with those in the literature, even those based on more representative samples, in terms of the implied returns to education and age, the size of wage increases at promotion, the curvature of wages in job levels and tenure, and the magnitude of wage

TABLE 5  
ESTIMATES OF TYPE-SPECIFIC INTERCEPTS IN WAGE EQUATION

| Parameters                                       | (1)                    | (2)                      | (3)                      | (4)                      |
|--------------------------------------------------|------------------------|--------------------------|--------------------------|--------------------------|
| A. Type-Specific Intercepts                      |                        |                          |                          |                          |
|                                                  | Type 1 ( <i>i</i> = 1) | Type 2 ( <i>i</i> = 2)   | Type 3 ( <i>i</i> = 3)   | Type 4 ( <i>i</i> = 4)   |
| Level 1: $\omega_{01}$                           | 8.805<br>(.100)        | 9.288<br>(.005)          | 9.213<br>(.011)          | 8.865<br>(.013)          |
| Level 2: $\omega_{02}$                           | 8.969<br>(.184)        | 9.359<br>(.004)          | 9.281<br>(.009)          | 8.945<br>(.012)          |
| Level 3: $\omega_{03}$                           | 9.534<br>(.005)        | 9.813<br>(.004)          | 9.738<br>(.007)          | 9.418<br>(.011)          |
| B. Coefficients on Age,<br>Education, and Tenure |                        |                          |                          |                          |
|                                                  | Value                  | Standard Error           |                          |                          |
| Levels 1 and 2:                                  |                        |                          |                          |                          |
| $\omega_1$ (age)                                 | .028                   | .003                     |                          |                          |
| $\omega_2$ (age <sup>2</sup> )                   | −.0003                 | .000001                  |                          |                          |
| $\omega_3$ (education)                           | .022                   | .00001                   |                          |                          |
| Level 1:                                         |                        |                          |                          |                          |
| $\omega_{412}$ (tenure)                          | .007                   | .0002                    |                          |                          |
| Level 3:                                         |                        |                          |                          |                          |
| $\omega_{13}$ (age)                              | .010                   | .001                     |                          |                          |
| $\omega_{23}$ (age <sup>2</sup> )                | −.0001                 | .00001                   |                          |                          |
| $\omega_{33}$ (education)                        | .021                   | .001                     |                          |                          |
| C. Prior Coefficients and Standard Deviations    |                        |                          |                          |                          |
|                                                  | Type 1 ( <i>i</i> = 1) | Type 2 ( <i>i</i> = 2)   | Type 3 ( <i>i</i> = 3)   | Type 4 ( <i>i</i> = 4)   |
| Prior: $\omega_{5i}$                             | 2.371<br>(.011)        | 1.833<br>(.027)          | 1.316<br>(.015)          | 1.364<br>(.010)          |
| Level 1: $\sigma_{\Lambda i1}$                   | .076<br>(.0005)        | .070<br>(.001)           | .057<br>(.001)           | .044<br>(.001)           |
| Level 2: $\sigma_{\Lambda i2}$                   | .063<br>(.00005)       | .047<br>(.001)           | .0302<br>(.0004)         | .0303<br>(.0004)         |
| Level 3: $\sigma_{\Lambda i3}$                   | .047<br>(.001)         | (as <i>i</i> = 1)<br>... | (as <i>i</i> = 1)<br>... | (as <i>i</i> = 1)<br>... |

NOTE.—Asymptotic standard errors are in parentheses.

growth on the job. These estimates reveal key features of the wage process at the firm.

To start, as shown in panel B of table 5, the human capital acquired before entry into the firm, as captured by a manager's age and education at entry, has a significant impact on wages. Namely, at levels 1 and 2, the (marginal) effect on wages of age at entry, evaluated at the average age at entry of 29.71 years, is a 1.0% increase in annual (log) wages, since  $\omega_1 + 2\omega_2(29.71) = 0.0102$ , and that of schooling is a 2.2% increase, since  $\omega_3 = 0.022$ . At level 3, the corresponding estimates are 0.4%, as  $\omega_{13} + 2\omega_{23}(29.71) = 0.0041$ , and 2.1%, as  $\omega_{33} = 0.021$ . These estimates are similar to those in the literature. For instance, Belzil and Bognanno (2008, their table 1) document coefficients of 0.0127 and 0.0494 for the impact of age and education on (log) wages from a large multifirm sample of US executives observed between 1981 and 1988. Perhaps not surprisingly, since I focus on highly educated individuals at lower managerial ranks, my estimate of the marginal impact of education on wages is smaller than theirs.

Next, the estimates of  $\{\omega_{0ik}\}$  in panel A of table 5 imply that promotions lead to sizable wage increases. That is, a promotion from level 1 to 2 entails an increase in (base) annual wages between \$590 for the fourth skill type and \$1,188 for the first one, whereas a promotion from level 2 to 3 entails an increase between \$4,638 for the fourth skill type and \$6,667 for the second one.<sup>47</sup> Since wages are on average much higher at higher levels, average wages are convex not only in levels—see the first three entries in column 1 of table 6 for the model and the note to table 3 for the data—but also in tenure, as higher levels are reached by managers only over time. These results are in line with the evidence from the literature on internal labor markets (Main, O'Reilly, and Wade 1993; Gibbons and Waldman 1999a; Belzil and Bognanno 2008; Waldman 2013).

In terms of the human capital acquired at the firm, I specify the tenure coefficient at level 1 as  $\omega_{41t} = \omega_{412}1\{t < 4\} + \omega_{414}1\{t \geq 4\}$  with  $\omega_{414} = -\omega_{412}$  to conserve on parameters, given the much smaller number of managers at level 1 after the third year at the firm—as  $\{\omega_{4kt}\}$  have proved to be relatively small at levels 2 and 3, they are omitted from the baseline specification and estimated instead as part of the robustness exercise discussed in appendix section B11. In light of the size of the estimate of  $\omega_{412}$  relative to, say, the size of the estimates of  $\omega_1$  and  $\omega_{13}$  in panel B of table 5, the deterministic component of the human capital acquired at the firm, in contrast to that acquired before entry into it, has a limited direct effect on

<sup>47</sup> These increases are calculated by differencing across job levels the estimates of  $\{\omega_{0ik}\}$  once converted from logs to levels, say,  $\$590 = e^{\omega_{012}} - e^{\omega_{011}}$ . The estimated year-of-entry effects in table B6 are consistent with the recessions between 1973 and 1982, which depressed the wages of entrants in the firm in those years. See app. sec. B7 for details.

TABLE 6  
COUNTERFACTUAL EXPERIMENTS ON IMPORTANCE OF LEARNING FOR WAGES

| STATISTIC                  | WAGES           |                       |                              |                  |                |
|----------------------------|-----------------|-----------------------|------------------------------|------------------|----------------|
|                            | Baseline<br>(1) | No<br>Learning<br>(2) | No<br>Experimentation<br>(3) | Fast Learning at |                |
|                            |                 |                       |                              | Level 1<br>(4)   | Level 2<br>(5) |
| Mean (US\$):               |                 |                       |                              |                  |                |
| Level 1                    | 39,584          | 39,706                | 39,763                       | 58,271           | 37,847         |
| Level 2                    | 43,179          | 43,070                | 42,600                       | 61,451           | 77,503         |
| Level 3                    | 48,963          | 48,454                | 48,818                       | 44,623           | 24,360         |
| Standard deviation (US\$): |                 |                       |                              |                  |                |
| Level 1                    | 6,936           | 6,791                 | 6,902                        | 35,961           | 8,668          |
| Level 2                    | 7,077           | 6,464                 | 6,831                        | 51,466           | 45,057         |
| Level 3                    | 8,046           | 6,534                 | 7,971                        | 45,784           | 4,281          |
| Cumulative growth (%):     |                 |                       |                              |                  |                |
| Tenure 2                   | 4.6             | 3.3                   | .9                           | 39.3             | 8.8            |
| Tenure 3                   | 8.9             | 6.8                   | 17.6                         | 48.5             | 51.5           |
| Tenure 4                   | 13.8            | 9.8                   | 20.5                         | 52.8             | 50.1           |
| Tenure 5                   | 15.9            | 11.1                  | 21.6                         | 55.4             | 50.3           |
| Tenure 6                   | 17.5            | 12.9                  | 22.1                         | 58.1             | 50.1           |
| Tenure 7                   | 18.5            | 14.6                  | 22.2                         | 60.6             | 49.8           |
| Tenure 7 (balanced; %)     | 19.4            | 15.4                  | 23.3                         | 62.5             | 51.2           |

NOTE.—No learning:  $\beta_k = \hat{\alpha}_k$ ,  $k = 1, 2, 3$ . Fast learning at level  $k$ :  $\alpha_k = 0.99$  and  $\beta_k = 0.01$ ,  $k = 1, 2$ .

wages. Acquired human capital still has an important indirect effect on wages through its impact on promotions, as illustrated in section VI. As for ability, since the estimated coefficients  $\{\omega_{5,i}\}$  in panel C of table 5 on the prior  $p_{it}$  in period  $t$  are positive and significant, average wages tend to increase with  $p_{it}$ : high-ability managers are valued—and so are compensated—more than low-ability managers.<sup>48</sup>

Finally, note that although the standard deviation  $\sigma_{A,ik}$  of wage shocks in panel C of table 5 approximately decreases with the job level  $k$ , the standard deviation of wages in the model is higher at higher levels, as in the data—see the fourth, fifth, and sixth entries in column 1 of table 6 for the model and the note to table 3 for the data. Thus, the greater wage dispersion at higher job levels is generated primarily by the endogenous mechanisms of the model of learning about ability, human capital acquisition, and job assignment rather than by idiosyncratic unexplained factors.

4. A Decomposition of Wage Growth

By table 6, the model implies an increase in average wages of approximately 20% over the first 7 years at the firm—18.5% for the unbalanced

<sup>48</sup> The parameters  $\{\omega_{5,ik}\}$  in (19) are estimated as part of the robustness exercise in app. sec. B11.

panel that includes separating managers and 19.4% for the balanced panel that excludes them—for an average yearly growth rate of 3.3%. Such growth is broadly consistent with that in Topel (1991, his table 2) from the Panel Study of Income Dynamics (PSID) and the lower end of the range estimated by Buchinsky et al. (2010, their fig. 2), also from the PSID.<sup>49</sup> By (19), the average wage of a manager of skill type  $i$  at level  $k$  can be decomposed into four terms: (i)  $\omega(h_1, k, i)$ , which captures the effect of the human capital at entry into the firm ( $h_1$ ), job assignment ( $k$ ), and other unmeasured manager characteristics ( $i$ ); (ii)  $\omega_{4kt} \cdot (t - 1)$ , which captures the direct effect of the deterministic component of the human capital acquired at the firm; (iii)  $(\omega_{5i} + \omega_{5kt})p_{it}$ , which captures the direct effect of learning about ability, as ability governs the systematic stochastic component of acquired human capital  $\{a_{jkt}^i\}$ ; and (iv) the negative (log) probability of assignment to level  $k$ .<sup>50</sup> Over the first 7 years, the first term accounts for nearly all the increase in wages. It would then seem that learning accounts for only a trivial percentage of wage growth.

But this conclusion is inaccurate: the term  $\omega(h_1, k, i)$  depends on a manager's assigned job  $k$ , which changes as information and human capital accumulate. Specifically, measuring the cumulative effect of learning on wages requires taking into account how a change in the prior  $p_{it}$  in  $t$  affects a manager's wage  $w_{it} = w_f(p_{it}, \kappa_t, i, \varepsilon_t)$  and job  $k_{it} = k_f(p_{it}, \kappa_t, i, \varepsilon_t)$  at the employing firm  $f$  in  $t$ , which determines a manager's prior  $p_{it+1} = p(p_{it}, k_{it})$  and human capital component  $\kappa_{t+1} = \kappa(\kappa_t, k_{it})$  in  $t + 1$ . A manager's prior and human capital component in  $t + 1$ , in turn, affect a manager's wage  $w_{it+1} = w_f(p_{it+1}, \kappa_{t+1}, i, \varepsilon_{t+1})$  and job  $k_{it+1} = k_f(p_{it+1}, \kappa_{t+1}, i, \varepsilon_{t+1})$  at firm  $f$  in  $t + 1$ , which determines a manager's prior  $p_{it+2} = p(p_{it+1}, k_{it+1})$  and human capital component  $\kappa_{t+2} = \kappa(\kappa_{t+1}, k_{it+1})$  in  $t + 2$ , and so on. That is, two additional processes need to be estimated to assess such a cumulative effect, one for beliefs about ability and one for job assignment, in addition to those for human capital and wages. Since wages substantially differ across job levels because of the term  $\omega(h_1, k, i)$ , the belief process can have a sizable impact on wage growth through its indirect effect on the path of promotions, although its direct effect through the term  $(\omega_{5i} + \omega_{5kt})p_{it}$  is small. As section VI shows, learning indeed leads managers progressively revealed to be of high ability to advance to higher job levels, at which they are paid higher wages. This indirect effect is large and accounts for almost all of the impact of learning on wage growth.

<sup>49</sup> Song et al. (2019) find that the median wages of employees with two to four and with five or more years of tenure in 1982, which is a year comparable to my sample, are \$29,763 and \$45,010 in 2013 dollars, respectively, which imply a cumulative growth rate of 51.2% (see their replication material). Although my sample consists of college-educated managers, the cumulative growth in their median wages is very similar (see Baker, Gibbs, and Holmström 1994b, their figure III).

<sup>50</sup> The estimates of  $\{\omega_{5kt}\}$ , which are only part of the robustness exercise in app. sec. B11, are relatively small.

## 5. Discussion: Structural Approach versus Reduced-Form Approach

The result that the direct or contemporaneous effect of learning on wages, captured here by  $(\omega_{5i} + \omega_{5it})p_{it}$  in (19), is small is consistent with the findings by Gibbons et al. (2005), Lluís (2005), and Hunnes (2012), who focus on only this direct effect and find it to be small or insignificant.<sup>51</sup> The approach of these papers, though, critically relies on the assumption that firms are symmetric. Without it, these methods cannot be in general applied. To see why, consider Lluís (2005), which is based on Gibbons and Waldman (1999b) and adapts the methods in Gibbons et al. (2005). In Lluís (2005), workers can be of high or low ability, and output signals about ability are i.i.d. normally distributed at each job, so jobs are equally informative about ability. Lluís (2005) denotes worker  $i$ 's rank (job, occupation, or sector)  $k$  in  $t$  by the rank dummy  $D_{ikt}$ . The worker's wage at rank  $k$  in  $t$  equals expected output up to measurement error,  $w_{it} = \sum_k D_{ikt} [d_k + X_{it}\beta_k + c_k\theta_{it}^e f(x_{it})] + \mu_{it}$ , where  $X_{it}$  captures observable worker characteristics;  $\theta_{it}^e$  is the mean of the unobserved posterior beliefs about the worker's ability in  $t$ ;  $f(\cdot)$  is the human capital function, which is normalized in estimation;  $x_{it}$  is labor market experience; and  $\mu_{it}$  is measurement error.

The parameters  $\{d_k\}$ ,  $\{\beta_k\}$ , and  $\{c_k\}$  of the wage equation can be estimated as follows. Observe first that expressions for  $\theta_{it}^e$  and  $\theta_{it-1}^e$  can be obtained by inverting those for a worker's wage in  $t$  and  $t-1$ . Substituting then these expressions for  $\theta_{it}^e$  and  $\theta_{it-1}^e$  into the martingale condition  $\theta_{it}^e = \theta_{it-1}^e + u_{it-1}$  for the posterior mean, where  $u_{it-1}$  is a noise term orthogonal to  $\theta_{it-1}^e$ , yields

$$\begin{aligned} \frac{w_{it}}{\sum_k D_{ikt} c_k f(x_{it})} &= \frac{\sum_k D_{ikt} (d_k + X_{it}\beta_k)}{\sum_k D_{ikt} c_k f(x_{it})} + \frac{w_{it-1} - \sum_k D_{ikt-1} (d_k + X_{it-1}\beta_k)}{\sum_k D_{ikt-1} c_k f(x_{it-1})} \\ &+ \underbrace{\frac{e_{it}}{u_{it-1} + \frac{\mu_{it}}{\sum_k D_{ikt} c_k f(x_{it})} - \frac{\mu_{it-1}}{\sum_k D_{ikt-1} c_k f(x_{it-1})}}}_{\text{.}} \end{aligned} \quad (20)$$

In the complete information case in which ability is known ( $\theta_{it}^e = \theta_{it-1}^e = \theta_i$ ), an expression similar to (20) holds, just without the term  $u_{it-1}$ . The equation in (20) is a random coefficient panel data model with unobserved individual time-varying effects, in which  $D_{ikt}$  is endogenous, as it depends on  $\theta_{it}^e$  and so is correlated with  $e_{it}$  through  $u_{it-1}$ , and  $w_{it-1}$  is correlated with  $\mu_{it-1}$ . Yet (20) can be estimated by nonlinear IV methods using as instruments for  $D_{ikt}$  and  $w_{it-1}$  past rank dummies, say,  $D_{ikt-1}$  and  $D_{ikt-2}$ , as well as interactions among them. Intuitively,  $D_{ikt-1}$  and  $D_{ikt-2}$  are uncorrelated with both

<sup>51</sup> By replicating the analysis of Lluís (2005) and Hunnes (2012) on my data, I found similar results.

$u_{it-1}$ , since they are functions of  $u_{it-2}$  and  $u_{it-3}$  that are independent of  $u_{it-1}$ , and  $\{\mu_{it}\}$ , since  $\mu_{it}$  is pure noise. In contrast to the complete information case, Lluís (2005) finds that the parameters of (20) in the incomplete information case are imprecisely estimated and their estimates are implausible, suggesting that learning is unimportant.<sup>52</sup>

Suppose now that although firms are symmetric, jobs differ in their informativeness. In this case, beliefs are updated differently depending on a worker's rank, so the above martingale condition becomes  $\theta_{it}^e = \theta_{it-1}^e + \Sigma_k D_{ikt-1} u_{ikt-1}$  and the first component of the error term  $e_{it}$  in (20) is  $\Sigma_k D_{ikt-1} u_{ikt-1}$  rather than  $u_{it-1}$ . Past rank dummies still provide valid instruments.<sup>53</sup> But when firms differ in their output, human capital, or informational technologies, as embedded in the informativeness of jobs, this IV approach no longer applies, since the inversion step necessary to derive the analog of (20) typically fails. To elaborate, similar to the case of the wage equation in (9), the expression for a worker's wage now contains an additional term. This term captures the difference between the expected continuation match surplus value of the best competitor of the worker's employing firm conditional on its offered job and that conditional on the job offered by the employing firm—see the negative (log) probability term in (9) and proposition 2 in the online appendix for the general case. When firms have identical technologies, this term cancels out, since firms offer the same jobs in each period in this case. In general, such an additional term depends on  $\theta_{it}^e$  in a nonlinear way through the best competitor's expected match surplus value, which is an unknown function. Hence, the initial inversion step required to obtain expressions for  $\theta_{it}^e$  and  $\theta_{it-1}^e$  is unwarranted, which implies that the IV approach described is infeasible.

## VI. Decomposing Job and Wage Mobility

In this section, I assess the role of learning about ability, human capital acquisition, the informativeness of jobs, and persistent uncertainty about ability for managers' job and wage paths.

<sup>52</sup> As Lluís (2005, 751) notes, none of the tests of equality of the coefficients  $\{\beta_k, c_k\}$  can reject the null hypothesis of no comparative advantage under learning, in which case job mobility is purely random. One reason for these results could be that they are based on German survey data. For instance, the low degree of worker mobility in these data makes it difficult to identify any pure rank effects or differential skill ( $X_{it}$ ) and ability ( $\theta_i$  or  $\theta_{it}^e$ ) effects across ranks. Using US data from the National Longitudinal Survey of Youth, though, Gibbons et al. (2005, 706) similarly find that "returns to skill and occupation" are "puzzlingly small" when learning is incorporated and that "introducing learning does not change the results substantially . . . once comparative advantage is properly accounted for."

<sup>53</sup> Although it now depends on  $D_{ikt-1}$  also through  $\Sigma_k D_{ikt-1} u_{ikt-1}$ ,  $e_{it}$  is uncorrelated with  $D_{ikt-1}$ , as  $\Sigma_k D_{ikt-1} u_{ikt-1}$  is orthogonal to any prior information or outcome including  $D_{ikt-1}$ . I thank Thomas Lemieux for helpful comments.



*Learning about ability.*—To evaluate the impact of learning, I compare the implications of the baseline model examined so far with those of a no learning version of it, in which jobs are assumed to be uninformative about ability. Namely, I set  $\beta_k = \alpha_k$  in the belief-updating rules in (4) for each level  $k$ , so prior beliefs are never updated, whereas all the other parameters are fixed at their estimated values.<sup>54</sup> Table 6 shows that learning generates faster wage growth and greater wage dispersion. Over the first 7 years at the firm, the additional wage growth due to learning is a large portion of the wage growth without learning, 26%–27%—wage growth increases from 14.6% to 18.5% for the unbalanced panel and from 15.4% to 19.4% for the balanced panel with learning. The additional wage dispersion due to learning is 23% at level 3—that is, wage dispersion increases from \$6,534 to \$8,046 with learning—with somewhat lower percentages at lower levels. Job assignment is the key channel through which learning affects wages, as is apparent from table B7, which displays the distributions of managers across levels, including the proportions of separating managers, by tenure in the baseline and no learning models. In the baseline model, by the third year of tenure, about 9% of managers have been promoted to level 3, whereas in the no learning model, fewer than 1% have. By the fifth year of tenure, the fraction of managers at level 3 is about 31% in the baseline model and about 16% in the no learning model. These results are consistent with the impact of learning on worker turnover reported by Nagypál (2007) and on occupational mobility found by Papageorgiou (2014). Overall, this experiment illustrates that learning about ability implies more rapid promotions for managers progressively revealed to be of high ability and thus leads to higher wage growth and dispersion, since wages are higher and more variable at higher levels. Hence, the effect of learning on wages is sizable but indirect, as it operates primarily through the impact of learning on the path of managers' job assignments at the firm.

*Human capital acquisition.*—In the previous experiment, managers' progression across job levels is due to only the human capital they acquire at the firm. Acquired human capital improves managers' expected output over time—in particular, at levels 2 and 3—and so makes it profitable for the firm to eventually assign managers to these levels, at which wages are higher and more dispersed. Moreover, acquired human capital is important to prevent demotions and thus reduce the wage loss that managers would otherwise experience after low performance. These findings

<sup>54</sup> When  $\beta_k = \alpha_k$ , I simulate the model assuming that initial priors about ability are never updated but managers of each ability experience high and low performance with the same probabilities as those in the baseline model. I follow the same approach for the remaining experiments that involve  $\{\alpha_k, \beta_k\}$ .

are mirrored by the results of the experiment in which I simulate the model assuming that human capital acquisition does not take place at the firm, by constraining the parameters of expected output to be constant over time, whereas all the other parameters are unchanged. In this case, almost no promotion occurs; when promotions do occur, they are followed by demotions; virtually no wage growth accrues; and wages in any tenure are nearly as dispersed as at entry into the firm. Since acquired human capital has a limited direct effect on wages, as implied by the value of the estimated coefficients on tenure  $\{\omega_{4kt}\}$  in the wage equation, these results suggest that, much like learning, human capital affects wages for the most part indirectly through its impact on the dynamics of job assignment. Also, as the absence of human capital mutes the impact of learning, the information and human capital acquired at the firm have a highly complementary indirect effect on wages.

*Learning by experimentation.*—To examine the role of the different informativeness of jobs about ability, I perform an exercise, referred to as *no experimentation*, in which levels 2 and 3 are made to be as informative as level 1 by setting  $\alpha_k$  and  $\beta_k$ ,  $k \geq 2$ , equal to their estimated values at level 1 in the belief-updating rules in (4). All the other parameters are fixed at their estimated values. Results for the analogous experiments of equal informativeness as levels 2 and 3 are reported in the online appendix. Table B8 shows that without experimentation, nearly all managers who do not separate from the firm are quickly promoted to level 3. In fact, after the first 2 years of tenure, the proportion of managers at level 1 is very small, although learning takes place slowly over time. Hence, the speed of promotions reflects not only the speed of learning at a job level but also the relative speed of learning across different job levels. Experimentation is also quantitatively important for wage growth. Indeed, fast promotions are accompanied by rapid wage growth, as table 6 shows. Without experimentation, by the third year of tenure, wages grow by 17.6%, nearly twice as much as in the baseline model. By the fourth year, managers experience a higher wage growth (20.5%) than they do over the first 7 years in the baseline model (approximately 20%). After that, wage growth is fairly flat. Overall, cumulative wage growth during the first 7 years at the firm would be 20% higher—it increases from 18.5% to 22.2% for the unbalanced panel and from 19.4% to 23.3% for the balanced panel. Wages at each level are only slightly less dispersed than in the baseline model. Thus, the differential informativeness of jobs is critical to accounting for the convexity of wages in job levels and tenure, as it leads to a much more gradual wage growth in early tenures than if jobs were equally informative.

*Persistent uncertainty about ability.*—As I estimate learning to be a gradual process, I conduct two experiments to assess the importance of persistent uncertainty about ability. In the fast learning at level 1 case, level 1 is made

to be nearly perfectly informative about ability ( $\alpha_1 = 0.99$  and  $\beta_1 = 0.01$ ), so that ability is mostly learned after one period of employment. The other parameters are fixed at their baseline values (for a similar approach, see Crawford and Shum 2005). Table 6 shows the implications for wages and table B7 for job assignments. During the first 7 years at the firm, faster learning at level 1 leads wages to grow more than 60% compared with approximately 20% in the baseline model. If managers could sort to the best jobs given their ability and human capital after just 1 year at the firm, wage growth over the first 2 years (39.3%) would already be more than twice as large as the wage growth over the first 7 years in the baseline model. Both the mean and the dispersion of wages at each level sharply increase: their standard deviation at any level is over five times larger than in the baseline model. As table B7 illustrates, promotions to level 3 occur much more rapidly. By the third year of tenure, 20% of managers are at level 3. Hence, persistent uncertainty about ability, like learning through experimentation, substantially compresses wage growth, especially in early tenures, thus helping the model account for the curvature of wages in job levels and tenure, and reduces wage dispersion. The case of fast learning at level 2 in appendix section B12 is similar.

## VII. Conclusion

This paper develops and estimates a model of learning about workers' ability, in which different jobs are differentially informative about ability, human capital acquisition with experience in the labor market, and job assignment to account for rich patterns of careers in firms. The empirical literature on careers focusing on comparative advantage and learning has argued that the impact of learning on wages is small but has investigated only the direct effect of learning, which is due to the impact of beliefs about ability in a period on the period wages. By explicitly estimating the joint dynamics of beliefs about ability, human capital, jobs, and wages, I can evaluate the total effect of learning on wages, which consists of both a direct and an indirect effect. Intuitively, learning also indirectly affects wages by influencing the dynamic selection process that leads managers who are progressively revealed to be of high ability to be promoted to higher levels of a firm's job hierarchy. Since wages at higher-level jobs tend to be higher and more variable, learning critically contributes to wage growth and dispersion by stimulating promotions. I estimate that this indirect effect is large and accounts for almost all of the impact of learning on wages. Since these results are based on data on one firm though, they should be subject to additional investigation to assess their validity. These findings nonetheless attest to the potential of learning and human capital acquisition models to account for salient features of careers in firms.

## Data Availability

The code replicating the estimates in this article can be found in Pastorino (2024) in the Harvard Dataverse, <https://doi.org/10.7910/DVN/L6EJZM>.

## Appendix A

### Model Appendix

#### A1. Proof of Proposition 1

There are two immediate implications of equilibrium. First, the worker chooses the firm,  $f$ , offering the highest present value of wages:  $W(s_t, \varepsilon_t|f) \geq W(s_t, \varepsilon_t|f')$ . Second, the employing firm,  $f$ , offers wages just sufficient to attract the worker in that it obtains the value  $\Pi^f(s_t, \varepsilon_t|f)$  subject to  $W(s_t, \varepsilon_t|f) \geq W(s_t, \varepsilon_t|f')$ , which, by profit maximization, reduces to

$$W(s_t, \varepsilon_t|f) = W(s_t, \varepsilon_t|f'). \quad (21)$$

Otherwise, firm  $f$  could lower its wage offer and still attract the worker. Thus, the worker must be made indifferent between working at firm  $f$  and working at firm  $f'$ . I refer to (21) as the worker's equilibrium indifference between firm A's and firm C's offers, or simply *worker indifference*.

I now show that a firm's job assignment problem in equilibrium simplifies to an autarky-type problem—that is, the problem of choosing the assignment for the worker that maximizes a firm's value of output,  $\bar{V}^f(s_t, \varepsilon_t)$ ,  $f = A, C$ . To start, note that maximizing profits for firm  $f$  is equivalent to maximizing the sum  $V^f(s_t, \varepsilon_t)$  of its own value and the worker's value by (21): the job that maximizes  $\Pi^f(s_t, \varepsilon_t)$  clearly also maximizes  $\Pi^f(s_t, \varepsilon_t) + W(s_t, \varepsilon_t|f')$ ,  $f' \neq f$ , since firm  $f$  takes as given the value of wages implied by the offer of firm  $f'$ . Combining (6), this result, and optimality for firm A implies that  $V^A(s_t, \varepsilon_t)$ , defined as  $\Pi^A(s_t, \varepsilon_t) + W(s_t, \varepsilon_t)$ , can be expressed as

$$\begin{aligned} V^A(s_t, \varepsilon_t) &= \max \{ V^A(s_t, \varepsilon_t|A), V^A(s_t, \varepsilon_t|C) \} \\ &= \max \left\{ \max_{k \in K^A} \left\{ y_A(s_t, k) + \varepsilon_{Akt} \right. \right. \\ &\quad \left. \left. + \delta \eta_{Ak}(\kappa_t) \int_{\varepsilon_{t+1}} EV^A(s_{t+1}, \varepsilon_{t+1}|s_t, k) dG \right\}, w_C(s_t, \varepsilon_t) \right. \\ &\quad \left. + \delta \eta_{Ck_C}(\kappa_t) \int_{\varepsilon_{t+1}} EV^A(s_{t+1}, \varepsilon_{t+1}|s_t, k_{Ct}) dG \right\}, \end{aligned} \quad (22)$$

since firm A's current profits are zero and the worker's wage is  $w_C(s_t, \varepsilon_t)$  if firm C employs the worker at state  $(s_t, \varepsilon_t)$ , in which case the state is updated conditional on the worker's job at C,  $k_{Ct}$ .

Consider first states at which firm A employs the worker. At these states, firm A must prefer employing to not employing the worker—that is,  $\Pi^A(s_t, \varepsilon_t|A) \geq \Pi^A(s_t, \varepsilon_t|C)$ —and pay a wage that makes the worker just indifferent between working at A and working at C—that is,  $W(s_t, \varepsilon_t|A) = W(s_t, \varepsilon_t|C)$  by (21). These two

facts imply that  $\Pi^A(s_t, \varepsilon_t|A) + W(s_t, \varepsilon_t|A) \geq \Pi^A(s_t, \varepsilon_t|C) + W(s_t, \varepsilon_t|C)$  or, equivalently,  $V^A(s_t, \varepsilon_t|A) \geq V^A(s_t, \varepsilon_t|C)$  by the definition of match surplus value.

Consider next states at which firm C employs the worker. Condition (7) requires that firm A be indifferent between employing and not employing the worker; that is,  $\Pi^A(s_t, \varepsilon_t|A) = \Pi^A(s_t, \varepsilon_t|C)$ . By the same logic as earlier, firm C must pay a wage that makes the worker just indifferent between working at A and working at C. These observations imply that  $V^A(s_t, \varepsilon_t|A) = V^A(s_t, \varepsilon_t|C)$ .

Since  $V^A(s_t, \varepsilon_t) = \max\{V^A(s_t, \varepsilon_t|A), V^A(s_t, \varepsilon_t|C)\}$ , this argument then yields that at all states,

$$V^A(s_t, \varepsilon_t) = V^A(s_t, \varepsilon_t|A) = \max_{k \in K^A} \left\{ \gamma_A(s_t, k) + \varepsilon_{Akt} + \delta \eta_{Ak}(\kappa_t) \int_{\varepsilon_{t+1}} EV^A(\cdot | s_t, k) dG \right\}. \quad (23)$$

Namely,  $V^A(s_t, \varepsilon_t)$  reduces to the value of firm A's problem of choosing the job that maximizes the value of its output,  $\bar{V}^A(s_t, \varepsilon_t)$ , defined recursively as

$$\bar{V}^A(s_t, \varepsilon_t) = \max_{k \in K^A} \left\{ \gamma_A(s_t, k) + \varepsilon_{Akt} + \delta \eta_{Ak}(\kappa_t) \int_{\varepsilon_{t+1}} E \bar{V}^A(s_{t+1}, \varepsilon_{t+1} | s_t, k) dG \right\}. \quad (24)$$

A similar argument applies to firm C so that  $V^C(s_t, \varepsilon_t)$  also reduces to  $\bar{V}^C(s_t, \varepsilon_t)$ . Hence,

$$V^f(s_t, \varepsilon_t) = V^f(s_t, \varepsilon_t|f) = \bar{V}^f(s_t, \varepsilon_t), \quad f = A, C. \quad (25)$$

I now derive (8). As just proved, when firm A employs the worker at state  $(s_t, \varepsilon_t)$ ,

$$V^A(s_t, \varepsilon_t|A) = \Pi^A(s_t, \varepsilon_t|A) + W(s_t, \varepsilon_t|A) \geq \Pi^A(s_t, \varepsilon_t|C) + W(s_t, \varepsilon_t|C) = V^A(s_t, \varepsilon_t|C).$$

Since  $\Pi^C(s_t, \varepsilon_t|A) = \Pi^C(s_t, \varepsilon_t|C)$  by (7), it follows that  $V^A(s_t, \varepsilon_t|A) \geq V^A(s_t, \varepsilon_t|C)$  implies that

$$\Pi^A(s_t, \varepsilon_t|A) + W(s_t, \varepsilon_t|A) + \Pi^C(s_t, \varepsilon_t|A) \geq \Pi^A(s_t, \varepsilon_t|C) + W(s_t, \varepsilon_t|C) + \Pi^C(s_t, \varepsilon_t|C). \quad (26)$$

Similarly, as argued, when firm C employs the worker at state  $(s_t, \varepsilon_t)$ ,

$$V^A(s_t, \varepsilon_t|C) = \Pi^A(s_t, \varepsilon_t|C) + W(s_t, \varepsilon_t|C) = \Pi^A(s_t, \varepsilon_t|A) + W(s_t, \varepsilon_t|A) = V^A(s_t, \varepsilon_t|A), \quad (27)$$

whereas it must be that  $\Pi^C(s_t, \varepsilon_t|C) \geq \Pi^C(s_t, \varepsilon_t|A)$ , since firm C is the employing firm. Thus,

$$\Pi^A(s_t, \varepsilon_t|C) + W(s_t, \varepsilon_t|C) + \Pi^C(s_t, \varepsilon_t|C) \geq \Pi^A(s_t, \varepsilon_t|A) + W(s_t, \varepsilon_t|A) + \Pi^C(s_t, \varepsilon_t|A) \quad (28)$$

by (27). Then, (26) and (28) establish that the employing firm is selected in equilibrium to maximize the sum of the values of the two firms and the worker conditional on the two firms' job offers. Equivalently, the employing firm is determined by the policy for the program with value (8), where  $S(s_t, \varepsilon_t) = \Pi^A(s_t, \varepsilon_t) + W(s_t, \varepsilon_t) + \Pi^C(s_t, \varepsilon_t)$ . QED

A2. *Proof of Proposition 2*

Note that the indifference condition in (21) can be rewritten as

$$w_A(s_t, \varepsilon_t) = W(s_t, \varepsilon_t | C) - \delta \eta_{A k_{At}}(\kappa_t) \int_{\varepsilon_{t+1}} EW(\cdot | s_t, k_{At}) dG. \quad (29)$$

Suppose without loss that firm A employs the worker. Then, condition (7) can be expressed as  $\delta \eta_{A k_{At}}(\kappa_t) \int_{\varepsilon_{t+1}} E \Pi^C(\cdot | s_t, k_{At}) dG = V^C(s_t, \varepsilon_t | C) - W(s_t, \varepsilon_t | C)$  or, equivalently, as

$$\begin{aligned} W(s_t, \varepsilon_t | C) = & V^C(s_t, \varepsilon_t | C) - \delta \eta_{A k_{At}}(\kappa_t) \int_{\varepsilon_{t+1}} E \Pi^C(\cdot | s_t, k_{At}) dG = \max_{k \in K^c} \left\{ \gamma_C(s_t, k) + \varepsilon_{Ckt} \right. \\ & \left. + \delta \eta_{Ck}(\kappa_t) \int_{\varepsilon_{t+1}} EV^C(\cdot | s_t, k) dG \right\} - \delta \eta_{A k_{At}}(\kappa_t) \int_{\varepsilon_{t+1}} E \Pi^C(\cdot | s_t, k_{At}) dG, \end{aligned}$$

where the second equality follows from the analog of (23) for firm C. By substituting this expression for  $W(s_t, \varepsilon_t | C)$  into (29) and combining terms using that  $EV^C(\cdot | s_t, k_{At}) = E \Pi^C(\cdot | s_t, k_{At}) + EW(\cdot | s_t, k_{At})$  by the definition of  $V^C(\cdot)$ , I can rewrite (29) as

$$\begin{aligned} w_A(s_t, \varepsilon_t) = & \max_{k \in K^c} \left\{ \gamma_C(s_t, k) + \varepsilon_{Ckt} + \delta \eta_{Ck}(\kappa_t) \int_{\varepsilon_{t+1}} EV^C(\cdot | s_t, k) dG \right\} \\ & - \delta \eta_{A k_{At}}(\kappa_t) \int_{\varepsilon_{t+1}} EV^C(\cdot | s_t, k_{At}) dG. \end{aligned} \quad (30)$$

Express the maximand in (30) as  $\gamma_C(s_t, k) + \delta \eta_{Ck}(\kappa_t) \int_{\varepsilon_{t+1}} E \bar{V}^C(\cdot | s_t, k) dG + \varepsilon_{Ckt}$  by (25) and denote it by  $\bar{v}^C(s_t, k) + \varepsilon_{Ckt}$ . By the properties of Gumbel distributions (Rust 1994), the first term on the right-hand side of (30),  $\bar{V}^C(s_t, \varepsilon_t) = \max_{k \in K^c} \{\bar{v}^C(s_t, k) + \varepsilon_{Ckt}\}$ , reduces to  $\bar{V}^C(s_t, \varepsilon_t) = \ln \sum_{k \in K^c} \exp\{\bar{v}^C(s_t, k)\} + \varepsilon_{Ct}$ , where  $\varepsilon_{Ct}$  is independent Gumbel distributed with mean zero and variance  $\pi^2/6$ , and so  $\bar{v}^C(s_t, k) = \gamma_C(s_t, k) + \delta \eta_{Ck}(\kappa_t) E(\ln \sum_{k' \in K^c} \exp\{\bar{v}^C(s_{t+1}, k')\} | s_t, k)$ .<sup>55</sup> By using (25) and following similar steps to rewrite the last term of (30), I obtain that

$$\begin{aligned} w_A(s_t, \varepsilon_t) = & \ln \sum_{k \in K^c} \exp\{\bar{v}^C(s_t, k)\} \\ & - \delta \eta_{A k_{At}}(\kappa_t) E(\ln \sum_{k \in K^c} \exp\{\bar{v}^C(s_{t+1}, k)\} | s_t, k_{At}) + \varepsilon_{Ct} \\ = & \bar{v}^C(s_t, k_{At}) - \ln \Pr(k_{Ct} = k_{At} | f_t = C, s_t) \\ & - \delta \eta_{A k_{At}}(\kappa_t) E(\ln \sum_{k \in K^c} \exp\{\bar{v}^C(s_{t+1}, k)\} | s_t, k_{At}) + \varepsilon_{Ct}, \end{aligned} \quad (31)$$

where the second equality uses that  $\Pr(k_{Ct} = k_{At} | f_t = C, s_t) = \exp\{\bar{v}^C(s_t, k_{At})\} / \sum_{k \in K^c} \exp\{\bar{v}^C(s_t, k)\}$ . If the law of motion of beliefs is the same across firms and  $\eta_{Ck}(\kappa_t) = \eta_{Ak}(\kappa_t)$  at each job  $k$ , then

<sup>55</sup> Recall that if  $\varepsilon_1$  and  $\varepsilon_2$  are independent Gumbel distributed with parameters  $(-\gamma, \mu)$ , where  $\gamma$  is Euler's constant, then  $y_{\max} = \max\{y_1 + \varepsilon_1, y_2 + \varepsilon_2\}$  is Gumbel distributed with parameters  $(-\gamma + \mu \ln(e^{\gamma/\mu} + e^{\gamma/\mu}), \mu)$ , mean  $-\gamma + \mu \ln(e^{\gamma/\mu} + e^{\gamma/\mu}) + \mu\gamma$ , and variance  $\pi^2 \mu^2 / 6$ . Thus,  $y_{\max}$  can be expressed as  $y_{\max} = \mu \ln(e^{\gamma/\mu} + e^{\gamma/\mu}) + \varepsilon^*$ , where  $\varepsilon^*$  is Gumbel distributed with parameters  $(-\gamma, \mu)$ . Here the productivity shocks  $\{\varepsilon_{\theta t}\}$  are assumed to be independent Gumbel distributed each with parameters  $(-\gamma, 1)$ .

$$\begin{aligned} \delta\eta_{Ck_t}(\kappa_t)E(\ln \sum_{k \in K^c} \exp\{\bar{v}^C(s_{t+1}, k)\} | s_t, k_{A_t}) \\ = \delta\eta_{Ak_t}(\kappa_t)E(\ln \sum_{k \in K^c} \exp\{\bar{v}^C(s_{t+1}, k)\} | s_t, k_{A_t}), \end{aligned}$$

where the left-hand side of this equality is the portion of  $\bar{v}^C(s_t, k_{A_t})$  corresponding to firm C's expected continuation match surplus value conditional on employing the worker at job  $k_{C_t} = k_{A_t}$  and the right-hand side of it is the next-to-last term in the last line of (31). Then, the claim follows, since  $w_A(s_t, \varepsilon_t)$  reduces to  $y_C(s_t, k_{A_t}) - \ln \Pr(k_{C_t} = k_{A_t} | f_t = C, s_t) + \varepsilon_{C_t}$ . QED

### A3. Proof of Lemma 1

Suppose that the standard deviation parameters  $\{\sigma_{Aik}\}$  are all distinct and assume without loss that  $\sigma_{A1k} > \sigma_{A2k} > \sigma_{A3k} > \sigma_{A4k}$  for each level  $k$ .<sup>56</sup> I show that the finite mixture distributions of the wages of managers of each skill type at firm A, conditional on their histories at the firm stated in the lemma, are identified at each tenure and level.

*First tenure year at A.*—The argument for managers in their first year at A is in the main text.

*Second tenure year at A.*—Consider the density of wages of managers with initial human capital  $h_1$  at level  $L_{A2}$  in their second year at A,

$$f(w_{A2} | L_{A1} = 1, L_{A2}, h_1) = \sum_{R_{A1}} \sum_i \Pr(R_{A1}, i | L_{A1} = 1, L_{A2}, h_1) f(w_{A2} | L_{A1} = 1, R_{A1}, L_{A2}, h_1, i), \quad (32)$$

which is a finite mixture of logistic densities. On the basis of the results in Holzmann, Munk, and Stratmann (2004, theorem 2.1), referred to as the *logistic mixture results*, the weights  $\{\Pr(R_{A1}, i | L_{A1} = 1, L_{A2}, h_1)\}$  and component densities  $\{f(w_{A2} | L_{A1} = 1, R_{A1}, L_{A2}, h_1, i)\}$  are identified up to their labeling by  $R_{A1}$  and  $i$  (see the online appendix for details). For later, it is useful to express each such weight as

$$\Pr(R_{A1}, i | L_{A1} = 1, L_{A2}, h_1) = \frac{\Pr(L_{A2} | L_{A1} = 1, R_{A1}, h_1, i) \Pr(R_{A1} | L_{A1} = 1, h_1, i) \Pr(i | L_{A1} = 1)}{\Pr(L_{A2} | L_{A1} = 1, h_1)}, \quad (33)$$

using that  $q_i = \Pr(i | L_{A1} = 1, h_1) = \Pr(i | L_{A1} = 1)$ . Since the standard deviations of the component densities satisfy  $\sigma_{A1k} > \sigma_{A2k} > \sigma_{A3k} > \sigma_{A4k}$  for each  $k$ , it is immediate to determine the value of  $i$  of each component density and weight. Moreover, summing the weights  $\{\Pr(R_{A1}, i | L_{A1} = 1, L_{A2}, h_1)\}$  given  $L_{A2}$ ,  $h_1$ , and  $i$  yields the probabilities  $\{\Pr(i | L_{A1} = 1, L_{A2}, h_1)\}$ , which are then identified.

By a similar logic, the density of wages of managers with initial human capital  $h_1$  at level  $L_{A2}$  in their second year at A and with recorded performance  $R_{A1}^o$  in their first year,

$$\begin{aligned} f(w_{A2} | L_{A1} = 1, R_{A1}^o, L_{A2}, h_1) &= \sum_{R_{A1}} \sum_i \Pr(R_{A1}, i | L_{A1} = 1, R_{A1}^o, L_{A2}, h_1) \\ &\cdot f(w_{A2} | L_{A1} = 1, R_{A1}^o, R_{A1}, L_{A2}, h_1, i), \end{aligned} \quad (34)$$

<sup>56</sup> If the parameters  $\{\sigma_{Aik}\}$  are not all distinct for some level  $k$ , mixture components can be labeled by  $i$  by ordering their weights or parameters according to  $i$  (see n. 43).



is a finite logistic mixture, whose weights  $\{\Pr(R_{A1}, i | L_{A1} = 1, R_{A1}^o, L_{A2}, h_1)\}$  and component densities  $\{f(w_{A2} | L_{A1} = 1, R_{A1}^o, R_{A1}, L_{A2}, h_1, i)\}$  are identified up to their labeling by  $R_{A1}$  and  $i$  by the logistic mixture results. Note that since these densities are independent of  $R_{A1}^o$ , the mixtures  $f(w_{A2} | L_{A1} = 1, L_{A2}, h_1)$  and  $f(w_{A2} | L_{A1} = 1, R_{A1}^o, L_{A2}, h_1)$  have identical component densities. Each weight  $\Pr(R_{A1}, i | L_{A1} = 1, R_{A1}^o, L_{A2}, h_1)$  can be expressed as

$$\Pr(R_{A1}, i | \cdot) = \frac{\Pr(L_{A2} | L_{A1} = 1, R_{A1}^o, R_{A1}, h_1, i) \Pr(R_{A1}^o, R_{A1} | L_{A1} = 1, h_1, i) \Pr(i | L_{A1} = 1)}{\Pr(R_{A1}^o, L_{A2} | L_{A1} = 1, h_1)}, \quad (35)$$

with  $\Pr(L_{A2} | L_{A1} = 1, R_{A1}^o, R_{A1}, h_1, i)$  independent of  $R_{A1}^o$ . Comparing the standard deviations of the component densities of the mixture makes it possible to label them and mixture weights by  $i$ .

To label the weights and component densities of the mixtures  $f(w_{A2} | L_{A1} = 1, L_{A2}, h_1)$  and  $f(w_{A2} | L_{A1} = 1, R_{A1}^o, L_{A2}, h_1)$  by  $R_{A1}$ , note first that  $\Pr(R_{A1}^o | L_{A1} = 1, R_{A1}, h_1, i)$ , which reduces to  $\Pr(R_{A1}^o | L_{A1} = 1, R_{A1}, 1)$  since it is independent of  $h_1$  and  $i$  by (11), is identified from the ratio of the identified expression on the right-hand side of (35) multiplied by the known probability  $\Pr(R_{A1}^o, L_{A2} | L_{A1} = 1, h_1)$  to the identified expression on the right-hand side of (33) multiplied by the known probability  $\Pr(L_{A2} | L_{A1} = 1, h_1)$ . Observe that the mixture weights in (33) and (35) can be correctly paired not only for each  $i$  but also for each  $R_{A1}$ , given  $R_{A1}^o$ ,  $L_{A2}$ , and  $h_1$  through their associated component densities, because the mixtures  $f(w_{A2} | L_{A1} = 1, L_{A2}, h_1)$  and  $f(w_{A2} | L_{A1} = 1, R_{A1}^o, L_{A2}, h_1)$  have identical component densities. Then, the value of  $R_{A1}$  of  $\Pr(R_{A1}^o | L_{A1} = 1, R_{A1}, 1)$  can be easily determined, since for any tenure  $t$  and level  $k$ ,

$$\Pr(R_{At}^o = 1 | L_{At} = k, R_{At} = 1, t) > \Pr(R_{At}^o = 1 | L_{At} = k, R_{At} = 0, t) \quad (36)$$

as  $\rho_1 > 0$  in (11), so  $\Pr(R_{A1}^o = 1 | L_{A1} = 1, R_{A1} = 1, 1) > \Pr(R_{A1}^o = 1 | L_{A1} = 1, R_{A1} = 0, 1)$  and the opposite inequality holds for  $R_{A1}^o = 0$ . Thus, the weights  $\{\Pr(R_{A1}, i | L_{A1} = 1, L_{A2}, h_1)\}$  and  $\{\Pr(R_{A1}, i | L_{A1} = 1, R_{A1}^o, L_{A2}, h_1)\}$  and their associated component densities can be labeled by  $R_{A1}$ .

*Third tenure year at A.*—Two cases arise, which correspond to the case in which, given an initial state  $s_{A1}$ , no two different histories of level assignments and performance lead to the same state  $s_{A3}$  at the beginning of  $t = 3$  and the case in which  $L_{A2} = 1$  so that the performance histories  $(R_{A1} = 1, R_{A2} = 0)$  and  $(R_{A1} = 0, R_{A2} = 1)$  lead to the same state  $s_{A3}$  at the beginning of  $t = 3$ . Since the former case is just a simpler version of the latter, I first consider in detail the latter case in which  $L_{A2} = 1$ . Note that the density of wages of managers with initial human capital  $h_1$  at level  $L_{A3}$  in their third year at A and at level  $L_{A2} = 1$  in their second year is a finite logistic mixture,

$$\begin{aligned} f(w_{A3} | L_{A1} = 1, L_{A2} = 1, L_{A3}, h_1) &= \sum_i \Pr(R_{A1} = 1, R_{A2} = 1, i | \cdot) f(w_{A3} | R_{A1} = 1, R_{A2} = 1, i, \cdot) \\ &+ [\Pr(R_{A1} = 1, R_{A2} = 0, i | \cdot) + \Pr(R_{A1} = 0, R_{A2} = 1, i | \cdot)] f(w_{A3} | R_{A1} = 1, R_{A2} = 0, i, \cdot) \\ &+ \Pr(R_{A1} = 0, R_{A2} = 0, i | \cdot) f(w_{A3} | R_{A1} = 0, R_{A2} = 0, i, \cdot), \end{aligned} \quad (37)$$

where  $f(w_{A3}|R_{A1} = 1, R_{A2} = 0, i, \cdot) = f(w_{A3}|R_{A1} = 0, R_{A2} = 1, i, \cdot)$ , since managers with the same initial human capital  $h_1$  of any skill type  $i$  at level  $L_{A2} = 1$  in their second year at A with a performance history of  $(R_{A1} = 1, R_{A2} = 0)$  and with a performance history of  $(R_{A1} = 0, R_{A2} = 1)$  reach the same state at the beginning of  $t = 3$ . The mixture weights,  $\Pr(R_{A1} = 1, R_{A2} = 1, i|\cdot)$ ,  $\Pr(R_{A1} = 1, R_{A2} = 0, i|\cdot) + \Pr(R_{A1} = 0, R_{A2} = 1, i|\cdot)$ , and  $\Pr(R_{A1} = 0, R_{A2} = 0, i|\cdot)$ , and component densities,  $f(w_{A3}|R_{A1} = 1, R_{A2} = 1, i, \cdot)$ ,  $f(w_{A3}|R_{A1} = 1, R_{A2} = 0, i, \cdot)$ , and  $f(w_{A3}|R_{A1} = 0, R_{A2} = 0, i, \cdot)$ , are identified up to their labeling by  $R_{A1}$ ,  $R_{A2}$ , and  $i$  by the logistic mixture results. The weights  $\Pr(R_{A1} = 1, R_{A2} = 1, i|\cdot)$  and  $\Pr(R_{A1} = 0, R_{A2} = 0, i|\cdot)$  can be expressed as

$$\begin{aligned} & \Pr(L_{A3}|L_{A1} = 1, R_{A1}, L_{A2} = 1, R_{A2}, h_1, i) \Pr(R_{A2}|L_{A1} = 1, R_{A1}, L_{A2} = 1, h_1, i) \\ \Pr(R_{A1}, R_{A2}, i|\cdot) &= \frac{\cdot \Pr(L_{A2} = 1|L_{A1} = 1, R_{A1}, h_1, i) \Pr(R_{A1}|L_{A1} = 1, h_1, i) \Pr(i|L_{A1} = 1)}{\Pr(L_{A2} = 1, L_{A3}|L_{A1} = 1, h_1)} \\ & \Pr(L_{A3}|L_{A1} = 1, R_{A1}, L_{A2} = 1, R_{A2}, h_1, i) \Pr(R_{A1}, R_{A2}|L_{A1} = 1, L_{A2} = 1, h_1, i) \\ &= \frac{\cdot \Pr(L_{A1} = 1, L_{A2} = 1, h_1, i)}{\Pr(L_{A1} = 1, L_{A2} = 1, L_{A3}, h_1)}. \end{aligned} \quad (38)$$

The weight  $\Pr(R_{A1} = 1, R_{A2} = 0, i|\cdot) + \Pr(R_{A1} = 0, R_{A2} = 1, i|\cdot)$  can be analogously expanded as

$$\begin{aligned} & \Pr(L_{A3}|L_{A1} = 1, R_{A1} = 1, L_{A2} = 1, R_{A2} = 0, h_1, i) \Pr(R_{A1} = 1, R_{A2} = 0|L_{A1} = 1, L_{A2} = 1, h_1, i) \\ & \cdot \Pr(L_{A1} = 1, L_{A2} = 1, h_1, i) + \Pr(L_{A3}|L_{A1} = 1, R_{A1} = 0, L_{A2} = 1, R_{A2} = 1, h_1, i) \\ & \cdot \Pr(R_{A1} = 0, R_{A2} = 1|L_{A1} = 1, L_{A2} = 1, h_1, i) \Pr(L_{A1} = 1, L_{A2} = 1, h_1, i) \\ & \frac{\cdot \Pr(R_{A1} = 0, R_{A2} = 1|L_{A1} = 1, L_{A2} = 1, h_1, i) \Pr(L_{A1} = 1, L_{A2} = 1, h_1, i)}{\Pr(L_{A1} = 1, L_{A2} = 1, L_{A3}, h_1)}, \end{aligned} \quad (39)$$

which simplifies to

$$\begin{aligned} & 2 \Pr(L_{A3}|L_{A1} = 1, R_{A1} = 1, L_{A2} = 1, R_{A2} = 0, h_1, i) \Pr(R_{A1} = 1, R_{A2} = 0|L_{A1} = 1, L_{A2} = 1, h_1, i) \\ & \frac{\cdot \Pr(L_{A1} = 1, L_{A2} = 1, h_1, i)}{\Pr(L_{A1} = 1, L_{A2} = 1, L_{A3}, h_1)}, \end{aligned} \quad (40)$$

since  $\Pr(L_{A3}|L_{A1} = 1, R_{A1} = 1, L_{A2} = 1, R_{A2} = 0, h_1, i) = \Pr(L_{A3}|L_{A1} = 1, R_{A1} = 0, L_{A2} = 1, R_{A2} = 1, h_1, i)$  and  $\Pr(R_{A1} = 1, R_{A2} = 0|L_{A1} = 1, L_{A2} = 1, h_1, i) = \Pr(R_{A1} = 0, R_{A2} = 1|L_{A1} = 1, L_{A2} = 1, h_1, i)$ . It then follows that the two products of probabilities in the numerator of (39) are identified, and so are the mixture weights  $\Pr(R_{A1} = 1, R_{A2} = 0, i|L_{A1} = 1, L_{A2} = 1, L_{A3}, h_1)$  and  $\Pr(R_{A1} = 0, R_{A2} = 1, i|L_{A1} = 1, L_{A2} = 1, L_{A3}, h_1)$ , since the probability  $\Pr(L_{A1} = 1, L_{A2} = 1, L_{A3}, h_1)$  is known from the data. As was the case earlier, it is possible to determine the value of  $i$  of each weight and component density of the mixture. Also, summing the mixture weights given  $L_{A3}$ ,  $h_1$ , and  $i$  yields the probabilities  $\{\Pr(i|L_{A1} = 1, L_{A2} = 1, L_{A3}, h_1)\}$ , which are therefore identified.

Similarly, the density of wages of managers with initial human capital  $h_1$  at level  $L_{A3}$  in their third year at A and at level  $L_{A2} = 1$  in their second year with recorded performance  $R_{A2}^o$ ,

$$\begin{aligned}
f(w_{A3}|L_{A1} = 1, L_{A2} = 1, R_{A2}^o, L_{A3}, h_1) &= \sum_i \Pr(R_{A1} = 1, R_{A2} = 1, i|\cdot) f(w_{A3}|R_{A1} = 1, R_{A2} = 1, i, \cdot) \\
&+ [\Pr(R_{A1} = 1, R_{A2} = 0, i|\cdot) + \Pr(R_{A1} = 0, R_{A2} = 1, i|\cdot)] f(w_{A3}|R_{A1} = 1, R_{A2} = 0, i, \cdot) \\
&+ \Pr(R_{A1} = 0, R_{A2} = 0, i|\cdot) f(w_{A3}|R_{A1} = 0, R_{A2} = 0, i, \cdot),
\end{aligned}$$

is a finite logistic mixture, with  $f(w_{A3}|R_{A1} = 1, R_{A2} = 0, i, \cdot) = f(w_{A3}|R_{A1} = 0, R_{A2} = 1, i, \cdot)$ . Its weights,  $\Pr(R_{A1} = 1, R_{A2} = 1, i|\cdot)$ ,  $\Pr(R_{A1} = 1, R_{A2} = 0, i|\cdot) + \Pr(R_{A1} = 0, R_{A2} = 1, i|\cdot)$ , and  $\Pr(R_{A1} = 0, R_{A2} = 0, i|\cdot)$ , and component densities,  $f(w_{A3}|R_{A1} = 1, R_{A2} = 1, i, \cdot)$ ,  $f(w_{A3}|R_{A1} = 1, R_{A2} = 0, i, \cdot)$ , and  $f(w_{A3}|R_{A1} = 0, R_{A2} = 0, i, \cdot)$ , are identified up to their labeling by  $R_{A1}$ ,  $R_{A2}$ , and  $i$  by the logistic mixture results. Since these densities are independent of  $R_{A2}^o$ , the mixtures  $f(w_{A3}|L_{A1} = 1, L_{A2} = 1, L_{A3}, h_1)$  and  $f(w_{A3}|L_{A1} = 1, L_{A2} = 1, R_{A2}^o, L_{A3}, h_1)$  have identical component densities. Observe that the weights  $\Pr(R_{A1} = 1, R_{A2} = 1, i|\cdot)$  and  $\Pr(R_{A1} = 0, R_{A2} = 0, i|\cdot)$  can be expanded as

$$\begin{aligned}
&\Pr(L_{A3}|L_{A1} = 1, R_{A1}, L_{A2} = 1, R_{A2}^o, R_{A2}, h_1, i) \\
\Pr(R_{A1}, R_{A2}, i|\cdot) &= \frac{\Pr(R_{A1}, R_{A2}^o, R_{A2}|L_{A1} = 1, L_{A2} = 1, h_1, i) \Pr(L_{A1} = 1, L_{A2} = 1, h_1, i)}{\Pr(L_{A1} = 1, L_{A2} = 1, R_{A2}^o, L_{A3}, h_1)}, \quad (41)
\end{aligned}$$

with  $\Pr(L_{A3}|L_{A1} = 1, R_{A1}, L_{A2} = 1, R_{A2}^o, R_{A2}, h_1, i)$  independent of  $R_{A2}^o$ . The weight  $\Pr(R_{A1} = 1, R_{A2} = 0, i|\cdot) + \Pr(R_{A1} = 0, R_{A2} = 1, i|\cdot)$  can be expressed as

$$\begin{aligned}
&\left[ \frac{\Pr(R_{A2}^o|L_{A2} = 1, R_{A2} = 0, 2) + \Pr(R_{A2}^o|L_{A2} = 1, R_{A2} = 1, 2)}{\Pr(R_{A2}^o|L_{A1} = 1, L_{A2} = 1, L_{A3}, h_1)} \right] \\
&\cdot \Pr(R_{A1} = 1, R_{A2} = 0, i|L_{A1} = 1, L_{A2} = 1, L_{A3}, h_1), \quad (42)
\end{aligned}$$

by using (11) to simplify  $\Pr(R_{A2}^o|L_{A1} = 1, R_{A1}, L_{A2} = 1, R_{A2}, h_1, i)$  with  $(R_{A1}, R_{A2}) = (1, 0)$  or  $(R_{A1}, R_{A2}) = (0, 1)$ , exploiting that  $\Pr(R_{A1} = 1, R_{A2} = 0, i|L_{A1} = 1, L_{A2} = 1, L_{A3}, h_1) = \Pr(R_{A1} = 0, R_{A2} = 1, i|L_{A1} = 1, L_{A2} = 1, L_{A3}, h_1)$  by the above observations, and collecting terms. The value of  $i$  of each weight and component density of the mixture can be determined in the way described earlier.

The weights and component densities of the mixtures  $f(w_{A3}|L_{A1} = 1, L_{A2} = 1, L_{A3}, h_1)$  and  $f(w_{A3}|L_{A1} = 1, L_{A2} = 1, R_{A2}^o, L_{A3}, h_1)$  can be labeled by  $R_{A1}$  and  $R_{A2}$  as follows. Note that the ratio of the identified expression on the right-hand side of (41) multiplied by the known probability  $\Pr(L_{A1} = 1, L_{A2} = 1, R_{A2}^o, L_{A3}, h_1)$  to the identified expression after the second equality in (38) multiplied by the known probability  $\Pr(L_{A1} = 1, L_{A2} = 1, L_{A3}, h_1)$  identifies the probability  $\Pr(R_{A2}^o|L_{A1} = 1, R_{A1}, L_{A2} = 1, R_{A2}, h_1, i)$ , which simplifies to  $\Pr(R_{A2}^o|L_{A2} = 1, R_{A2}, 2)$  by (11), with  $R_{A1} = R_{A2} = 1$  or  $R_{A1} = R_{A2} = 0$ . Any such ratio can be correctly calculated because the mixtures  $f(w_{A3}|L_{A1} = 1, L_{A2} = 1, L_{A3}, h_1)$  and  $f(w_{A3}|L_{A1} = 1, L_{A2} = 1, R_{A2}^o, L_{A3}, h_1)$  are characterized by the same component densities and so can be properly matched through them, component by component.

This argument applies to the two components of the mixtures  $f(w_{A3}|L_{A1} = 1, L_{A2} = 1, L_{A3}, h_1)$  and  $f(w_{A3}|L_{A1} = 1, L_{A2} = 1, R_{A2}^o, L_{A3}, h_1)$  arising from the performance histories  $(R_{A1}, R_{A2}) = (1, 1)$  and  $(R_{A1}, R_{A2}) = (0, 0)$ . As for the component of the two mixtures arising from the performance histories  $(R_{A1}, R_{A2}) = (1, 0)$  and  $(R_{A1}, R_{A2}) = (0, 1)$ , observe that the ratio of the probability in (42) multiplied by the known probability  $\Pr(R_{A2}^o|L_{A1} = 1, L_{A2} = 1, L_{A3}, h_1)$  to the probability  $\Pr(R_{A1} = 1, R_{A2} = 0, i|L_{A1} = 1, L_{A2} = 1, L_{A3}, h_1)$ , which is recovered as argued,

identifies the sum of probabilities  $\Pr(R_{A2}^o | L_{A2} = 1, R_{A2} = 0, 2) + \Pr(R_{A2}^o | L_{A2} = 1, R_{A2} = 1, 2)$ .

Let now  $R_{A2}^o = 1$ . Once the three ratios of probabilities described have been calculated for the three components of the mixtures  $f(w_{A3} | L_{A1} = 1, L_{A2} = 1, L_{A3}, h_1)$  and  $f(w_{A3} | L_{A1} = 1, L_{A2} = 1, R_{A2}^o = 1, L_{A3}, h_1)$ , the second-lowest and lowest such ratios identify  $\Pr(R_{A2}^o = 1 | L_{A2} = 1, R_{A2} = 1, 2)$  and  $\Pr(R_{A2}^o = 1 | L_{A2} = 1, R_{A2} = 0, 2)$ , respectively, and so the values of  $(R_{A1}, R_{A2})$  of  $(1, 1)$  and  $(0, 0)$  of the corresponding mixture weights and component densities. The remaining ratio identifies  $\Pr(R_{A2}^o = 1 | L_{A2} = 1, R_{A2} = 0, 2) + \Pr(R_{A2}^o = 1 | L_{A2} = 1, R_{A2} = 1, 2)$  and thus the values of  $(R_{A1}, R_{A2})$  of  $(1, 0)$  or  $(0, 1)$  of the corresponding mixture weight and component density. A similar argument applies when  $R_{A2}^o = 0$ .

Consider next the case in which no two different histories of level assignments and performance lead to the same state  $s_{\beta}$  at the beginning of  $t = 3$ , given a common initial state  $s_{\beta 1}$ . This case is analogous to the one just considered, except that all possible combinations of managers' first-year performance, second-year level assignments, second-year performance, initial human capital, and skill types lead to distinct mixture components, as in the case of the second tenure year at A. The labeling of the components of these mixtures by  $R_{A1}$  and  $R_{A2}$  proceeds as detailed in the main text.

*Remaining tenure years at A.*—The logic applied so far extends to any tenures.<sup>57</sup>  
QED

#### A4. Proof of Proposition 3

The proof consists of two parts. In part 1, I show that the parameters of the distribution of classification error in performance ratings are identified from the weights of the finite mixture distributions of wages of managers with any level of initial human capital between  $t = 2$  and  $t = 4$ , conditional on their histories of level assignments and performance ratings at A, which are identified by lemma 1. In part 2, I show that once this error distribution is identified, the probabilities of high performance by manager ability at each level are identified from repeated observations on the distribution of managers' performance ratings at the corresponding level.

##### A4.1. Part 1: Identification of the Parameters of the Distribution of Classification Error

By lemma 1,  $\{\Pr(R_{A1}^o | L_{A1} = 1, R_{A1}, 1)\}$  and  $\{\Pr(R_{A2}^o | L_{A2} = 1, R_{A2}, 2)\}$  are identified, so the error rates  $E_0(1, t)$  and  $E_1(1, t)$ ,  $t = 1, 2$ , are identified by (11). The parameters  $(\rho_0, \rho_1, \rho_2(1))$  of the distribution of classification error at level 1 can then be recovered from  $E_0(1, 1)$ ,  $E_1(1, 1)$ , and either  $E_0(1, 2)$  or  $E_1(1, 2)$  by (11). As for the parameter  $\rho_2(2)$  for level 2,  $E_0(2, 2)$  and  $E_1(2, 2)$  are identified from  $\{\Pr(R_{A2}^o | L_{A2} = 2, R_{A2}, 2)\}$ , which implies that  $\rho_2(2)$  is identified from either  $E_0(2, 2)$  or  $E_1(2, 2)$ , once  $\rho_0$  and  $\rho_1$  are recovered. By extending this argument in the natural

<sup>57</sup> Mixture components can be labeled by performance by ordering the means of the component densities according to  $p_{\mu}$ . Note, e.g., that based on the recursive characterization of these means in (16), their monotonicity in  $p_{\mu}$  follows by usual contraction mapping arguments under standard regularity conditions.

way, it is easy to show that the parameter  $\rho_2(3)$  for level 3 is identified from the distribution of wages at any level of A in  $t = 4$  of managers assigned to level 3 in  $t = 3$  with a high or low performance rating in  $t = 3$ .

#### A4.2. Part 2: Identification of the Parameters of the Distribution of Performance

A summary of the argument is as follows. Consider managers with initial human capital  $h_1$  at level 1 of A (at least) up to period  $t = 3$ . Their probabilities of high performance ratings  $\Pr(R_{A3}^o = 1|L_{A1} = 1, L_{A2} = 1, L_{A3} = 1, h_1)$ ,  $\Pr(R_{A2}^o = 1, R_{A3}^o = 1|L_{A1} = 1, L_{A2} = 1, L_{A3} = 1, h_1)$ , and  $\Pr(R_{A1}^o = 1, R_{A2}^o = 1, R_{A3}^o = 1|L_{A1} = 1, L_{A2} = 1, L_{A3} = 1, h_1)$  identify the corresponding probabilities of high performance  $\Pr(R_{A3} = 1|L_{A1} = 1, L_{A2} = 1, L_{A3} = 1, h_1)$ ,  $\Pr(R_{A2} = 1, R_{A3} = 1|L_{A1} = 1, L_{A2} = 1, L_{A3} = 1, h_1)$ , and  $\Pr(R_{A1} = 1, R_{A2} = 1, R_{A3} = 1|L_{A1} = 1, L_{A2} = 1, L_{A3} = 1, h_1)$ , which I denote by  $\nu_1(h_1)$ ,  $\nu_2(h_1)$ , and  $\nu_3(h_1)$ , respectively, once the distribution of classification error in performance ratings at level 1 is recovered. Similarly, the probability of any other sequence of high and low performance at level 1 is also identified. From this recovered distribution of performance, the probabilities  $(\alpha_1, \beta_1)$ ,  $\alpha_1 > \beta_1$ , are identified by known results on the identifiability of finite mixtures of binomial distributions. An analogous argument applies to  $(\alpha_2, \beta_2)$  and  $(\alpha_3, \beta_3)$ . The formal argument proceeds in five steps. The first four steps focus on managers with initial human capital  $h_1$  at level 1 of A (at least) up to  $t = 3$ . The last step focuses first on managers with initial human capital  $h_1$  promoted to level 2 in  $t = 2$  and retained at this level for at least two more years and then on managers with initial human capital  $h_1$  promoted to level 3 in  $t = 3$  and retained at this level for at least two more years.

*Step 1:  $\nu_1(h_1)$  is identified.*—Let  $q_{i3}(h_1) = \Pr(i|L_{A1} = 1, L_{A2} = 1, L_{A3} = 1, h_1)$  and note that

$$\begin{aligned} & \Pr(R_{A3}^o = 1|L_{A1} = 1, L_{A2} = 1, L_{A3} = 1, h_1) \\ &= \sum_{R_{A3}} \sum_i \Pr(R_{A3}^o = 1, R_{A3}, i|L_{A1} = 1, L_{A2} = 1, L_{A3} = 1, h_1) \\ &= \sum_i q_{i3}(h_1) \sum_{R_{A3}} \Pr(R_{A3}^o = 1|L_{A1} = 1, L_{A2} = 1, L_{A3} = 1, R_{A3}) \\ & \quad \cdot \Pr(R_{A3}|L_{A1} = 1, L_{A2} = 1, L_{A3} = 1, h_1, i), \end{aligned}$$

where  $\Pr(R_{A1}^o|L_{A1}, \dots, L_{A4}, R_{A4}, h_1, i)$  is independent of  $h_1$  and  $i$  by (11). Observe that

$$\begin{aligned} & \Pr(R_{A1}, R_{A2}, R_{A3}|L_{A1} = 1, L_{A2} = 1, L_{A3} = 1, h_1, \theta, i) = \pi_{A1}(\theta)^{R_{A1}} \pi_{A1}(\theta)^{R_{A2}} \pi_{A1}(\theta)^{R_{A3}} \\ & \quad \cdot [1 - \pi_{A1}(\theta)]^{1-R_{A1}} [1 - \pi_{A1}(\theta)]^{1-R_{A2}} [1 - \pi_{A1}(\theta)]^{1-R_{A3}}, \end{aligned}$$

where  $\pi_{A1}(\bar{\theta}) = \alpha_1$  and  $\pi_{A1}(\underline{\theta}) = \beta_1$ , which will be useful in what follows. Hence,

$$\begin{aligned} & \Pr(R_{A3}|L_{A1} = 1, L_{A2} = 1, L_{A3} = 1, h_1, i) \\ &= \sum_{R_{A1}, R_{A2}} \Pr(R_{A1}, R_{A2}, R_{A3}|\cdot, h_1, \bar{\theta}, i) p_{i3}(h_1) \\ & \quad + \sum_{R_{A1}, R_{A2}} \Pr(R_{A1}, R_{A2}, R_{A3}|\cdot, h_1, \underline{\theta}, i) [1 - p_{i3}(h_1)] \\ &= \alpha_1^{R_{A3}} (1 - \alpha_1)^{1-R_{A3}} p_{i3}(h_1) + \beta_1^{R_{A3}} (1 - \beta_1)^{1-R_{A3}} [1 - p_{i3}(h_1)], \end{aligned}$$

where  $p_{i3}(h_1)$  is defined as  $\Pr(\bar{\theta}|L_{A1} = 1, L_{A2} = 1, L_{A3} = 1, h_1, i)$ . From (11), the expression for  $\Pr(R_{A3}|L_{A1} = 1, L_{A2} = 1, L_{A3} = 1, h_1, i)$ , and the property that  $\sum_i q_{i3}(h_1) = 1$ , it follows that

$$\begin{aligned} \Pr(R_{A3}^o = 1|L_{A1} = 1, L_{A2} = 1, L_{A3} = 1, h_1) \\ = [1 - E_1(1, 3)][(\alpha_1 - \beta_1)\sum_i q_{i3}(h_1)p_{i3}(h_1) + \beta_1] \\ + E_0(1, 3)[1 - (\alpha_1 - \beta_1)\sum_i q_{i3}(h_1)p_{i3}(h_1) - \beta_1]. \end{aligned} \quad (43)$$

Since the left-hand side of (43) is known for each  $h_1$  and  $E_0(1, 3)$  and  $E_1(1, 3)$  are identified by part 1,  $\nu_1(h_1) = (\alpha_1 - \beta_1)\sum_i q_{i3}(h_1)p_{i3}(h_1) + \beta_1$  is identified for each  $h_1$  by (43).

*Step 2:  $\nu_2(h_1)$  is identified.*—Analogously to step 1, observe that

$$\begin{aligned} \Pr(R_{A2}^o = 1, R_{A3}^o = 1|L_{A1} = 1, L_{A2} = 1, L_{A3} = 1, h_1) \\ = \sum_{R_{A2}} \sum_{R_{A3}} \sum_i \Pr(R_{A2}^o = 1, R_{A2}, R_{A3}^o = 1, R_{A3}, i|L_{A1} = 1, L_{A2} = 1, L_{A3} = 1, h_1) = \sum_i q_{i3}(h_1) \\ \cdot \sum_{R_{A2}} \sum_{R_{A3}} \Pr(R_{A2}^o = 1|L_{A1} = 1, L_{A2} = 1, R_{A2}) \Pr(R_{A3}^o = 1|L_{A1} = 1, L_{A2} = 1, L_{A3} = 1, R_{A3}) \\ \cdot \Pr(R_{A2}, R_{A3}|L_{A1} = 1, L_{A2} = 1, L_{A3} = 1, h_1, i), \end{aligned} \quad (44)$$

where the second equality uses that by (11),  $\Pr(R_{A2}^o = 1|L_{A1} = 1, L_{A2} = 1, R_{A2}, L_{A3} = 1, R_{A3}, h_1, i)$  is independent of  $L_{A3}, R_{A3}, h_1$ , and  $i$  and that  $\Pr(R_{A3}^o = 1|L_{A1} = 1, L_{A2} = 1, R_{A2}^o = 1, R_{A2}, L_{A3} = 1, R_{A3}, h_1, i)$  is independent of  $R_{A2}^o, R_{A2}, h_1$ , and  $i$ . Define  $B_2 = [1 - E_1(1, 2)][1 - E_1(1, 3)]$ ,  $C_2 = [1 - E_1(1, 2)]E_0(1, 3) + E_0(1, 2)[1 - E_1(1, 3)]$ , and  $D_2 = E_0(1, 2)E_0(1, 3)$ , which are known functions of the identified classification error rates at level 1 in  $t = 2, 3$ . It is easy to show that

$$\begin{aligned} \Pr(R_{A2}^o = 1, R_{A3}^o = 1|L_{A1} = 1, L_{A2} = 1, L_{A3} = 1, h_1) = B_2[(\alpha_1^2 - \beta_1^2)\sum_i q_{i3}(h_1)p_{i3}(h_1) + \beta_1^2] \\ + C_2[(\alpha_1 - \beta_1)\sum_i q_{i3}(h_1)p_{i3}(h_1) + \beta_1 - (\alpha_1^2 - \beta_1^2)\sum_i q_{i3}(h_1)p_{i3}(h_1) - \beta_1^2] \\ + D_2[1 - 2(\alpha_1 - \beta_1)\sum_i q_{i3}(h_1)p_{i3}(h_1) - 2\beta_1 + (\alpha_1^2 - \beta_1^2)\sum_i q_{i3}(h_1)p_{i3}(h_1) + \beta_1^2]. \end{aligned} \quad (45)$$

Since the left-hand side of (45) is known for each  $h_1$  and  $\nu_1(h_1)$  is identified for each  $h_1$  by step 1,  $\nu_2(h_1) = (\alpha_1^2 - \beta_1^2)\sum_i q_{i3}(h_1)p_{i3}(h_1) + \beta_1^2$  is also identified for each  $h_1$  by (45).

*Step 3:  $\nu_3(h_1)$  is identified.*—By proceeding as in the earlier steps, note that

$$\begin{aligned} \Pr(R_{A1}^o = 1, R_{A2}^o = 1, R_{A3}^o = 1|L_{A1} = 1, L_{A2} = 1, L_{A3} = 1, h_1) \\ = \sum_{R_{A1}} \sum_{R_{A2}} \sum_{R_{A3}} \sum_i \Pr(R_{A1}^o = 1, R_{A1}, R_{A2}^o = 1, R_{A2}, R_{A3}^o = 1, R_{A3}, i|L_{A1} = 1, L_{A2} = 1, L_{A3} = 1, h_1) \\ = \sum_i q_{i3}(h_1) \sum_{R_{A1}} \sum_{R_{A2}} \sum_{R_{A3}} \Pr(R_{A1}^o = 1|L_{A1} = 1, R_{A1}) \Pr(R_{A2}^o = 1|L_{A1} = 1, L_{A2} = 1, R_{A2}) \\ \cdot \Pr(R_{A3}^o = 1|L_{A1} = 1, L_{A2} = 1, L_{A3} = 1, R_{A3}) \Pr(R_{A1}, R_{A2}, R_{A3}|L_{A1} = 1, L_{A2} = 1, L_{A3} = 1, h_1, i), \end{aligned}$$

where the second equality factors the joint probability  $\Pr(R_{A1}^o = 1, R_{A1}, R_{A2}^o = 1, R_{A2}, R_{A3}^o = 1, R_{A3}, i|L_{A1} = 1, L_{A2} = 1, L_{A3} = 1, h_1)$  into a product of conditional probabilities that satisfy conditional independence assumptions analogous to those invoked in steps 1 and 2. Let  $B_3 = [1 - E_1(1, 1)][1 - E_1(1, 2)][1 - E_1(1, 3)]$ ,  $C_3 = [1 - E_1(1, 1)][1 - E_1(1, 2)]E_0(1, 3) + [1 - E_1(1, 1)]E_0(1, 2)[1 - E_1(1, 3)] + E_0(1, 1)[1 - E_1(1, 2)][1 - E_1(1, 3)]$ ,  $D_3 = [1 - E_1(1, 1)]E_0(1, 2)E_0(1, 3) + E_0(1, 1)[1 - E_1(1, 2)]E_0(1, 3) + E_0(1, 1)E_0(1, 2)[1 - E_1(1, 3)]$ , and  $E_3 = E_0(1, 1)E_0(1, 2)E_0(1, 3)$ , which are known functions of the identified classification error rates at level 1 in  $t = 1, 2, 3$ . Thus,

$$\begin{aligned}
\Pr(R_{A1}^o = 1, R_{A2}^o = 1, R_{A3}^o = 1 | \cdot) &= B_3[(\alpha_1^3 - \beta_1^3) \sum_i q_{i3}(h_1) p_{i3}(h_1) + \beta_1^3] \\
&+ C_3[(\alpha_1^2 - \beta_1^2) \sum_i q_{i3}(h_1) p_{i3}(h_1) + \beta_1^2 - (\alpha_1^3 - \beta_1^3) \sum_i q_{i3}(h_1) p_{i3}(h_1) - \beta_1^3] \\
&+ D_3[(\alpha_1 - \beta_1) \sum_i q_{i3}(h_1) p_{i3}(h_1) + \beta_1 - 2(\alpha_1^2 - \beta_1^2) \sum_i q_{i3}(h_1) p_{i3}(h_1) - 2\beta_1^2] \\
&+ (\alpha_1^3 - \beta_1^3) \sum_i q_{i3}(h_1) p_{i3}(h_1) + \beta_1^3] + E_3[1 - 3(\alpha_1 - \beta_1) \sum_i q_{i3}(h_1) p_{i3}(h_1) - 3\beta_1 \\
&+ 3(\alpha_1^2 - \beta_1^2) \sum_i q_{i3}(h_1) p_{i3}(h_1) + 3\beta_1^2 - (\alpha_1^3 - \beta_1^3) \sum_i q_{i3}(h_1) p_{i3}(h_1) - \beta_1^3].
\end{aligned} \tag{46}$$

Since the left-hand side of (46) is known for each  $h_1$  and  $v_1(h_1)$  and  $v_2(h_1)$  are identified for each  $h_1$  by steps 1 and 2,  $v_3(h_1) = (\alpha_1^3 - \beta_1^3) \sum_i q_{i3}(h_1) p_{i3}(h_1) + \beta_1^3$  is also identified for each  $h_1$  by (46).

*Step 4:  $\alpha_1, \beta_1$ , and  $\sum_i q_{i3}(h_1) p_{i3}(h_1)$  are identified.*—By steps 1–3, the distribution of performance in  $t \leq 3$  of managers with initial human capital  $h_1$  at level 1 of A (at least) up to  $t = 3$  is identified from the corresponding distribution of their performance ratings. Such a distribution of performance is a finite binomial mixture with component probabilities  $(\alpha_1, \beta_1)$ ,  $\alpha_1 > \beta_1$ , and mixing weights  $(\sum_i q_{i3}(h_1) p_{i3}(h_1), 1 - \sum_i q_{i3}(h_1) p_{i3}(h_1))$ . A necessary and sufficient condition for the identifiability of a finite mixture of binomial distributions is that the number of trials  $n$  equals at least twice the number of components  $r$  minus 1, namely,  $n \geq 2r - 1$  (Blischke 1964, 1978). Since the number of trials ( $n = 3$ ) equals twice the number of components ( $r = 2$  with parameters  $\alpha_1$  and  $\beta_1$ ) minus 1 for the binomial mixture of interest here, the mixture is identified.<sup>58</sup>

*Step 5:  $\{\alpha_k, \beta_k\}_{k \geq 1}$  are identified.*—The arguments in the previous steps can be adapted to show that  $\{\alpha_k, \beta_k\}_{k \geq 1}$  are also identified. In particular, information on the performance ratings of managers promoted to level 2 in their second year at A and assigned to level 2 for at least two more years can be used to identify  $\alpha_2$  and  $\beta_2$ . Information on the performance ratings of managers promoted to level 2 in their second year at A, then promoted to level 3 in their third year at A, and assigned to level 3 for at least two more years can be used to identify  $\alpha_3$  and  $\beta_3$ . QED

#### A5. Proof of Proposition 4

By the proof of proposition 3,  $\alpha_1$  and  $\beta_1$  are identified. By lemma 1, the probabilities  $\{\Pr(R_{A1}, i | L_{A1} = 1, L_{A2}, h_1)\}$  in (33) are also identified, since they are the weights of the identified finite mixture density of wages of managers with initial human capital  $h_1$  at level  $L_{A2}$  in their second year at A in (32) and their values of  $R_{A1}$  and  $i$  can be determined. Then, multiplying the right-hand side of (33) by the known probability  $\Pr(L_{A2} | L_{A1} = 1, h_1)$ , dividing this product by  $q_i = \Pr(i | L_{A1} = 1)$ , and summing the resulting ratio over  $L_{A2}$  given  $R_{A1}, h_1$ , and  $i$  yields  $\Pr(R_{A1} | L_{A1} = 1, h_1, i)$ , which is then identified, since  $\sum_{L_{A2}} \Pr(L_{A2} | L_{A1} = 1, R_{A1}, h_1, i) = 1$ . From, say,  $\Pr(R_{A1} = 1 | L_{A1} = 1, h_1, i) = (\alpha_1 - \beta_1) p_{i1} + \beta_1$ , it follows that  $p_{i1}$  is identified for each  $i$ . QED

<sup>58</sup> If  $\alpha_k = \beta_k$  or  $\alpha_k = 1 - \beta_k$  for some  $k$ , then two periods of observations on the distribution of managers' performance ratings at level  $k$  are sufficient to identify  $\alpha_k$ .

A6. *Proof of Corollary 1*

Note first that by the proof of lemma 1, the probabilities  $\{\Pr(L_{A1} = 1, R_{A1}, L_{A2}, R_{A2}, L_{A3}, R_{A3}, L_{A4}, h_1, i)\}$  are identified from the weights of the identified finite mixture density of wages of managers in their fifth year at A with initial human capital  $h_1$  at level  $L_{A4}$  in their fourth year at A, at level  $L_{A3}$  in their third year, and at level  $L_{A2}$  in their second year, and so are the probabilities  $\{\Pr(L_{A1} = 1, R_{A1}, L_{A2}, R_{A2}, L_{A3}, R_{A3}, h_1, i)\}$  from them. Then, the probabilities  $\{\Pr(R_{A1}, R_{A2}, R_{A3} | L_{A1} = 1, L_{A2}, L_{A3}, h_1, i)\}$  are also identified, since  $\{\Pr(i | L_{A1} = 1, L_{A2}, L_{A3}, h_1)\}$  are identified by the proof of lemma 1 and  $\{\Pr(L_{A1} = 1, L_{A2}, L_{A3}, h_1)\}$  are known from the data. In turn, the probabilities  $\{\Pr(R_{A2}, R_{A3} | L_{A1} = 1, L_{A2}, L_{A3}, h_1, i)\}$  and  $\{\Pr(R_{A3} | L_{A1} = 1, L_{A2}, L_{A3}, h_1, i)\}$  can be recovered by appropriately summing  $\{\Pr(R_{A1}, R_{A2}, R_{A3} | L_{A1} = 1, L_{A2}, L_{A3}, h_1, i)\}$  over  $R_{A1}$  and over  $R_{A1}$  and  $R_{A2}$ , respectively, given  $L_{A2}, L_{A3}, h_1$ , and  $i$ . Hence, letting  $L_{A2} = L_{A3} = 1$ , by an argument analogous to that in step 4 of the proof of proposition 3 applied to managers of skill type  $i$ , it follows that  $\alpha_1$  and  $\beta_1$  are identified. Namely, the distribution of performance of managers with any level of initial human capital  $h_1$  of skill type  $i$  at level 1 is an identifiable binomial mixture of the distribution of performance of such managers of high and low ability with component probabilities  $(\alpha_1, \beta_1)$ ,  $\alpha_1 > \beta_1$ , and mixing weights  $(p_{i3}(h_1), 1 - p_{i3}(h_1))$ , where  $p_{i3}(h_1) = \Pr(\theta | L_{A1} = 1, L_{A2} = 1, L_{A3} = 1, h_1, i)$ . Thus,  $(\alpha_1, \beta_1)$  are identified. Observe next that once  $(\alpha_1, \beta_1)$  are recovered, the probabilities  $\{\Pr(R_{A1} | L_{A1} = 1, h_1, i)\}$ , which are identified as shown in the proof of proposition 4, pin down  $\{p_{i1}\}$ , since  $\Pr(R_{A1} = 1 | L_{A1} = 1, h_1, i) = (\alpha_1 - \beta_1)p_{i1} + \beta_1$ . Finally, a similar argument applied to managers of skill type  $i$  at levels 2 and 3, which follows the logic of step 5 of proposition 3, establishes that  $\{\alpha_k, \beta_k\}_{k>1}$  are identified as well. QED

A7. *Proof of Proposition 5*

The argument consists of two parts. In part 1, I show that managers' assignment probabilities at firm A, conditional on their past assignments and performance at the firm, are identified for each skill type, tenure, and level. In part 2, I show that exogenous separation rates are identified. The rest of the proof is as in the main text. In particular, the identification of A's expected output at levels 1 and 3 proceeds analogously to the case without exogenous separations.

## A7.1. Part 1: Identification of Conditional Assignment Probabilities at Firm A by Skill Type

Managers' conditional assignment probabilities  $\Pr(L_{A2} | L_{A1} = 1, R_{A1}, h_1, i)$ ,  $\Pr(L_{A3} | L_{A1} = 1, R_{A1}, L_{A2}, R_{A2}, h_1, i)$ , and so on are identified at each tenure and level from the weights of the conditional finite mixture distributions of wages of managers of each skill type with initial human capital  $h_1$  and all possible histories of performance at the corresponding tenures and levels, which are identified by lemma 1. Formally, recall from the proof of lemma 1 that the probabilities  $\{\Pr(R_{A1}, i | L_{A1} = 1, L_{A2}, h_1)\}$  in (33), which are the weights of the identified finite mixture density of wages  $f(w_{A2} | L_{A1} = 1, L_{A2}, h_1)$  in (32), and the probabilities  $\{\Pr(i | L_{A1} = 1, \dots, L_{At}, h_1)\}$  for any  $t$  are identified. Also, the product  $\Pr(L_{A2} | L_{A1} = 1, R_{A1}, h_1, i) \Pr(R_{A1} | L_{A1} = 1, h_1, i) \Pr(i | L_{A1} = 1)$  on the right-hand



side of (33) is identified from the product of  $\Pr(R_{A1}, i | L_{A1} = 1, L_{A2}, h_1)$  and the known probability  $\Pr(L_{A2} | L_{A1} = 1, h_1)$ . As  $\Pr(R_{A1} | L_{A1} = 1, h_1, i)$  is identified by the proof of proposition 4, so is  $\Pr(L_{A2} | L_{A1} = 1, R_{A1}, h_1, i)$ .

By analogous steps, it is possible to show that  $\Pr(L_{A3} | L_{A1} = 1, R_{A1}, L_{A2}, R_{A2}, h_1, i)$  is identified from the weights of the identified finite mixture density of wages  $f(w_{A3} | L_{A1} = 1, L_{A2}, L_{A3}, h_1)$ . For instance, consider the case in which  $L_{A2} = 1$  and either  $R_{A1} = R_{A2} = 1$  or  $R_{A1} = R_{A2} = 0$ . The product of the identified expression after the first equality in (38) and the known probability  $\Pr(L_{A2} = 1, L_{A3} | L_{A1} = 1, h_1)$  identifies

$$\begin{aligned} & \Pr(L_{A3} | L_{A1} = 1, R_{A1}, L_{A2} = 1, R_{A2}, h_1, i) \Pr(R_{A2} | L_{A1} = 1, R_{A1}, L_{A2} = 1, h_1, i) \\ & \cdot \Pr(L_{A2} = 1 | L_{A1} = 1, R_{A1}, h_1, i) \Pr(R_{A1} | L_{A1} = 1, h_1, i) \Pr(i | L_{A1} = 1). \end{aligned} \quad (47)$$

Once the product  $\Pr(L_{A2} = 1 | L_{A1} = 1, R_{A1}, h_1, i) \Pr(R_{A1} | L_{A1} = 1, h_1, i) \Pr(i | L_{A1} = 1)$  is recovered as just argued, the product  $\Pr(L_{A3} | L_{A1} = 1, R_{A1}, L_{A2} = 1, R_{A2}, h_1, i)$  is also identified. This latter product—once summed over  $L_{A3}$  given  $R_{A1}, R_{A2}, h_1$ , and  $i$ —pins down  $\Pr(R_{A2} | L_{A1} = 1, R_{A1}, L_{A2} = 1, h_1, i)$ , which implies that  $\Pr(L_{A3} | L_{A1} = 1, R_{A1}, L_{A2} = 1, R_{A2}, h_1, i)$  is identified from (47). This logic extends to any other histories of level assignments and performance and to the remaining tenures.

## A7.2. Part 2: Identification of Exogenous Separation Rates

By (12),  $Ew_{Ait}(k)$  is given by

$$\begin{aligned} Ew_{Ait}(k) &= y_C(s_{it}, k) - \ln \Pr(L_{Ct} = k | f_t = C, s_{it}) \\ &= y_C(s_{it}, k) + \ln \sum_{k'} \exp\{\bar{v}^C(s_{it}, k')\} - \bar{v}^C(s_{it}, k) \\ &= \ln \sum_{k'} \exp\{\bar{v}^C(s_{it}, k')\} - \delta \eta_{it} E(\ln \sum_{k'} \exp\{\bar{v}^C(s_{it+1}, k')\} | s_{it}, k), \end{aligned} \quad (48)$$

as  $\Pr(L_{Ct} = k | f_t = C, s_{it}) = \exp\{\bar{v}^C(s_{it}, k)\} / \sum_{k'} \exp\{\bar{v}^C(s_{it}, k')\}$ . So, for any levels  $k$  and  $k''$  at A,

$$\begin{cases} \ln \sum_{k'} \exp\{\bar{v}^C(s_{it}, k')\} = Ew_{Ait}(k) + \delta \eta_{it} E(\ln \sum_{k'} \exp\{\bar{v}^C(s_{it+1}, k')\} | s_{it}, k), \\ \ln \sum_{k'} \exp\{\bar{v}^C(s_{it}, k')\} = Ew_{Ait}(k'') + \delta \eta_{it} E(\ln \sum_{k'} \exp\{\bar{v}^C(s_{it+1}, k')\} | s_{it}, k''), \end{cases} \quad (49)$$

for each  $s_{it}$  and  $t$ . Suppose that  $t = T$ . If, say,  $\eta_{2t} = \eta_{3t}$ , then subtracting the second equation in (49) from the first one, term by term, with  $k = 2$  and  $k'' = 3$ , yields that

$$\begin{aligned} \eta_{2t} &= \frac{Ew_{Ait}(2) - Ew_{Ait}(3)}{\delta E(\ln \sum_{k'} \exp\{\bar{v}^C(s_{it+1}, k')\} | s_{it}, 3) - \delta E(\ln \sum_{k'} \exp\{\bar{v}^C(s_{it+1}, k')\} | s_{it}, 2)} \\ &= \frac{Ew_{Ait}(2) - Ew_{Ait}(3)}{\delta E[Ew_{Ait+1}(2) + \delta \eta_{2t+1} E(\ln \sum_{k'} \exp\{\bar{v}^C(s_{it+2}, k')\} | s_{it+1}, 2) | s_{it}, 3] \\ &\quad - \delta E[Ew_{Ait+1}(3) + \delta \eta_{2t+1} E(\ln \sum_{k'} \exp\{\bar{v}^C(s_{it+2}, k')\} | s_{it+1}, 3) | s_{it}, 2]} \end{aligned} \quad (50)$$

where the second equality follows from  $\eta_{2t+1} = \eta_{3t+1}$  and expanding  $\ln \sum_k \exp\{\bar{v}^C(s_{it+1}, k')\}$  using (49). If the process for  $s_{it}$  is level symmetric in  $\tau \geq T + 1$ , then (50) reduces to

$$\eta_{2t} = \frac{Ew_{\Lambda it}(2) - Ew_{\Lambda it}(3)}{\delta E[Ew_{\Lambda it+1}(2)|s_{it}, 3] - \delta E[Ew_{\Lambda it+1}(3)|s_{it}, 2]}, \quad (51)$$

which implies that  $\eta_{2t}$  and  $\eta_{3t} = \eta_{2t}$  are identified—as mean wages by skill type at each state, tenure, and level are identified—and so are  $\{\eta_{2t}\}$  and  $\{\eta_{3t}\}$  from  $T + 1$  on, since  $\eta_{2t} = \eta_{2T}$  and  $\eta_{3t} = \eta_{2t}$  in  $t \geq T + 1$ . Now, if  $\{y_C(s_{it}, 2)\}$  are known in  $t \geq T + 1$ , then  $\{\Pr(L_{Ct} = 2|f_t = C, s_{it})\}$  are identified from  $\{Ew_{\Lambda it}(2)\}$  in each such period by (12). As a result,  $\bar{v}^C(s_{it}, 2)$  in  $t \geq T + 1$  is identified for each  $s_{it}$  from

$$\bar{v}^C(s_{it}, 2) = y_C(s_{it}, 2) + \delta \eta_{2t} E[\bar{v}^C(s_{it+1}, 2) - \ln \Pr(L_{Ct+1} = 2|f_{t+1} = C, s_{it+1})|s_{it}, 2],$$

once recursively expanded, and so is  $\ln \sum_k \exp\{\bar{v}^C(s_{it}, k')\}$  in  $t \geq T + 1$  for each  $s_{it}$ , since it equals  $\bar{v}^C(s_{it}, 2) - \ln \Pr(L_{Ct} = 2|f_t = C, s_{it})$ . But then  $\{\ln \sum_k \exp\{\bar{v}^C(s_{it}, k')\}\}$  are also identified in  $t = T$  by the first equation in (49) with  $k = 2$  or  $k = 3$ , once  $\eta_{2T}$  and  $\eta_{3T}$  are identified.

Observe next that the two equations in (49) with  $t = T - 1$  imply the condition  $Ew_{\Lambda T-1}(k) - Ew_{\Lambda T-1}(k') = \delta \eta_{k'T-1} E(\ln \sum_k \exp\{\bar{v}^C(s_{iT}, k')\}|s_{iT-1}, k') - \delta \eta_{kT-1} E(\ln \sum_k \exp\{\bar{v}^C(s_{iT}, k')\}|s_{iT-1}, k)$ . When evaluated at two distinct states, this condition pins down  $\eta_{kT-1}$  and  $\eta_{k'T-1}$  for any  $k$  and  $k'$ , once  $\{\ln \sum_k \exp\{\bar{v}^C(s_{iT}, k')\}\}$  are recovered. Then,  $\{\ln \sum_k \exp\{\bar{v}^C(s_{iT-1}, k')\}\}$  are identified by (49). Letting  $k = 2$  and  $k' = 3$  and repeating this argument for the remaining periods establishes that  $\{\eta_{2t}\}$ ,  $\{\eta_{3t}\}$ , and  $\{\ln \sum_k \exp\{\bar{v}^C(s_{it}, k')\}\}$  are identified in those periods too.

Finally, rearranging the first equation in (49) with  $k = 1$  yields that

$$\eta_{1t} = \frac{\ln \sum_k \exp\{\bar{v}^C(s_{it}, k')\} - Ew_{\Lambda it}(1)}{\delta E(\ln \sum_k \exp\{\bar{v}^C(s_{it+1}, k')\}|s_{it}, 1)}, \quad (52)$$

so that  $\{\eta_{1t}\}$  are identified as well, since they are functions of known objects.<sup>59</sup>  
QED

#### A8. Proof of Proposition 6

To start, note that the third equality in (48) derived from (12) can be expressed as  $\ln \sum_k \exp\{\bar{v}^C(s_{it}, k')\} = Ew_{\Lambda it}(k) + \delta \eta_{kt} E(\ln \sum_k \exp\{\bar{v}^C(s_{it+1}, k')\}|s_{it}, k)$ , which is a functional equation that identifies firm C's expected match surplus value  $\ln \sum_k \exp\{\bar{v}^C(s_{it}, k')\}$  at each state, once the process of the state,  $\{Ew_{\Lambda it}(k)\}$ , and  $\{\eta_{kt}\}$  are recovered. Next, I guess and verify that  $\bar{v}^C(s_{it}, k) = \bar{v}^A(s_{it}, k) + \phi(s_{it})$  for

<sup>59</sup> With a panel of length  $T$ ,  $\{\eta_{kt}\}$  until period  $T$  are identified up to  $\{\ln \sum_k \exp\{\bar{v}^C(s_{iT+1}, k')\}\}$ . Similarly,  $\{y_A(s_{it}, k)\}$  at two levels until  $T$  are identified up to  $\{\ln \sum_k \exp\{\bar{v}^A(s_{iT+1}, k')\}\}$  by (15). In estimation, as described in app. secs. B4–B6, I maintain that the environment is stationary in  $t \geq T = 8$  and recover  $\{\eta_{kt}\}$  from the probabilities of managers' separation from firm A up to  $t = T - 1$ , on the basis of the market-wide employment and assignment problem characterized in lemma 2. I estimate  $\{\ln \sum_k \exp\{\bar{v}^A(s_{it}, k')\}\}$  in  $t \geq T$  by assuming that  $\{y_A(s_{it}, 1)\}$  are known in  $t \geq T - 1$  so that the only unknown parameters of these match surplus values are  $e_{A28}$  and  $e_{A38}$ .

each state and level  $k$ . Observe that from this conjecture and the definition of  $\gamma(s_{it}, k)$ , it follows that for any  $k$ ,

$$\begin{aligned}
 \bar{v}^C(s_{it}, k) &= y_C(s_{it}, k) + \delta \eta_{ht} E(\ln \sum_{k'} \exp\{\bar{v}^C(s_{it+1}, k')\} | s_{it}, k) \\
 &= y_C(s_{it}, k) + \delta \eta_{ht} E(\ln \sum_{k'} \exp\{\bar{v}^A(s_{it+1}, k') + \phi(s_{it+1})\} | s_{it}, k) \\
 &= y_A(s_{it}, k) + \phi(s_{it}) - \delta \eta_{ht} E[\phi(s_{it+1}) | s_{it}, k] \\
 &\quad + \delta \eta_{ht} E[\ln(\exp\{\phi(s_{it+1})\} \sum_{k'} \exp\{\bar{v}^A(s_{it+1}, k')\}) | s_{it}, k] \\
 &= y_A(s_{it}, k) + \phi(s_{it}) + \delta \eta_{ht} E(\ln \sum_{k'} \exp\{\bar{v}^A(s_{it+1}, k')\} | s_{it}, k) \\
 &= \bar{v}^A(s_{it}, k) + \phi(s_{it}),
 \end{aligned} \tag{53}$$

where the second equality uses the conjecture, the third equality uses that  $y_C(s_{it}, k) = y_A(s_{it}, k) + \gamma(s_{it}, k)$  and the definition of  $\gamma(s_{it}, k)$ , the fourth equality simplifies terms, and the fifth equality uses the form of  $\bar{v}^A(s_{it}, k)$ . Since  $\bar{v}^A(s_{it}, k)$  is the unique fixed point to the Bellman's equation in (15) defining  $\bar{v}^A(s_{it}, k)$ , with  $\delta$  replaced by  $\delta \eta_{ht}$ , it follows that  $\bar{v}^A(s_{it}, k) + \phi(s_{it})$  is the unique fixed point to (53),  $k \geq 1$ . Thus, the conjecture  $\bar{v}^C(s_{it}, k) = \bar{v}^A(s_{it}, k) + \phi(s_{it})$  is the unique fixed point to the Bellman's equation defining  $\bar{v}^C(s_{it}, k)$ ,  $k \geq 1$ . Conditional assignment probabilities at C then satisfy  $\Pr(L_{C,t} = k | f_i = C, s_{it}) = \exp\{\bar{v}^A(s_{it}, k) + \phi(s_{it})\} / \sum_{k'} \exp\{\bar{v}^A(s_{it}, k') + \phi(s_{it})\} = \exp\{\bar{v}^A(s_{it}, k)\} / \sum_{k'} \exp\{\bar{v}^A(s_{it}, k')\} = \Pr(L_{A,t} = k | f_i = A, s_{it})$ . With  $\{\Pr(L_{A,t} = k | f_i = A, s_{it})\}$  identified by proposition 5,  $y_C(s_{it}, k)$  is in turn identified from  $Ew_{A,it}(k)$  for each state, tenure, and level by (12). QED

## Appendix B

### Empirical Appendix

See sections 3–5 in the online appendix for omitted details about the data and estimation.

#### B1. Construction of the BGH Firm's Job Hierarchy

As explained in Baker, Gibbs, and Holmström (1994a), the authors infer the firm's job hierarchy by grouping managers' original job titles into levels as follows. The raw data contain 276 job titles, but just 14 of these, each representing at least 0.5% of employee-years, account for about 90% of all observations and 93% of those with titles coded. To fill the job ladder from the bottom to the top of the hierarchy, Baker, Gibbs, and Holmström (1994a) add to these 14 titles the title of chairman-CEO and the only two titles observed in transitions from the 14 major titles to the title of chairman-CEO for a total of 17 titles. Using managers' transitions across titles for both individual years and the entire sample period, Baker, Gibbs, and Holmström (1994a) construct eight hierarchical levels ranging from level 1 to level 8, which is the highest one and consists of only the title of chairman-CEO. Level 1 consists of the three titles that employ almost only new

hires. Most transitions from level 1 within the firm are to six other titles, categorized as level 2. Transitions out of level 2 are almost exclusively to three other titles, classified as level 3, and so on. After assigning major titles to levels, Baker, Gibbs, and Holmström (1994a) allocate less common titles on the basis of transitions between them and titles already assigned, until all 17 titles are assigned to a level. Note that this procedure, which is common in the literature, is independent of information on managers' compensation.

## B2. Human Capital Process

Instances of the human capital functions in (1) that are special cases of (17) can be derived from laws of motion of human capital of the form

$$H_{jkt}^j = C_{jkt}^j [(1 - \sigma^j) H_{jkt-1}^j + z_{jkt-1}^j (H_{jkt-1}^j)^{\lambda_j} (i_{k_{t-1}}^j)^{\mu_{jk_{t-1}}}], j = G, S, \quad (54)$$

in the spirit of Heckman, Lochner, and Taber (1998), where  $H_{jkt}^j = C_{jkt}^j$ ,  $C_{jkt}^j$  denotes the efficiency of the accumulated human capital in period  $t$ ,  $\sigma^j$  is the human capital depreciation rate, and  $\lambda_j \in \{0, 1\}$ . Although I abstract from managers' skill types, analogous expressions can be obtained when managers differ in their skill types.

*Case 1:*  $\lambda_j = 0$ .—In this case, (54) becomes  $H_{jkt}^j = C_{jkt}^j [(1 - \sigma^j) H_{jkt-1}^j + z_{jkt-1}^j (i_{k_{t-1}}^j)^{\mu_{jk_{t-1}}}]$ . Letting  $z_{jkt-1}^j = C_{jkt-1}^j \cdots C_{jkt-1}^j$  with  $z_{jkt-1}^j = C_{jkt-1}^j$  yields that

$$\begin{aligned} H_{jkt}^j &= C_{jkt}^j [(1 - \sigma^j) C_{jkt-1}^j + C_{jkt-1}^j (i_{k_{t-1}}^j)^{\mu_{jk_{t-1}}}], \\ H_{jkt}^j &= C_{jkt}^j [(1 - \sigma^j)^2 C_{jkt-2}^j C_{jkt-1}^j + (1 - \sigma^j) C_{jkt-1}^j C_{jkt-2}^j (i_{k_{t-1}}^j)^{\mu_{jk_{t-1}}} + C_{jkt-1}^j C_{jkt-2}^j (i_{k_{t-1}}^j)^{\mu_{jk_{t-1}}}], \end{aligned}$$

and so on. With  $A_{jkt}^j \xi_{jkt}^j = C_{jkt-1}^j \cdots C_{jkt-1}^j$ , it follows that

$$H_{jkt}^j = A_{jkt}^j [(1 - \sigma^j)^{t-1} + (1 - \sigma^j)^{t-2} (i_{k_{t-1}}^j)^{\mu_{jk_{t-1}}} + \cdots + (i_{k_{t-1}}^j)^{\mu_{jk_{t-1}}}] \xi_{jkt}^j. \quad (55)$$

Consider first  $H_{jkt}^G$  and recall that  $i_{k_t}^G = \bar{i}$ ,  $1 \leq t \leq t-1$ , for managers in the market in  $t-1$ . Let  $a_{jkt}^G = \ln A_{jkt}^G$ ,  $\mu_{Gk_t} = \mu_G$  for  $\tau \geq 1$ , and  $\varepsilon_{jkt}^G = \ln \xi_{jkt}^G$ . When  $\sigma^G > 0$ , defining  $\eta_G(t-1)$  as the function  $\ln\{(1 - \sigma^G)^{t-1} + [1 - (1 - \sigma^G)^{t-1}] \bar{i}^{\mu_G} / \sigma^G\}$ ,  $h_{jkt}^G = \ln H_{jkt}^G$  can be expressed as

$$\begin{aligned} h_{jkt}^G &= \ln A_{jkt}^G + \ln\{(1 - \sigma^G)^{t-1} + [1 - (1 - \sigma^G)^{t-1}] \bar{i}^{\mu_G} / \sigma^G\} + \ln \xi_{jkt}^G \\ &= a_{jkt}^G + \eta_G(t-1) + \varepsilon_{jkt}^G \end{aligned}$$

by (55). When  $\sigma^G = 0$ , redefining  $\eta_G$  as  $\bar{i}^{\mu_G}$ ,  $h_{jkt}^G$  can instead be expressed as

$$h_{jkt}^G = \ln A_{jkt}^G + \ln[1 + (t-1) \bar{i}^{\mu_G}] + \ln \xi_{jkt}^G \approx a_{jkt}^G + \eta_G \cdot (t-1) + \varepsilon_{jkt}^G. \quad (56)$$

Consider now  $H_{jkt}^S$ . Recall that  $i_{k_{t-1}}^S = \bar{i}_{k_{t-1}}$  for managers at level  $k_{t-1}$  in  $t-1$ . If  $\sigma^S = 1$ , then (55) reduces to  $H_{jkt}^S = A_{jkt}^S (\bar{i}_{k_{t-1}})^{\mu_{S k_{t-1}}} \xi_{jkt}^S$ . Letting  $a_{jkt}^S = \ln A_{jkt}^S$ ,  $\eta_{S k_{t-1}} = \mu_{S k_{t-1}} \ln \bar{i}_{k_{t-1}}$ , and  $\varepsilon_{jkt}^S = \ln \xi_{jkt}^S$ , it follows that  $h_{jkt}^S = \ln H_{jkt}^S$  is given by

$$h_{jkt}^S = \ln A_{jkt}^S + \mu_{S k_{t-1}} \ln \bar{i}_{k_{t-1}} + \ln \xi_{jkt}^S = a_{jkt}^S + \eta_{S k_{t-1}} + \varepsilon_{jkt}^S. \quad (57)$$

The expressions in (56) and (57), augmented with manager skill types, are instances of (17).

*Case 2:*  $\lambda_j = 1$ .—In this case, (54) becomes  $H_{jkt}^j = C_{jkt}^j [1 - \sigma^j + z_{jkt-1}^j (i_{k-1}^j)^{\mu_{jkt-1}}] H_{jkt-1}^j$  so that  $H_{jkt}^j = C_{jkt}^j C_{jkt-1}^j [1 - \sigma^j + z_{jkt-1}^j (i_{k-1}^j)^{\mu_{jkt-1}}]$  and so on. With  $A_{jkt}^j \xi_{jkt}^j = C_{jkt}^j \cdots C_{jkt-1}^j$  as in case 1, then

$$H_{jkt}^j = A_{jkt}^j [1 - \sigma^j + z_{jkt-1}^j (i_{k-1}^j)^{\mu_{jkt-1}}] \cdots [1 - \sigma^j + z_{jkt-1}^j (i_{k-1}^j)^{\mu_{jkt-1}}] \xi_{jkt}^j.$$

Consider first  $H_{jkt}^G$  and recall that  $i_{k\tau}^G = \bar{i}$ ,  $1 \leq \tau \leq t-1$ , for managers in the market in  $t-1$ . With  $a_{jkt}^G = \ln A_{jkt}^G$ ,  $z_{jkt}^G$  independent of  $f, k$ , and  $\tau$ ,  $\mu_{Gk} = \mu_G$  for  $\tau \geq 1$ ,  $\eta_G$  defined as  $\ln(1 - \sigma^G + z^G \bar{i}^{\mu_G})$ , and  $\varepsilon_{jkt}^G = \ln \xi_{jkt}^G$ , it follows that  $h_{jkt}^G = \ln H_{jkt}^G$  can be expressed similarly to (56) as

$$h_{jkt}^G = \ln A_{jkt}^G + (t-1) \ln(1 - \sigma^G + z^G \bar{i}^{\mu_G}) + \ln \xi_{jkt}^G = a_{jkt}^G + \eta_G \cdot (t-1) + \varepsilon_{jkt}^G.$$

Consider now  $H_{jkt}^S$ . Recall that  $i_{k-1}^S = \bar{i}_{k-1}$ , for managers at level  $k_{t-1}$  in  $t-1$ . If  $\sigma^S = 1$ ,  $C_{jkt} = \tilde{C}_{jkt}/H_{jkt-1}$ , and  $z_{jkt-1}^S = \tilde{C}_{jkt-1}^S \cdots \tilde{C}_{jkt-1}^S$ , then (54) becomes  $H_{jkt}^S = A_{jkt}^S \cdot (\bar{i}_{k-1}^S)^{\mu_{Skt-1}} \xi_{jkt}^S$  with  $A_{jkt}^S \xi_{jkt}^S = \tilde{C}_{jkt}^S \cdots \tilde{C}_{jkt-1}^S$  similarly to case 1. Then,  $h_{jkt}^S = \ln H_{jkt}^S$  reduces to (57).

### B3. Specification of Expected Output

Since  $h_{jkt}^p = b_k(h_1)$  and  $a_{jkt}^j(i) = [a_{jkt}^j + a_j^j(i)]/(\alpha_k - \beta_k)$ , with  $a_{jkt}^j \in \{\bar{a}_{jkt}^j, \underline{a}_{jkt}^j\}$  and  $a_j^j(i) \in \{\bar{a}_j^j(i), \underline{a}_j^j(i)\}$ , it follows that

$$\begin{aligned} \bar{y}_{jkt}(i) &= b_k(h_1) + \sum_j \left[ \frac{\bar{a}_{jkt}^j + \bar{a}_j^j(i)}{\alpha_k - \beta_k} + g_k^j(\cdot) \right] \text{ and} \\ \underline{y}_{jkt}(i) &= b_k(h_1) + \sum_j \left[ \frac{\underline{a}_{jkt}^j + \underline{a}_j^j(i)}{\alpha_k - \beta_k} + g_k^j(\cdot) \right], \end{aligned}$$

where  $g_k^G(\cdot) = g_k^G(t-1)$  and  $g_k^S(\cdot) = g_k^S(i_{k-1}^S)$ . Then, expected output in (10) can be expressed as

$$\begin{aligned} y_f(s_{it}, k) &= b_k(h_1) + \sum_j \left[ \frac{a_{jkt}^j + a_j^j(i)}{\alpha_k - \beta_k} + g_k^j(\cdot) \right] + \frac{\beta_k}{\alpha_k - \beta_k} \sum_j [\bar{a}_{jkt}^j - \underline{a}_{jkt}^j + \bar{a}_f^j(i) - \underline{a}_f^j(i)] \\ &\quad + \sum_j [\bar{a}_{jkt}^j - \underline{a}_{jkt}^j + \bar{a}_f^j(i) - \underline{a}_f^j(i)] p_{it} = b_k(h_1) + d_{jkt}(k_{t-1}, i) + e_{jkt}(i) p_{it}, \end{aligned} \quad (58)$$

as in (18). Observe that  $d_{jkt}(k_{t-1}, i)$  denotes the sum of the terms after  $b_k(h_1)$  in the first line of (58) and  $e_{jkt}(i)$  denotes the term multiplying  $p_{it}$  before the second equality in (58).

### B4. Market-Wide Assignment Problem

Let firm entry into the labor market for any manager be free. Specifically, suppose that if firm  $f$  employs a manager in period  $t$ , given its job offer  $k_{ft}$ , then firm  $f' \neq f$  incurs the cost  $c_{f'it} = c_{f'it}(s_{it}, k_{ft})$  before output is realized to compete for the manager in period  $t+1$ . If firm  $f'$  is the only other firm submitting a wage and job offer in  $t+1$ , then its flow cost of entry  $c_{f'it}$  must equal the expected present discounted value of profits after entry, as entry takes place until the

marginal firm can just break even. In this case, the market-wide employment and job assignment problem characterized in proposition 1 admits a simpler representation, as shown in lemma 2 under the notation and assumptions maintained from section IV.A onward.

**LEMMA 2.** Suppose that entry into the labor market for a manager is free and that firm  $f' \neq f$  incurs the cost  $c_{f'il}$  to enter in  $t + 1$  if firm  $f$  employs the manager in  $t$ . Then, the employing firm and assigned level are determined by the policy that solves

$$S(s_{it}, \varepsilon_t) = \max_{f \in \{A, C\}, k_f} \left\{ y_f(s_{it}, k_f) + \varepsilon_{\beta t} - c_{f'it}(s_{it}, k_f) + \delta \eta_{k_f t} \int_{\varepsilon_{t+1}} ES(s_{it+1}, \varepsilon_{t+1} | s_{it}, k_f) dG \right\}. \quad (59)$$

*Proof.* Consider first states at which firm A employs a manager so that  $S(s_{it}, \varepsilon_t) = S(s_{it}, \varepsilon_t | A) = V^A(s_{it}, \varepsilon_t | A) + \Pi^C(s_{it}, \varepsilon_t | A)$ . Then, by the free entry assumption,  $\Pi^C(s_{it}, \varepsilon_t | A) = -c_{Cit}(s_{it}, k_{At}) + \delta \eta_{k_{At} t} \int_{\varepsilon_{t+1}} E \Pi^C(s_{it+1}, \varepsilon_{t+1} | s_{it}, k_{At}) dG = 0$ . Since A's choice of  $k_{At}$  maximizes  $\Pi^A(s_{it}, \varepsilon_t | A)$  by definition as well as  $V^A(s_{it}, \varepsilon_t | A)$  by proposition 1, it follows that  $k_{At}$  also maximizes  $S(s_{it}, \varepsilon_t | A)$ . Then,  $S(s_{it}, \varepsilon_t | A) = \max_{k_A} \{y_A(s_{it}, k_A) + \varepsilon_{A k_A t} - c_{Cit}(s_{it}, k_A) + \delta \eta_{k_A t} \int_{\varepsilon_{t+1}} ES(s_{it+1}, \varepsilon_{t+1} | s_{it}, k_A) dG\}$ . Off the equilibrium path, C is the employing firm, so  $S(s_{it}, \varepsilon_t) = S(s_{it}, \varepsilon_t | C) = \Pi^A(s_{it}, \varepsilon_t | C) + V^C(s_{it}, \varepsilon_t | C)$ , and by the free entry assumption,  $\Pi^A(s_{it}, \varepsilon_t | C) = 0$ . Hence, C's choice of job  $k_{Ct}$ —which, by definition, maximizes  $\Pi^C(s_{it}, \varepsilon_t | C)$  and, by proposition 1,  $V^C(s_{it}, \varepsilon_t | C)$ —also maximizes  $S(s_{it}, \varepsilon_t | C) = \max_{k_C} \{y_C(s_{it}, k_C) + \varepsilon_{C k_C t} - c_{Cit}(s_{it}, k_C) + \delta \eta_{k_C t} \int_{\varepsilon_{t+1}} ES(s_{it+1}, \varepsilon_{t+1} | s_{it}, k_C) dG\}$ .

Consider now states at which firm C employs a manager so that  $S(s_{it}, \varepsilon_t) = S(s_{it}, \varepsilon_t | C) = \Pi^A(s_{it}, \varepsilon_t | C) + V^C(s_{it}, \varepsilon_t | C)$ . Since, by the free entry assumption,  $\Pi^A(s_{it}, \varepsilon_t | C) = 0$ , C's choice of  $k_{Ct}$  maximizes not only  $\Pi^C(s_{it}, \varepsilon_t | C)$  and, by proposition 1,  $V^C(s_{it}, \varepsilon_t | C)$  but also  $S(s_{it}, \varepsilon_t | C)$ . Off the equilibrium path, A is the employing firm, and  $S(s_{it}, \varepsilon_t) = S(s_{it}, \varepsilon_t | A) = V^A(s_{it}, \varepsilon_t | A) + \Pi^C(s_{it}, \varepsilon_t | A)$ , with  $\Pi^C(s_{it}, \varepsilon_t | A) = 0$  by the free entry assumption. Thus, A's choice of  $k_{At}$ , which maximizes  $\Pi^A(s_{it}, \varepsilon_t | A)$  and, by proposition 1,  $V^A(s_{it}, \varepsilon_t | A)$ , also maximizes  $S(s_{it}, \varepsilon_t | A)$ .

Recall that  $S(s_{it}, \varepsilon_t) = S(s_{it}, \varepsilon_t | A) \geq S(s_{it}, \varepsilon_t | C)$  at all states at which firm A employs a manager and  $S(s_{it}, \varepsilon_t) = S(s_{it}, \varepsilon_t | C) > S(s_{it}, \varepsilon_t | A)$  at all states at which firm C employs a manager by proposition 1. Combining this result with the above observations yields (59), as desired. QED

A more convenient form for  $S(s_{it}, \varepsilon_t)$  can be obtained under the assumption of no recall of previously employed managers. In this case, for the sample paths of interest, in which a manager is either employed by firm A or first employed by firm A and then by firm C,  $S(s_{it}, \varepsilon_t)$  reduces to

$$S(s_{it}, \varepsilon_t) = \max \left\{ \max_{k_A} \left\{ y_A(s_{it}, k_A) + \varepsilon_{A k_A t} + \delta \eta_{k_A t} \int_{\varepsilon_{t+1}} ES(s_{it+1}, \varepsilon_{t+1} | s_{it}, k_A) dG \right\}, \bar{V}^C(s_{it}, \varepsilon_t) \right\}, \quad (60)$$

if  $c_{Cit}(s_{it}, k_A) \approx 0$ . In (60),  $\bar{V}^C(s_{it}, \varepsilon_t)$  is firm C's value of (net) output—see proposition 1—and firm C can be thought to represent all other firms in the labor market. To see how C can be formally interpreted as such, note that when the labor market is populated by multiple firms in some set  $\mathcal{F}$ , (59) becomes  $S(s_{it}, \varepsilon_t) = \max_{f \in \mathcal{F}, k_f} \{S(s_{it}, \varepsilon_t | f, k_f)\} = \max_{f \in \mathcal{F}} \{\max_{k_f} \{S(s_{it}, \varepsilon_t | f, k_f)\}\} = \max_{f \in \mathcal{F}} \{S(s_{it}, \varepsilon_t | f)\}$ , where  $S(s_{it}, \varepsilon_t | f) = \max_{k_f} \{S(s_{it}, \varepsilon_t | f, k_f)\}$  and

$$S(s_{it}, \varepsilon_t | f, k_f) = y_f(s_{it}, k_f) + \varepsilon_{\beta_f t} - \sum_{f' \neq f} c_{f'it}^*(s_{it}, k_f) + \delta \eta_{k_f t} \int_{\varepsilon_{t+1}} ES(s_{it+1}, \varepsilon_{t+1} | s_{it}, k_f) dG. \quad (61)$$

The value  $S(s_{it}, \varepsilon_t)$  can also be expressed as the value of the choice between firm A and the firm offering the largest value of (net) output among all other firms. Namely, if  $f_t^*$  is defined as  $f_t^* = \arg \max_{f \neq A} \{S(s_{it}, \varepsilon_t | f)\}$ , then  $S(s_{it}, \varepsilon_t) = \max\{S(s_{it}, \varepsilon_t | A), S(s_{it}, \varepsilon_t | f_t^*)\}$ . By interpreting firm  $f_t^*$  as firm C and the cost functions  $\sum_{f' \neq A} c_{f'it}^*(s_{it}, k_A)$  and  $\sum_{f' \neq f_t^*} c_{f'it}^*(s_{it}, k_{f_t^*})$  in  $S(s_{it}, \varepsilon_t | A)$  and  $S(s_{it}, \varepsilon_t | f_t^*)$  as the cost functions  $c_{Cit}(s_{it}, k_A)$  and  $c_{Ait}(s_{it}, k_C)$ , respectively, it is easy to see that (59) and, under the assumptions described, (60) follow.

Since the productivity shocks  $\{\varepsilon_{\beta_{it}}\}$  are assumed to be independent Gumbel distributed each with mean zero and variance  $\pi^2/6$ ,  $S(s_{it}, \varepsilon_t)$  in (60) specializes to

$$S(s_{it}, \varepsilon_t) = \max\{\max_k \{\zeta^A(s_{it}, k) + \varepsilon_{Ait}\}, \max_k \{\zeta^C(s_{it}, k) + \varepsilon_{Ckt}\}\}, \quad (62)$$

where  $\max_k \{\zeta^C(s_{it}, k) + \varepsilon_{Ckt}\} = \ln \sum_k \exp\{\zeta^C(s_{it}, k)\} + \varepsilon_{Ckt}$ ,  $\varepsilon_{Ckt}$  is independent Gumbel distributed with mean zero and variance  $\pi^2/6$ , and  $\zeta^C(s_{it}, k) = y_C(s_{it}, k) + \delta \eta_{kt} E(\ln \sum_{k'} \exp\{\zeta^C(s_{it+1}, k')\} | s_{it}, k)$ . In estimating the employment, assignment, and separation probabilities  $\{\Pr(f_i = A, L_{At} = k | s_{it})\}$  and  $\{\Pr(f_i = C | s_{it})\}$  on the basis of (62), I normalize firm C's expected output at each state and level, which pins down  $\{\ln \sum_k \exp\{\zeta^C(s_{it}, k)\}\}$ .<sup>60</sup>

### B5. Human Capital and Output Parameters at the Firm

As described in section V.A, to conserve on parameters due to the rapid promotions at the firm, the (near) absence of demotions, and the high separation rate from the firm in each tenure, I estimate only  $\{d_{A1t}(L2)\}$  and  $\{d_{A3t}(L3)\}$  among the parameters  $\{d_{Ait}(k_{t-1})\}$  for firm A. I also allow for common components across the parameters  $\{d_{A1t}(L2)\}$  and  $\{e_{A2t}\}$  by specifying them as  $d_{A1t}(L2) = -d_t$  between  $t = 4$  and  $t = 7$  and  $e_{A2t} = d_t + e_t$  between  $t = 2$  and  $t = 8$ , with  $d_t = d_2 1\{2 \leq t \leq 4\} + d_5 1\{5 \leq t \leq 8\}$ .

Consider first  $\{d_{Ait}(k_{t-1})\}$ . At level 1, I set  $d_{A11}(L0) = d_{A11}$  to 1,000 (and  $e_{A11}$  to zero), since all managers in the sample are first observed at this level. For  $t > 1$ , I estimate  $d_{A14}(L2)$  and  $d_{A15}(L2)$  and maintain that  $d_{A1t}(L2) = d_{A15}(L2)$  for  $t = 6, 7$  and  $d_{A1t}(L2) = 0$  otherwise. At level 2, I did not estimate any such parameter. At level 3, I estimate  $d_{A3t}(L3)$  only after  $t = 3$ , since no manager is at level 3 before this tenure, and maintain that  $d_{A36}(L3) = d_{A35}(L3)$  and  $d_{A38}(L3) = 0$ .

Consider now  $\{e_{Ait}\}$ . At level 1, I did not estimate any such parameter in  $t \geq 3$ , because of the small number of managers observed at level 1 in those tenures. At

<sup>60</sup> I compute values and employment and assignment probabilities in  $t \geq 8$  by assuming that managers no longer acquire human capital then, which is not implausible. Indeed, compared with those in earlier tenures, managers' employment and assignment profiles in  $t \geq 8$  in the data display much less variation with tenure, previous assignments, and performance. Then, for a given vector of parameter values, I calculate terminal values in  $t = 8$  as the values of the infinite-horizon market-wide employment and assignment problem characterized in lemma 2 from  $t = 8$  on. Given these terminal values, I solve for the values of the finite-horizon market-wide employment and assignment problem in each tenure between  $t = 1$  and  $t = 7$  by backward induction.

level 2, I estimate  $e_{A22}$ ,  $e_{A23}$ ,  $e_{A25}$ , and  $e_{A26}$ ; maintain that  $e_{A24} = e_{A22}$  and  $e_{A27} = e_{A26}$ ; and anchor  $e_{A28}$  by setting  $e_8 = 0$ . Since no manager is assigned to level 2 at entry, I did not estimate  $e_{A21}$ . At level 3, I restrict  $e_{A31} = e_{A32} = e_{A33} < 0$ , as no manager is at this level before  $t = 3$ , and estimate the parameters  $e_{A31}$ ,  $e_{A34}$ ,  $e_{A37}$ , and  $e_{A38}$ , with  $e_{A36} = e_{A35} = e_{A33} + e_{A34}$ .

### B6. Exogenous Separation Rates

To conserve on parameters, I allow only for variation in the exogenous separation rates  $\{1 - \eta_{it}\}$  that proves statistically significant, whenever setting these parameters equal across levels or tenures does not affect any other parameter estimates. As a result, the parameters of exogenous separations that I estimate for levels 1, 2, and 3, respectively, are as follows:  $\eta_{11}$ ,  $\eta_{21}$ , and  $\eta_{31}$  for tenure 1; none for tenure 2, since  $\eta_{12} = \eta_{11}$ ,  $\eta_{22} = \eta_{21}$ , and  $\eta_{32} = \eta_{31}$ ;  $\eta$  for tenure 3, since  $\eta_{13} = \eta_{14} + \eta$ ,  $\eta_{23} = \eta_{22} + \eta$ , and  $\eta_{33} = \eta_{32}$ ;  $\eta_{14}$  and  $\eta_{24}$  for tenure 4, since  $\eta_{34} = \eta_{24}$ ;  $\eta_{25}$  for tenure 5, since  $\eta_{15} = \eta_{14}$  and  $\eta_{35} = \eta_{25}$ ;  $\eta_{26}$  for tenure 6, since  $\eta_{16} = \eta_{15}$  and  $\eta_{36} = \eta_{26}$ ;  $\eta_{27}$  for tenure 7, since  $\eta_{17} = \eta_{16}$  and  $\eta_{37} = \eta_{27}$ ; and none from tenure 8 onward, since  $\eta_{1t} = \eta_{17}$ ,  $\eta_{2t} = \eta_{27}$ , and, given the high probability of retention at the firm at level 3 in the data,  $\eta_{3t} = 0$ ,  $t \geq 8$ . So, I estimate the parameters  $(\eta_{11}, \eta, \eta_{14})$  for level 1,  $(\eta_{21}, \eta, \eta_{24}, \eta_{25}, \eta_{26}, \eta_{27})$  for level 2, and  $e_{31}$  for level 3.

### B7. Estimated Wage Equation

Recall that  $h_{fjt}^p = b_k(h_1)$  and  $a_{fjt}^j(i) = [a_{fjt}^j + a_f^j(i)]/(\alpha_k - \beta_k)$ , with  $a_{fjt}^j \in \{\bar{a}_{fjt}^j, \underline{a}_{fjt}^j\}$  and  $a_f^j(i) \in \{\bar{a}_f^j(i), \underline{a}_f^j(i)\}$ , so that by the results in appendix section B3,

$$\begin{aligned} y_C(s_{it}, k) = & \sum_j \left[ \frac{a_{fjt}^j(i)}{\alpha_k - \beta_k} + g_k^j(\cdot) \right] + \underbrace{\frac{\beta_k}{\alpha_k - \beta_k} \sum_j [\bar{a}_{fjt}^j(i) - \underline{a}_{fjt}^j(i)]}_{\omega_{0k}} + \underbrace{b_k(h_1)}_{b_k^x x_1 + b_k^{xx} x_1^2 + b_k^e e_1 + \sum_m b_{ym} 1\{y_1 = m\}} \\ & + \underbrace{\frac{[1/(\alpha_k - \beta_k)] \sum_j [\bar{a}_{fjt}^j(i) + \beta_k(\bar{a}_{fjt}^j(i) - \underline{a}_{fjt}^j(i))]}{t-1}}_{\omega_{4kt}} \cdot (t-1) + \underbrace{\left\{ \sum_j [\bar{a}_{fjt}^j(i) - \underline{a}_{fjt}^j(i)] \right\}}_{\omega_{5i}} + \underbrace{\left\{ \sum_j [\bar{a}_{fjt}^j(i) - \underline{a}_{fjt}^j(i)] \right\}}_{\omega_{5kt}} p_{it}. \end{aligned}$$

By (12) and the condition in proposition 6, the expression in (19) follows:

$$w_{\lambda it} = \omega(h_1, k, i) + \omega_{4kt} \cdot (t-1) + (\omega_{5i} + \omega_{5kt}) p_{it} - \ln \Pr(L_{\lambda t} = k | f_i = A, s_{it}) + \varepsilon_{\lambda it}, \quad (63)$$

with  $\omega(h_1, k, i) = \omega_{0k} + \omega_{1k} x_1 + \omega_{2k} x_1^2 + \omega_{3k} e_1 + \sum_m \omega_{ym} 1\{y_1 = m\}$  and  $(\omega_{1k}, \omega_{2k}, \omega_{3k}, \omega_{ym}) = (b_k^x, b_k^{xx}, b_k^e, b_{ym})$ . As mentioned in the main text, I maintain that the parameters  $\omega_{1k}$ ,  $\omega_{2k}$ , and  $\omega_{3k}$  are common between levels 1 and 2 and denote their values by  $\omega_1$ ,  $\omega_2$ , and  $\omega_3$ . I set  $\omega_{ym} = 0$ ,  $0 \leq m \leq 3$ , and  $\omega_{y4} = \omega_{y5}$  for the recession years 1974 and 1975. I estimate the coefficient on tenure only at level 1 and specify it as  $\omega_{41t} = \omega_{412} 1\{t < 4\} - \omega_{412} 1\{t \geq 4\}$ , which helps the model account for the progressively greater proportion of managers at level 1 who are paid relatively low wages in high tenures. Thus, the baseline wage parameters are  $\{\omega_{0i1}, \omega_{0i2}, \omega_{0i3}\}$ ,  $\omega_1$ ,  $\omega_2$ ,  $\omega_3$ ,  $\omega_{13}$ ,  $\omega_{23}$ ,  $\omega_{33}$ ,  $\{\omega_{ym}\}_{m=5}^9$ ,  $\omega_{412}$ ,  $\{\omega_{5i1}\}$ ,  $\{\sigma_{\lambda i1}, \sigma_{\lambda i2}\}$ , and, as argued,  $\sigma_{\lambda 3}$ . See appendix section B11 for a more general version of the model that includes the parameters  $\{\omega_{4kt}\}$  for levels 2 and 3 and  $\{\omega_{5kt}\}$ .

When parameter estimates are modified to evaluate the importance of uncertainty about ability, learning about it, and human capital acquisition in section VI,



wages are recomputed for any new sequences of states, employing firms, and job assignments for each manager skill type. To see that the mapping between wage and expected output parameters is invariant to these parameter changes, note that in the experiments that assess the role of uncertainty about ability and learning about it, expected output at each job level is kept at its baseline value for each possible state  $s_{jt}$ . The parameters  $\{\alpha_h, \beta_h\}$  are adjusted only in the updating of beliefs about ability to capture that job levels are more or less informative than in the baseline model, depending on the experiment considered. In the experiment that evaluates the role of human capital acquisition, the parameters of expected output at each job level are simply assumed to be constant over time.

#### *B8. Estimates of Classification Error*

To limit parameter proliferation based on model diagnostics and fit, I maintain that  $\rho_1 = \rho_0$  in (11). As discussed, I estimate classification error parameters only at levels 1 and 2—namely,  $\rho_0$  and  $\{\rho_2(k)\}_{k=1,2}$ . Since the probability of a recorded rating, conditional on a manager's performance, is not a linear function of the probability of high performance by (11), the error in ratings in general leads to bias. Moreover, the greater in absolute value  $\rho_2(k)$  is, the greater the persistence in misreporting is. The estimates in table B3 imply that performance is measured with error, and over time, this error decreases for managers whose performance is low and increases for managers whose performance is high, as  $E_0(k, t)$  decreases with tenure, whereas  $E_1(k, t)$  increases with tenure when  $\rho_2(k) < 0$  in (11). This result conforms with the idea that the performance of unsuccessful managers is assessed more and more thoroughly—and that of successful managers less and less thoroughly—over time, since poor performance may eventually lead to disciplining actions by a firm.

#### *B9. The BGH Firm's Retention and Job Assignment Policy*

The policy that emerges by simulating the model at the estimated parameter values implies that, on average, managers at higher levels have higher ability, acquired human capital, and beliefs that their ability is high. These results support the intuition by Baker, Gibbs, and Holmström (1994a, 1994b), among others, that a firm acts as an information acquisition filter in the labor market by producing information about managers' ability, which helps improve the matching of managers to jobs and firms over time. Within the firm, this process of information acquisition leads to the endogenous selection of managers to higher-level jobs through promotions, which is the key channel through which learning affects wages. As for separations, the estimated exogenous separation rates in table B5 are fairly close to the hazard rates of separation in the data in table B1, so most separations are indeed exogenous. Crucial features of the data that explain this finding are that separations are largely uncorrelated with performance—which governs the process of learning about ability and so the systematic stochastic component of the process of human capital acquisition at the firm in the model—and approximately constant with tenure and past job assignments—which govern the deterministic component of the process of human capital acquisition at the firm in the model. See also Baker, Gibbs, and Holmström (1994a) on this point. In

particular, there is no evidence that separations mask a tendency for managers to be laid off or move to other firms in response to poor evaluations of performance that would otherwise lead to demotions. That some separations are endogenous is confirmed by the higher wage growth of managers retained by the firm throughout the sample years relative to those who separate, as discussed in the main text.

*B10. Goodness of Fit*

To assess the fit of the model to the data, I simulate 4,000 initial prior realizations per manager, drawn from the estimated distribution of initial priors. Table 1 shows that the model tracks remarkably well the profiles of managers' level assignments, which are nonlinear and nonmonotone in tenure, as well as the tenure pattern of managers' separations. Table B1 further shows that the model also accurately reproduces the feature that managers' flows out of levels 1 and 2 result from an essentially constant hazard rate of separation and a hazard rate of promotion that first increases then decreases or plateaus with tenure. As table 2 illustrates, the model successfully fits the distributions of performance ratings both at level 1 and at level 2 in nearly all tenures. Last, table 3 shows that the model matches quite well the distribution of wages at each level and tenure, except for some discrepancies at level 3 in the highest tenures, which are partly due to the smaller number of observations on managers in those tenures because of the high separation rate from the firm. In fact, the model fits the distributions of managers' wages from the larger sample that includes entrants in the firm at the original levels 1–4 substantially better, as discussed in section 5 in the online appendix.

One criterion to formally evaluate model fit is the Pearson's  $\chi^2$  goodness-of-fit test. I perform it using the statistic  $N \sum_{r=1}^R \{[x(r) - \hat{x}(r)]^2 / \hat{x}(r)\}$ , where  $x(\cdot)$  denotes the empirical density function of a given endogenous variable,  $\hat{x}(\cdot)$  denotes the maximum likelihood estimate of the density function of that variable,  $N$  denotes the number of observations, and  $R$  denotes the number of categories considered (not taking into account that the parameters of the model are estimated). I compare the observed and predicted distributions of managers across levels, their performance ratings at levels 1 and 2, and their wages at each level in each of the first 7 years of tenure. Overwhelmingly, the test does not reject the model at conventional significance levels with respect to the distribution of managers across levels and their probability of retention at the firm, the hazard rates of separation, retention at a level, and promotion to the next level and the distributions of performance ratings and wages at levels 1 and 2. As for the distributions of wages at level 3, the test does reject the model in medium to high tenures. One reason for this latter result is the small number of managers at level 3 in high tenures, as confirmed by the improved fit of the model to the larger sample of entrants in the firm; see the online appendix for details.

*B11. Robustness of Estimates*

To examine the robustness of results to the estimated specification of the wage process, I also consider a more general version of the wage equation in (19), which includes the tenure coefficient  $\omega_{akt} = \omega_{ak}$  for levels 2 and 3 and the prior coefficient  $\omega_{5kt}$  specified as  $\omega_{5kt} = \omega_{5k}^l + \omega_{5k}^g \cdot (t - 1) + \omega_5^* x_1$ , with additional

parameters ( $\omega_{42}, \omega_{43}, \omega_{52}^l, \omega_{53}^l, \omega_{51}^g, \omega_{52}^g, \omega_{53}^g, \omega_5^x$ ). The online appendix shows that the estimates of the parameters of this more general version of the model that are common to the baseline model are broadly similar to those of the baseline model, except for the output parameters, which are somewhat larger. Some estimates, though, are less significant, especially  $\alpha_3$  and  $e_{A38}$ , which cannot be estimated with any precision. The estimates of the additional wage parameters are fairly small, particularly the prior coefficients  $\omega_{52}^l$  and  $\omega_{53}^l$  for levels 2 and 3, respectively, relative to the prior coefficient  $\omega_{5i}$  estimated for the baseline model. Thus, allowing the coefficient on the prior  $p_u$  in period  $t$  in the wage equation to vary with a manager's job level and tenure does not substantially affect the estimation results.

*B12. The Role of Persistent Uncertainty about Ability*

Recall the experiment of fast learning at level 1 in section VI. In the case of fast learning at level 2, level 2 is assumed to be nearly perfectly informative about ability ( $\alpha_2 = 0.99$  and  $\beta_2 = 0.01$ ), whereas the other parameters are fixed at their baseline values. See tables 6 and B7 for the distributions of managers' wages and their job paths at the firm, respectively. Similar to the case of fast learning at level 1, fast learning at level 2 yields higher wage growth, larger wage dispersion at levels 1 and 2, and faster promotions. Additionally, relative to the baseline model, it leads to much lower wage dispersion at level 3 and a higher percentage of managers at level 2 and, accordingly, a lower percentage of managers at level 3 in high tenures. Intuitively, the greater informativeness of level 2 makes it the preferred assignment over levels 1 and 3 for a larger set of states, including higher priors and tenures, than that in the baseline model. As managers tend to reach level 3 with higher and less dispersed priors than in the baseline model, the standard deviation of wages at level 3 is much lower than both that in the baseline model and that at level 2 in this experiment.

TABLE B1  
HAZARD RATES OF SEPARATION, RETENTION AT LEVEL, AND PROMOTION (PERCENTAGES)

| LEVEL AND TENURE | SEPARATION |       | RETENTION |       | PROMOTION |       |
|------------------|------------|-------|-----------|-------|-----------|-------|
|                  | Data       | Model | Data      | Model | Data      | Model |
| Level 1:         |            |       |           |       |           |       |
| 1 to 2           | 14.5       | 14.5  | 45.6      | 45.7  | 39.9      | 39.8  |
| 2 to 3           | 14.6       | 14.5  | 36.8      | 20.3  | 48.6      | 55.2  |
| 3 to 4           | 11.7       | 8.4   | 45.6      | 46.9  | 42.7      | 27.8  |
| 4 to 5           | 11.9       | 5.0   | 60.6      | 65.0  | 27.5      | 19.9  |
| 5 to 6           | 9.1        | 5.1   | 62.1      | 64.8  | 28.8      | 21.6  |
| 6 to 7           | 12.2       | 5.0   | 73.2      | 79.1  | 14.6      | 11.4  |
| Level 2:         |            |       |           |       |           |       |
| 2 to 3           | 16.3       | 13.6  | 61.9      | 55.5  | 21.8      | 10.9  |
| 3 to 4           | 15.6       | 16.9  | 47.0      | 51.7  | 37.4      | 31.4  |
| 4 to 5           | 14.7       | 14.2  | 54.8      | 56.9  | 30.5      | 28.9  |
| 5 to 6           | 12.8       | 12.1  | 60.5      | 62.6  | 26.7      | 25.3  |
| 6 to 7           | 15.4       | 11.5  | 59.4      | 62.9  | 25.1      | 25.6  |

TABLE B1 (*Continued*)

| LEVEL AND TENURE | SEPARATION |       | RETENTION |       | PROMOTION |       |
|------------------|------------|-------|-----------|-------|-----------|-------|
|                  | Data       | Model | Data      | Model | Data      | Model |
| Level 3:         |            |       |           |       |           |       |
| 3 to 4           | 10.5       | 12.2  | 89.5      | 87.8  | ...       | ...   |
| 4 to 5           | 12.2       | 14.2  | 87.8      | 85.8  | ...       | ...   |
| 5 to 6           | 10.1       | 12.1  | 89.9      | 87.9  | ...       | ...   |
| 6 to 7           | 10.0       | 11.5  | 90.0      | 88.5  | ...       | ...   |

NOTE.—Promotions are by one level.

TABLE B2  
PERCENTAGE DISTRIBUTIONS OF CHANGES IN LOG WAGES BY TENURE

| Tenure | -.15 to .00 | .00 to .15 | .15 to .30 | Growth Rate |
|--------|-------------|------------|------------|-------------|
| 1 to 2 | 22.9        | 69.9       | 7.2        | 5.2         |
| 2 to 3 | 22.6        | 70.4       | 6.6        | 5.1         |
| 3 to 4 | 24.9        | 70.3       | 4.3        | 3.9         |
| 4 to 5 | 23.6        | 70.1       | 5.9        | 2.2         |
| 5 to 6 | 22.5        | 70.5       | 6.9        | .7          |
| 6 to 7 | 21.9        | 68.5       | 8.3        | 1.8         |

TABLE B3  
ESTIMATES OF PARAMETERS OF CLASSIFICATION ERROR  
IN PERFORMANCE RATINGS

| Parameters               | Level 1 ( $k = 1$ ) | Level 2 ( $k = 2$ )            |
|--------------------------|---------------------|--------------------------------|
| Base error: $\rho_0$     | .521<br>(.009)      | (as $k = 1$ )<br>(as $k = 1$ ) |
| Persistence: $\rho_2(k)$ | -.703<br>(.001)     | -.544<br>(.031)                |

NOTE.—Recall that  $\rho_1 = \rho_0$ ; see app. sec. B8 for a description of this parameterization. Asymptotic standard errors are in parentheses.

TABLE B4  
ESTIMATES OF PARAMETERS OF EXPECTED OUTPUT AT FIRM

| Parameters    | Value   | Standard Error |
|---------------|---------|----------------|
| Level 1:      |         |                |
| $d_{A14}(L2)$ | -14.095 | .073           |
| $d_{A15}(L2)$ | -9.592  | .067           |
| $d_{A16}(L2)$ | -9.592  | .067           |
| $d_{A17}(L2)$ | -9.592  | .067           |
| $e_{A12}$     | 59.210  | 2.935          |
| Level 2:      |         |                |
| $e_{A22}$     | 51.269  | .083           |
| $e_{A23}$     | 44.202  | .731           |
| $e_{A24}$     | 51.269  | .083           |
| $e_{A25}$     | 43.438  | .184           |
| $e_{A26}$     | 44.496  | .080           |
| $e_{A27}$     | 44.496  | .080           |
| $e_{A28}$     | 9.592   | .067           |

TABLE B4 (Continued)

| Parameters    | Value  | Standard Error |
|---------------|--------|----------------|
| Level 3:      |        |                |
| $d_{A34}(L3)$ | 17.070 | .377           |
| $d_{A35}(L3)$ | 4.056  | .014           |
| $d_{A36}(L3)$ | 4.056  | .014           |
| $d_{A37}(L3)$ | 4.561  | .054           |
| $e_{A31}$     | -7.999 | .171           |
| $e_{A32}$     | -7.999 | .171           |
| $e_{A33}$     | -7.999 | .171           |
| $e_{A34}$     | 45.173 | .176           |
| $e_{A35}$     | 37.174 | .040           |
| $e_{A36}$     | 37.174 | .040           |
| $e_{A37}$     | 43.814 | .021           |
| $e_{A38}$     | 40.067 | 5.608          |

NOTE.—All parameters are expressed in thousands. Unreported parameters are normalized to zero. See app. sec. B5 for a description of the parameterization of  $\{d_{Akt}(Lk)\}$  and  $\{e_{Akt}\}$ .

TABLE B5  
ESTIMATES OF EXOGENOUS SEPARATION RATES

| Parameters             | Value (%) | Standard Error |
|------------------------|-----------|----------------|
| Level 1:               |           |                |
| $1 - \eta_{11}$        | 14.5      | .002           |
| $1 - \eta_{14} - \eta$ | 8.4       | .0003          |
| $1 - \eta_{14}$        | 5.0       | .0002          |
| Level 2:               |           |                |
| $1 - \eta_{21}$        | 13.6      | .004           |
| $1 - \eta_{22} - \eta$ | 16.9      | .004           |
| $1 - \eta_{24}$        | 14.2      | .0001          |
| $1 - \eta_{25}$        | 12.1      | .029           |
| $1 - \eta_{26}$        | 11.5      | .006           |
| $1 - \eta_{27}$        | 11.1      | .0003          |
| Level 3:               |           |                |
| $1 - \eta_{31}$        | 12.2      | .001           |

NOTE.—See app. sec. B6 for a description of the parameterization of  $\{1 - \eta_{ht}\}$ .

TABLE B6  
ESTIMATES OF YEAR DUMMIES IN WAGE EQUATION (BASELINE: 1970–73)

| Parameters    | 1975<br>( $m = 5$ ) | 1976<br>( $m = 6$ ) | 1977<br>( $m = 7$ ) | 1978<br>( $m = 8$ ) | 1979<br>( $m = 9$ ) |
|---------------|---------------------|---------------------|---------------------|---------------------|---------------------|
| $\omega_{ym}$ | -.063<br>(.0005)    | -.107<br>(.005)     | -.140<br>(.006)     | -.208<br>(.005)     | -.169<br>(.003)     |

NOTE.—Recall that  $\omega_{ym} = 0, 0 \leq m \leq 3$ , and  $\omega_{y4} = \omega_{y5}$ ; see app. sec. B7 for a description of this parameterization. Asymptotic standard errors are in parentheses.

TABLE B7  
COUNTERFACTUAL EXPERIMENTS ON IMPORTANCE OF LEARNING AND PERSISTENT UNCERTAINTY ABOUT ABILITY FOR LEVEL ASSIGNMENTS

| TENURE | SEPARATION |      |             |      |               |      | LEVEL 1  |       |             |       |               |       | LEVEL 2  |      |             |      |               |      | LEVEL 3  |      |             |      |               |      |
|--------|------------|------|-------------|------|---------------|------|----------|-------|-------------|-------|---------------|-------|----------|------|-------------|------|---------------|------|----------|------|-------------|------|---------------|------|
|        | Baseline   |      | No Learning |      | Fast Learning |      | Baseline |       | No Learning |       | Fast Learning |       | Baseline |      | No Learning |      | Fast Learning |      | Baseline |      | No Learning |      | Fast Learning |      |
|        | at 1       | at 2 | at 1        | at 2 | at 1          | at 2 | at 1     | at 2  | at 1        | at 2  | at 1          | at 2  | at 1     | at 2 | at 1        | at 2 | at 1          | at 2 | at 1     | at 2 | at 1        | at 2 | at 1          | at 2 |
| 1      | .0         | .0   | .0          | .0   | .0            | .0   | 100.0    | 100.0 | 100.0       | 100.0 | 100.0         | 100.0 | .0       | .0   | .0          | .0   | .0            | .0   | .0       | .0   | .0          | .0   | .0            | .0   |
| 2      | 14.5       | 14.5 | 14.5        | 14.5 | 14.5          | 14.5 | 45.7     | 57.7  | 57.7        | 40.5  | .0            | 39.8  | 27.8     | 45.0 | .0          | 85.4 | .0            | .0   | .0       | .0   | .0          | .0   | .0            | .0   |
| 3      | 26.5       | 26.6 | 26.5        | 26.1 | 26.6          | 26.1 | 17.2     | 20.6  | 20.6        | 14.6  | 5.0           | 47.3  | 52.0     | 38.9 | .0          | 35.0 | 8.9           | .8   | 20.0     | .0   | .8          | 20.0 | 33.8          | 33.8 |
| 4      | 37.1       | 37.3 | 36.7        | 36.6 | 36.7          | 36.6 | 8.1      | 11.6  | 11.6        | 7.3   | 2.0           | 29.2  | 40.3     | 25.9 | .0          | 29.1 | 25.6          | 10.8 | 30.1     | .0   | 10.8        | 30.1 | 32.3          | 32.3 |
| 5      | 45.3       | 45.1 | 45.0        | 45.4 | 45.0          | 45.4 | 5.3      | 8.2   | 8.2         | 4.9   | 1.3           | 18.3  | 30.3     | 17.7 | .0          | 24.7 | 31.2          | 16.4 | 32.4     | .0   | 16.4        | 32.4 | 28.6          | 28.6 |
| 6      | 51.5       | 51.2 | 51.3        | 51.9 | 51.3          | 51.9 | 3.4      | 5.6   | 5.6         | 3.2   | 1.0           | 12.6  | 23.3     | 13.1 | .0          | 21.7 | 32.5          | 19.9 | 32.3     | .0   | 19.9        | 32.3 | 25.4          | 25.4 |
| 7      | 56.9       | 56.4 | 56.7        | 57.4 | 56.7          | 57.4 | 2.7      | 4.5   | 4.5         | 2.6   | .9            | 8.3   | 15.4     | 8.7  | .0          | 19.2 | 32.1          | 23.6 | 32.0     | .0   | 23.6        | 32.0 | 22.6          | 22.6 |

NOTE.—No learning:  $\beta_k = \hat{\alpha}_k$ ,  $k = 1, 2, 3$ . Fast learning at level  $k$ :  $\alpha_k = 0.99$  and  $\beta_k = 0.01$ ,  $k = 1, 2$ .

TABLE B8  
COUNTERFACTUAL EXPERIMENT ON IMPORTANCE OF EXPERIMENTATION FOR LEVEL ASSIGNMENTS

| TENURE | SEPARATION |                 |      | LEVEL 1  |                 |       | LEVEL 2  |                 |      | LEVEL 3  |                 |      |
|--------|------------|-----------------|------|----------|-----------------|-------|----------|-----------------|------|----------|-----------------|------|
|        | Baseline   | Experimentation | No   | Baseline | Experimentation | No    | Baseline | Experimentation | No   | Baseline | Experimentation | No   |
|        |            |                 |      |          |                 |       |          |                 |      |          |                 |      |
| 1      | .0         | .0              | .0   | 100.0    | 100.0           | 100.0 | .0       | .0              | .0   | .0       | .0              | .0   |
| 2      | 14.5       | 14.5            | 14.5 | 45.7     | 84.6            | 84.6  | 39.8     | .9              | .0   | .0       | .0              | .0   |
| 3      | 26.5       | 26.9            | 26.9 | 17.2     | 6.2             | 6.2   | 47.3     | 18.3            | 8.9  | 8.9      | 48.6            | 48.6 |
| 4      | 37.1       | 36.4            | 36.4 | 8.1      | 2.0             | 2.0   | 29.2     | 7.7             | 25.6 | 25.6     | 53.8            | 53.8 |
| 5      | 45.3       | 45.3            | 45.3 | 5.3      | 1.1             | 1.1   | 18.3     | 3.5             | 31.2 | 31.2     | 50.2            | 50.2 |
| 6      | 51.5       | 51.8            | 51.8 | 3.4      | .6              | .6    | 12.6     | 2.0             | 32.5 | 32.5     | 45.6            | 45.6 |
| 7      | 56.9       | 57.3            | 57.3 | 2.7      | .5              | .5    | 8.3      | 1.3             | 32.1 | 32.1     | 40.9            | 40.9 |

NOTE.—No experimentation:  $\alpha_k = \hat{\alpha}_1$  and  $\beta_k = \hat{\beta}_1$ ,  $k = 2, 3$ .

## Appendix C

### Alternative Explanations

The analysis so far has focused on the role of symmetric learning about ability and human capital acquisition for the dynamics of managers' jobs and wages at the firm in the BGH data. I now briefly discuss the potential role of three alternative mechanisms: asymmetric learning about ability, performance incentives provided through either bonus pay or promotion tournaments, and complementarities among managers in production.

#### *C1. Asymmetric Learning*

I have maintained that firms and managers share the same information about managers' ability, and so their learning about it is symmetric. To see that this assumption is plausible for my data, first note that the analysis in Baker, Gibbs, and Holmström (1994a), as the authors discuss, supports the idea that careers at the firm are governed by a process of learning about ability and that managers' performance is public information.<sup>61</sup> Second, DeVaro and Waldman (2012) perform a formal test of asymmetric information and learning on the BGH data, based on the positive (negative) correlation between a manager's education level and probability of promotion (size of the wage increase due to promotion) under asymmetric learning. With asymmetric learning, such correlations should also be stronger for first than for subsequent promotions. Although DeVaro and Waldman (2012) argue that their results are more consistent with asymmetric than with symmetric learning, the authors do not find conclusive evidence in support of asymmetric learning. For related work on asymmetric information, see Waldman (1984), Greenwald (1986), Bernhardt and Scoones (1993), Bernhardt (1995), Waldman (1996), and Ekinci, Kauhanen, and Waldman (2019).<sup>62</sup>

#### *C2. Performance Incentives through Bonus Pay*

The model proposed abstracts from the possibility that firms may induce managers to exert effort on the job by making their pay contingent on their performance. When this is the case, fixed pay (salary or, as it is referred to here, wage) and variable pay (bonus) may be linked directly to current performance, not just

<sup>61</sup> As Baker, Gibbs, and Holmström (1994a, 903) argue, their "findings suggest that . . . career dynamics are driven by the firm's learning about and selecting on ability" and note that the fastest promoted managers at the firm tend to exit more often, which is consistent with their ability being perceived as relatively high also by other firms and so with information about ability being public.

<sup>62</sup> A general model in which the flow of information about managers across firms is imperfect and possibly affected by a firm's retention and job assignment decisions is beyond the scope of this paper. Allowing for firms' private information about managers' ability—and so for different information sets across firms—would render the identification of the model prohibitively difficult in light of my data from one firm. The model developed in this paper provides just a first attempt at measuring the importance of learning about ability and human capital acquisition for job and wage mobility under assumptions commonly maintained in the empirical literature on careers.



indirectly to past performance through beliefs about ability. If incentives matter, then current wages may be expected to be more strongly correlated with current performance than with past performance. As Kahn and Lange (2014) also note, the BGH data do not support the notion that the correlation of wages with current performance exceeds that with past performance. As for bonus pay, Frederiksen, Kriechel, and Lange (2017) analyze bonus information in the BGH data, available from 1981, and find evidence consistent with the use of bonuses to set incentives for performance (see also Gibbs 1995). Bonuses, though, represent a small fraction of total pay: they are paid to 25% of managers, mainly at the highest levels, and do not account for a large portion of pay. The median bonus of managers receiving one at (the original) levels 1–3 is less than 10% of salary; at (the original) level 4, it is less than 15%. Hence, this evidence does not seem to suggest that performance incentives alone are a more critical determinant of pay than the mechanisms I focus on.<sup>63</sup>

### *C3. Performance Incentives through Promotion Tournaments*

Firms can provide workers with incentives for performance through the implicit promise of promotion. In tournament models (Lazear and Rosen 1981), this mechanism links workers' future promotions and pay to their current performance, which encourages workers to exert effort. Tournament models with homogeneous workers and without sorting are easily distinguishable from models of learning and job assignment, since they imply that all workers exert the same amount of effort, so the winner of a tournament is determined merely by luck. Tournament models with heterogeneous workers and sorting instead are difficult to empirically distinguish from models of learning and job assignment (Rosen 1986). For instance, both tournament and learning models with sorting predict serial correlation in promotion rates and wage increases. One aspect of the BGH data at odds with the idea that tournaments are central to the dynamics of jobs and wages at the BGH firm is the high frequency of (real) wage decreases. A tournament model does not naturally lead to such decreases, unlike a learning model. Also, Baker and Holmström (1995) document that the difference in average wages between adjacent levels of the BGH firm's job hierarchy ranges from 18% to 47%, yet the wage premium upon promotion is around only 7%. Thus, wages at each job level are highly dispersed. The authors interpret these findings as evidence against the hypothesis that the wage structure at the firm is governed by a promotion tournament.

### *C4. Production Complementarities*

I model managers as imperfect substitutes in production by allowing a manager's expected output to depend on the manager's previous job assignment, as

<sup>63</sup> The incidence, size, and variability of bonuses at the firm tend to be higher at higher levels (Gibbs 1995). According to my estimates, learning contributes to wage growth and dispersion primarily through its impact on promotions to higher levels, at which wages are higher and more dispersed. Thus, the effect of learning on wage growth and dispersion that I estimate excluding bonuses can be conjectured to be a lower bound on the effect of learning on total pay growth and dispersion.

discussed in section V.A. I abstract from explicitly modeling complementarities in production or capacity constraints for three reasons. First, the goal of my analysis is to investigate whether a model that combines learning about ability, human capital acquisition, and job assignment can account for the main patterns of individual job and wage mobility in firms. To this end, I formulate assumptions that make my model comparable to those I integrate into my framework, which ignore such complementarities. Second, the variability of the size of job levels over the sample years suggests that the firm may not be subject to too stringent capacity constraints on the employment of managers or their assignment to job levels. Additionally, the correlation of performance or wages across managers who are part of, say, a same cost center, which is a proxy for a production unit within the firm, is on average fairly low, which is consistent with a low degree of complementarity among managers. Third, work that allows for complementarities in production among workers typically addresses questions that are different from those I focus on here—such as the importance of organizational capital for the growth rate and the size distribution of firms (Prescott and Visscher 1980)—or, because of its complexity, confines the analysis to essentially static environments (Ferrall 1997; Ferrall, Salvanes, and Sørensen 2009) or stylized cases (Davis 1997). In contrast to the literature, I allow for a fully dynamic interaction between forward-looking firms and workers, which leads firms and workers to face complex intertemporal trade-offs between the opportunity cost of investing in information and human capital and the benefit of doing so. Given the challenges that the estimation of such a model entails, the assumption of imperfect substitutability of managers in production that I formulate can be considered as a first approximation to developing a rich yet tractable empirical framework for careers in firms. By closely matching assignment and wage paths in the BGH data, the framework proposed, despite its simple production structure, seems to be nonetheless promising at capturing key features of careers.

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